

CHEMISTRY OF ADVANCED ENERGY STORAGE DEVICES FOR E-MOBILITY(18G6E14)

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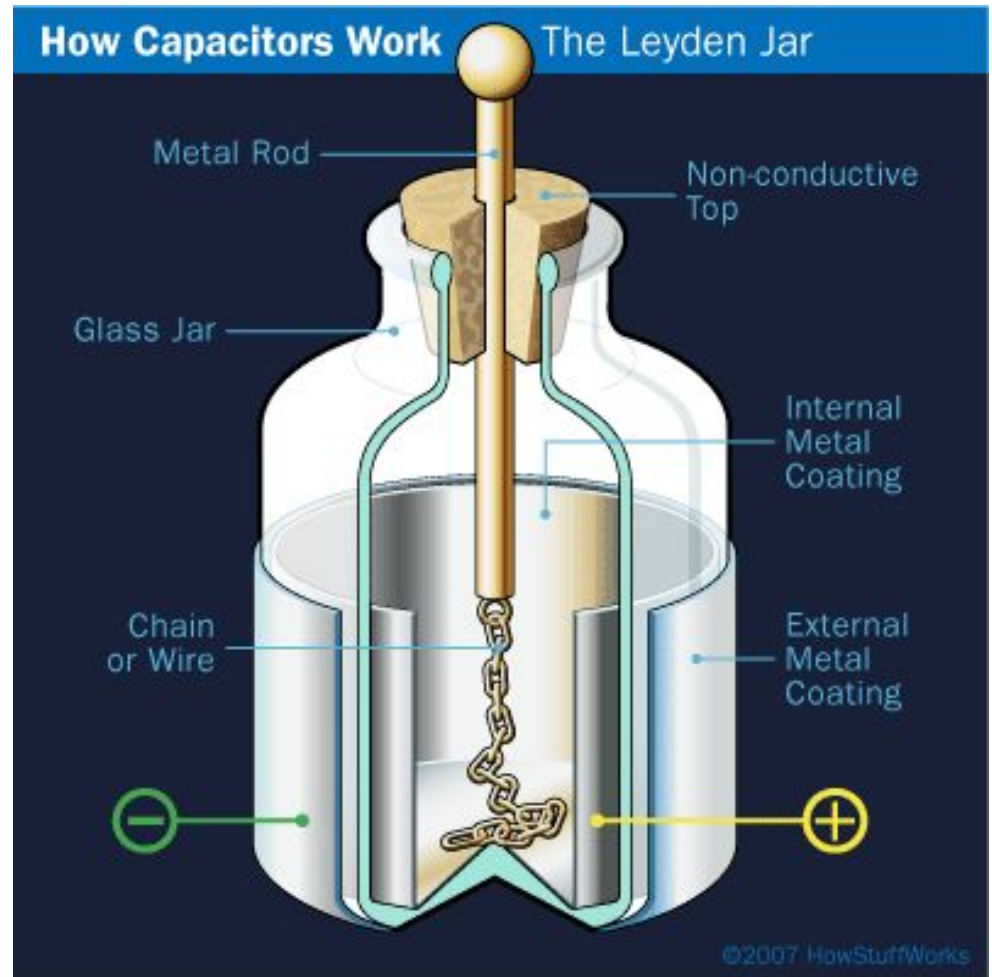




Introduction to super capacitors

History of Capacitors

- In 1740, Ewald Georg von Kleist constructed the first capacitor.
- In the same year Pieter von Musschenboek invented the Leyden Jar.
- Ben Franklin soon found out a flat piece of glass can be used in place of the jar model

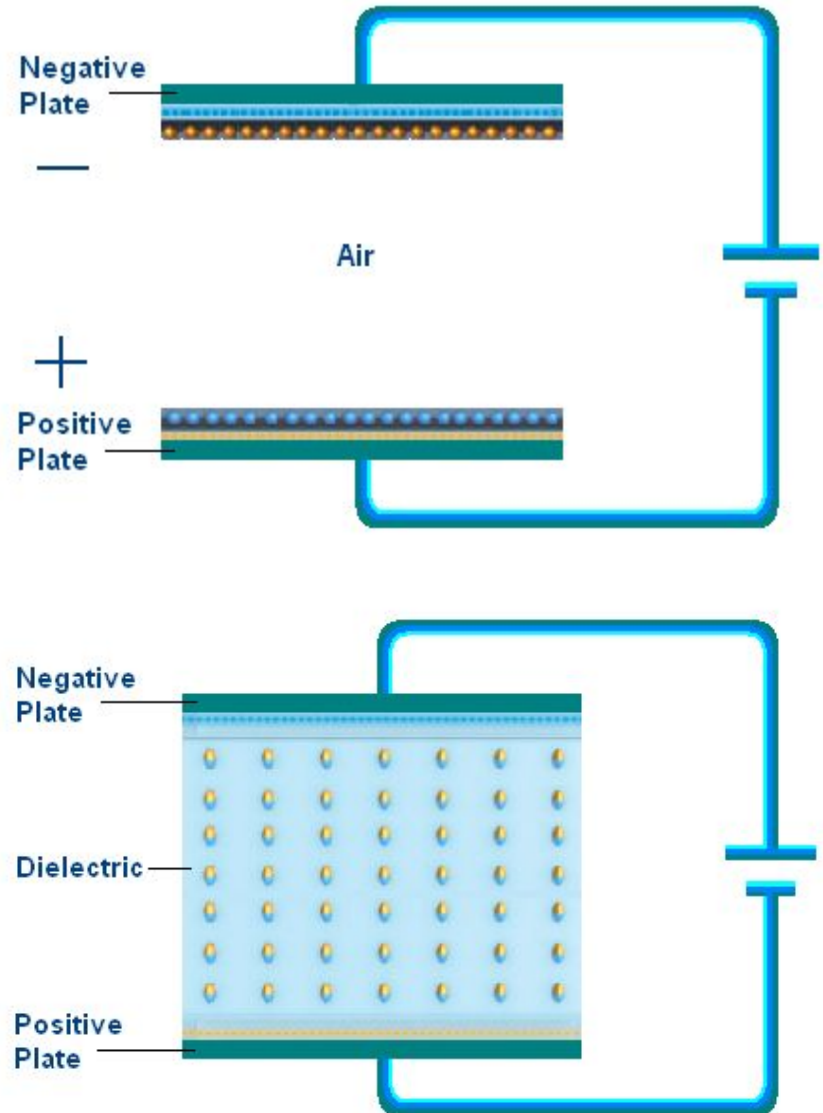


Capacitors

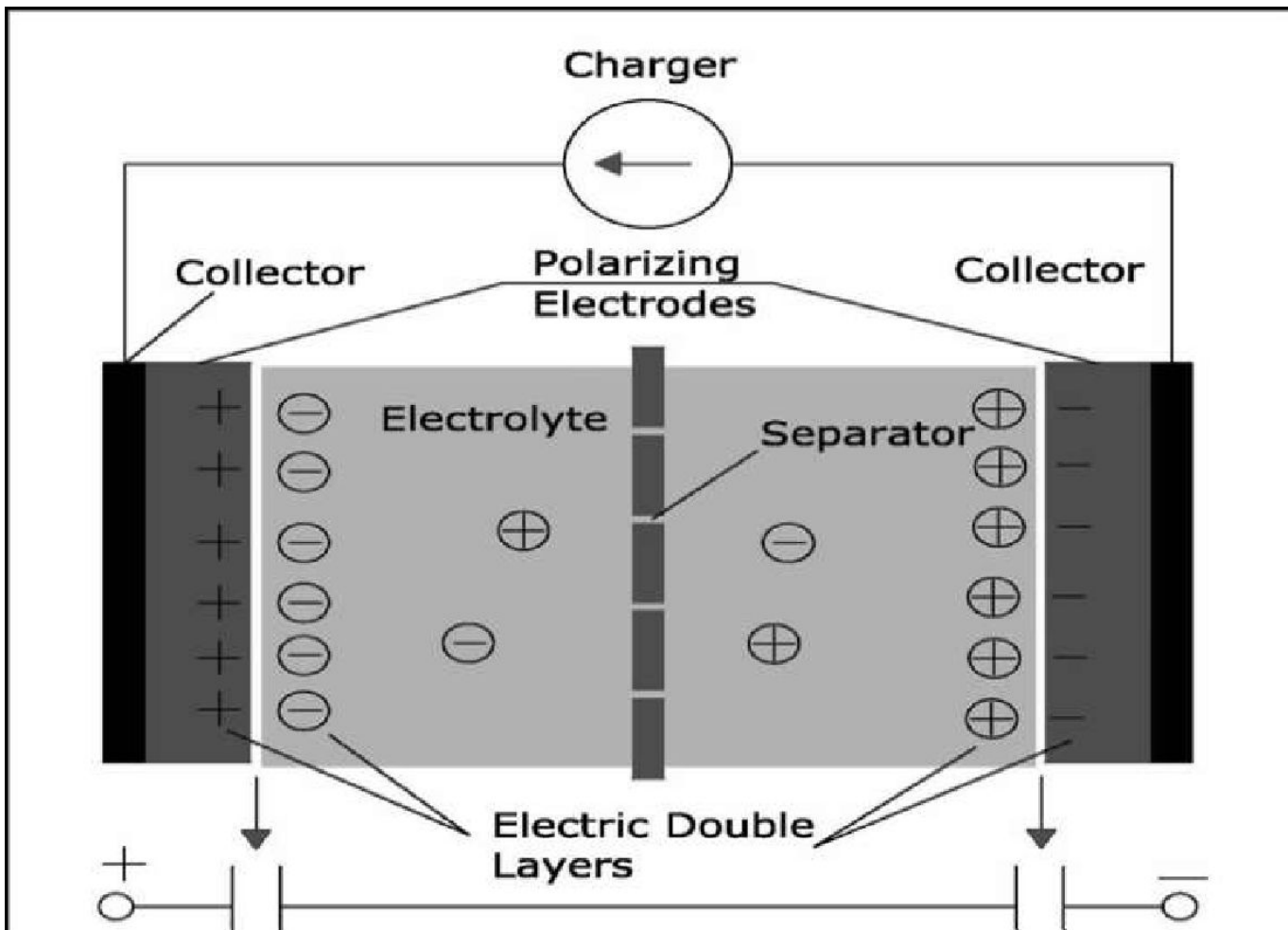
- Two parallel plates with dielectric in between
- Capacitance limited by flat surface area and dielectric properties

$$C = \epsilon_r \epsilon_0 \frac{A}{d}$$

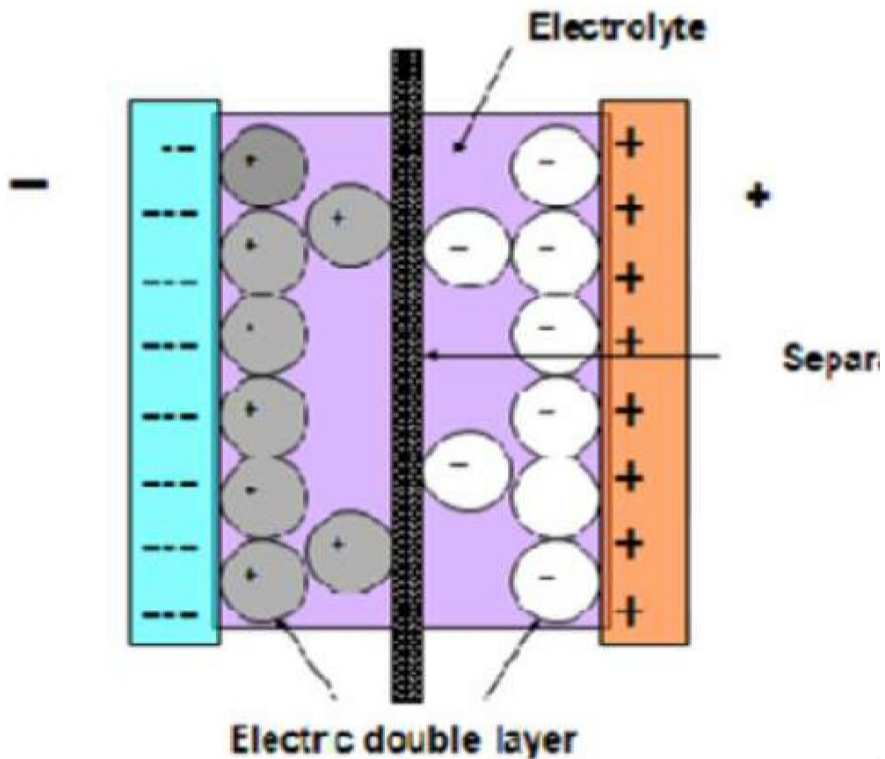
- C is the capacitance
- A is the area
- ϵ_r is the relative static permittivity (dielectric constant)
- ϵ_0 is the permittivity of free space (8.854×10^{-12} F/m)
- d is distance



Introduction to super capacitors



Introduction to super capacitors



- ❖ The purpose of having separator is to prevent the charges moving across the electrodes.
- ❖ The amount of energy stored is very large as compared to standard capacitor because of the enormous surface area created by the porous carbon electrodes and the small charge separation created by the dielectric separator.
- ❖ The distance between the plates is in the order of angstroms.



SUPERCAPACITOR

□ A **supercapacitor** is an electrochemical capacitor that has an very high energy density as compared to common capacitors, about 100 times greater.

□ Supercapacitor is also known as Electric Double Layer Capacitor(EDLC) or Ultracapacitor.

□ The capacitance range is From 100 Farad to 5K Farad



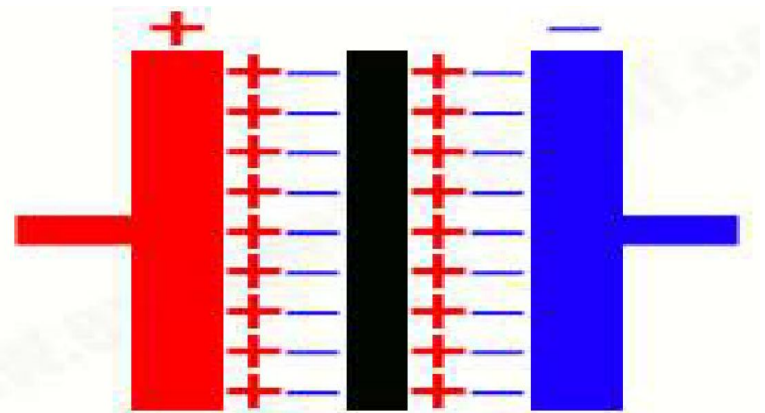
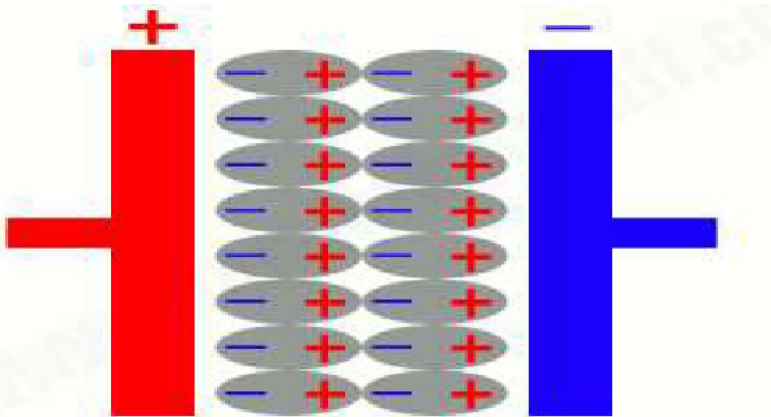
History of Super Capacitors

- The Electric Double Layer Capacitor effect was first noticed in 1957 by General Electric.
- In 1957 H. Becker developed a "Low voltage electrolytic capacitor with porous carbon electrodes".
- Standard Oil of Ohio re-discovered this effect in 1966.
- Standard Oil of Ohio gave the licensing to NEC, which in 1978 marketed the product as a “supercapacitor”.

Basic Design

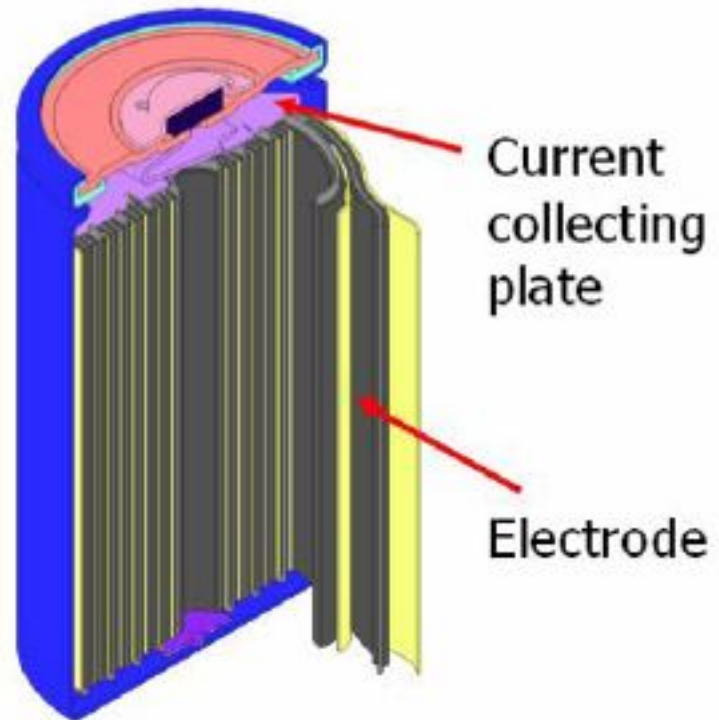
□ Electrochemical capacitors (supercapacitors) consist of two electrodes separated by an ion permeable membrane (separator), and an electrolyte electrically connecting both electrodes. When the voltage is applied, ions in the electrolyte form electric double layers of opposite polarity to the electrode's polarity.

□ For example, positive electrodes will have a layer of negative ions and negative electrodes will have a layer of positive ions.



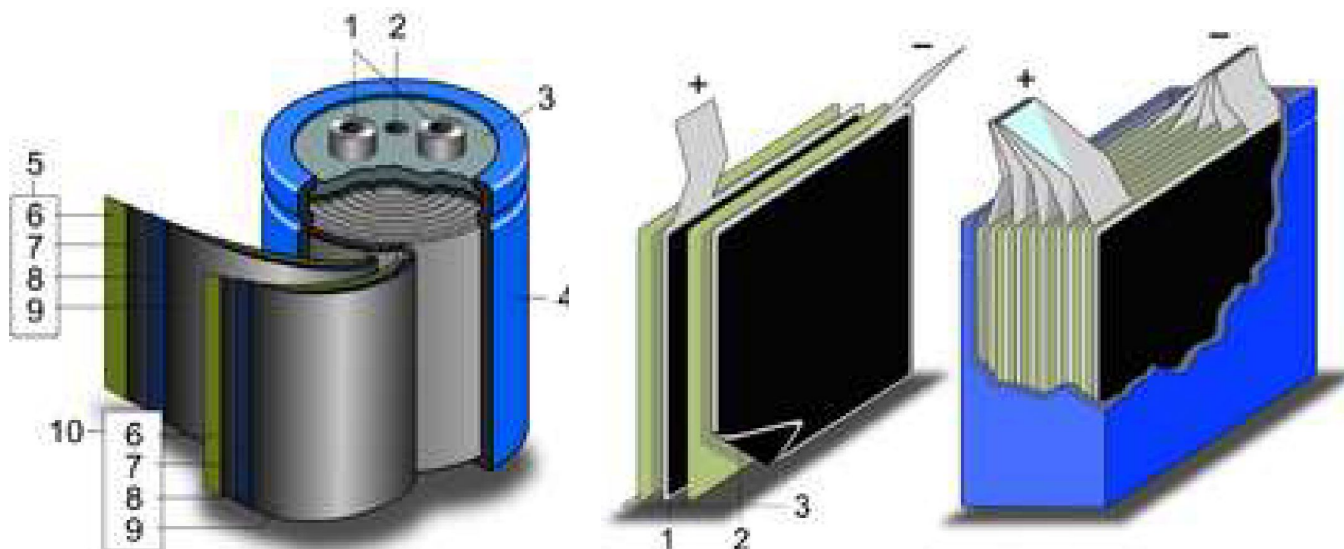
Super/Ultracapacitors

- Ultracapacitors technology: construction



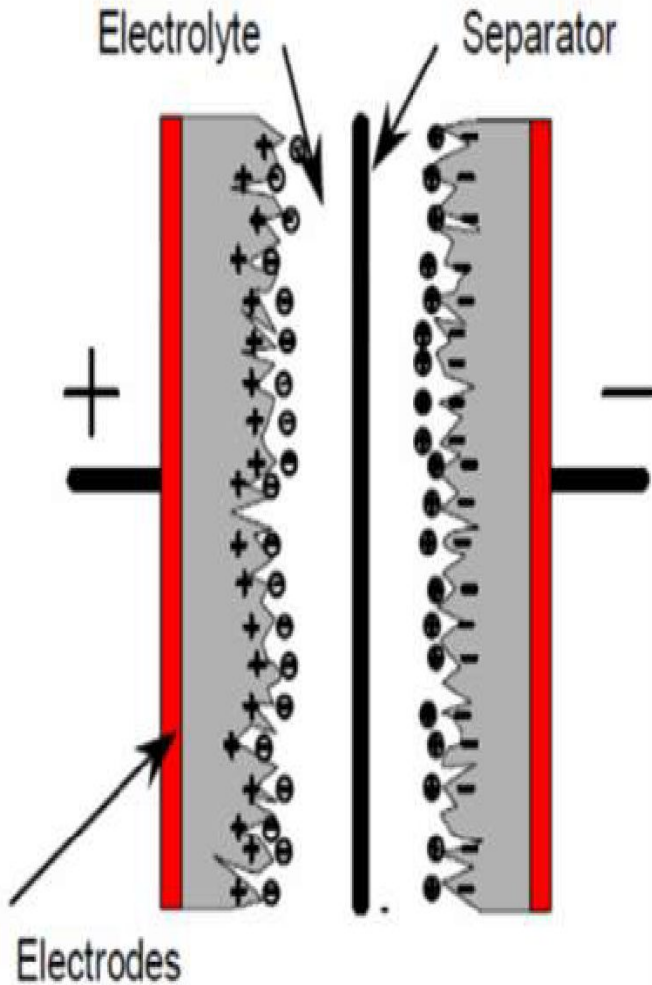
CONSTRUCTION

- Supercapacitors are constructed with two metal foils, each coated with an electrode material such as activated carbon.
- The electrodes are kept apart by an ion permeable membrane (separator) used as an insulator to protect the electrodes against short circuits.
- The construction is subsequently rolled or folded into a cylindrical or rectangular shape and is packed in an aluminium can.



1. Positive electrode
2. Negative electrode
3. Separator
4. Aluminum can
5. Positive pole
6. Separator
7. Carbon electrode
8. Collector
9. Carbon electrode
10. Negative pole

Cell Construction



- ❑ An ultracapacitor cell basically consists of two electrodes, a separator, and an electrolyte.
- ❑ Electrodes are made up of a metallic collector, which is the high conducting part, and of an active material, which is the high surface area part.
- ❑ The two electrodes are separated by a membrane, the separator, which allows the mobility of the charged ions but forbids the electronic conductance. Then the system is impregnated with an electrolyte.
- ❑ Working voltage is determined by decomposition voltage of electrolyte and depends mainly on environment temperature, current intensity and required lifetime.

Super/Ultracapacitors

- Some typical Maxwell's ultracapacitor packages:



- At 2.7 V, a BCAP2000 capacitor can store more than 7000 J in the volume of a soda can.
- In comparison a 1.5 mF, 500 V electrolytic capacitor can store less than 200 J in the same volume.



ELECTRODES

- ⦿ Electrochemical inert materials with the highest specific surface area are utilized for electrodes in order to form a double layer with a maximum number of electrolyte ions.
- ⦿ The main difficulties are to find cheap materials, which are chemically and electrically compatible with the electrolyte.
- ⦿ As high surface active materials, metal oxides, carbon and graphite are the most interesting.
- ⦿ Capacitors for high energy applications require electrodes made of high surface area activated carbon with appropriate surface and pore geometry. The electrode capacitance increases linearly with the carbon surface area.

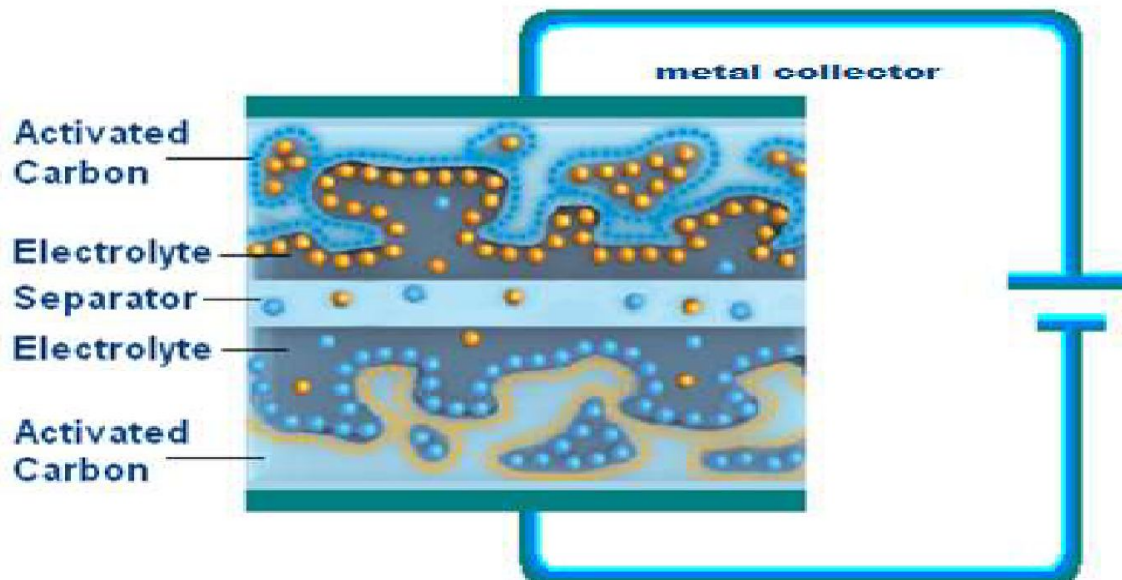


ELECTROLYTE

- ⦿ The electrolyte may be of the solid, organic or aqueous type.
- ⦿ Organic electrolytes are produced by dissolving quaternary salts in organic solvents. Their dissociation voltage may be greater than 2.5 V.
- ⦿ Aqueous electrolytes are typically KOH or H₂SO₄, presenting a dissociation voltage of only 1.23 V.
- ⦿ As a consequence of the quadratic dependence of the energy density of the capacitor on the capacitor's voltage use of an organic electrolyte would be desirable.
- ⦿ However, if power density is important, the increase in the internal resistance (ESR) due to the lower electrolyte conductivity has to be considered as well.

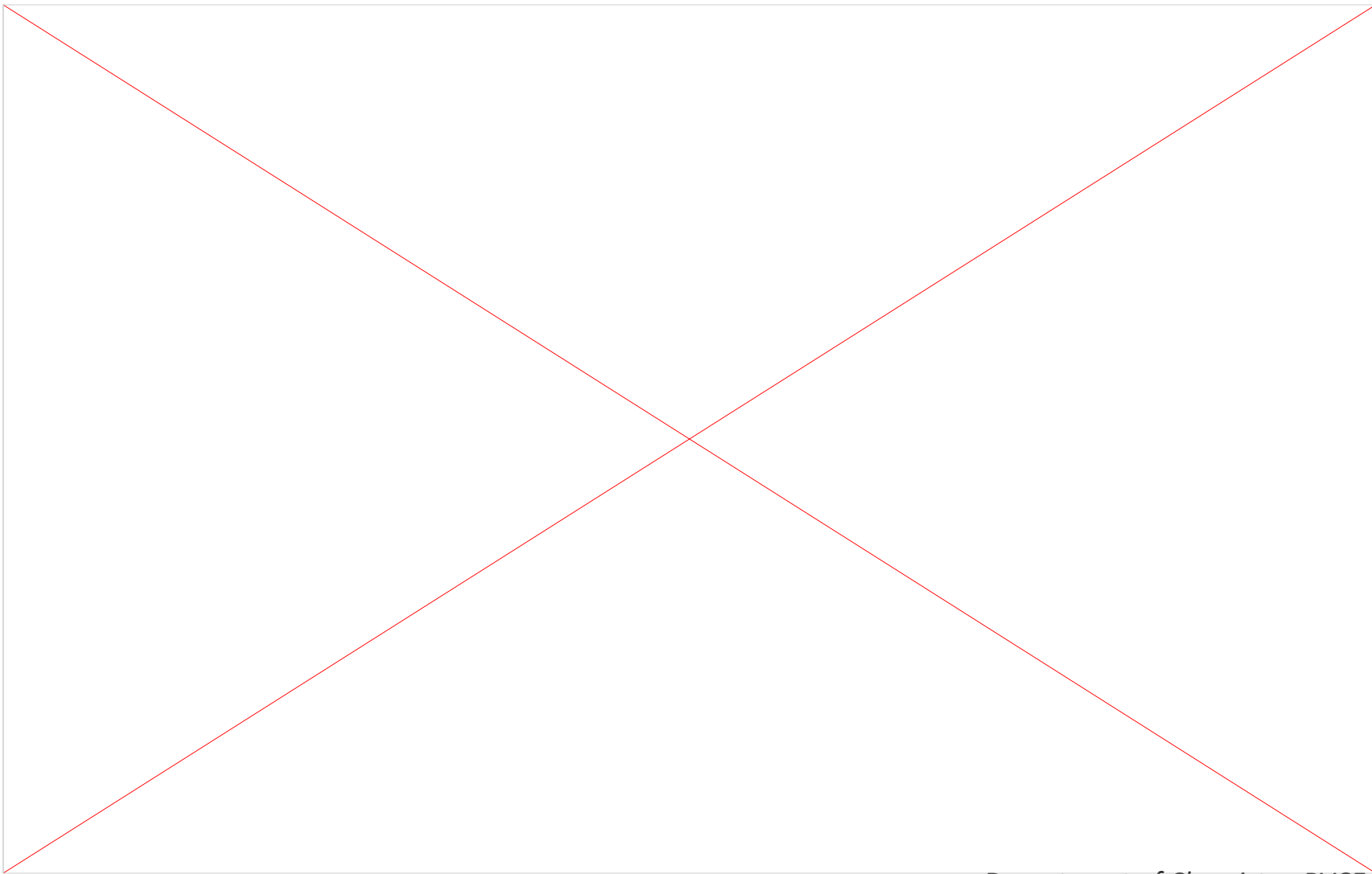
WORKING OF SUPERCAPACITOR

● In a supercapacitor, there is no conventional dielectric. Both plates are soaked in an electrolyte and separated by a very thin insulator. When the plates are charged, an opposite charge forms on either side of the separator, creating what's called an electric **doublelayer**. This is why supercapacitors are often referred to as double-layer capacitors



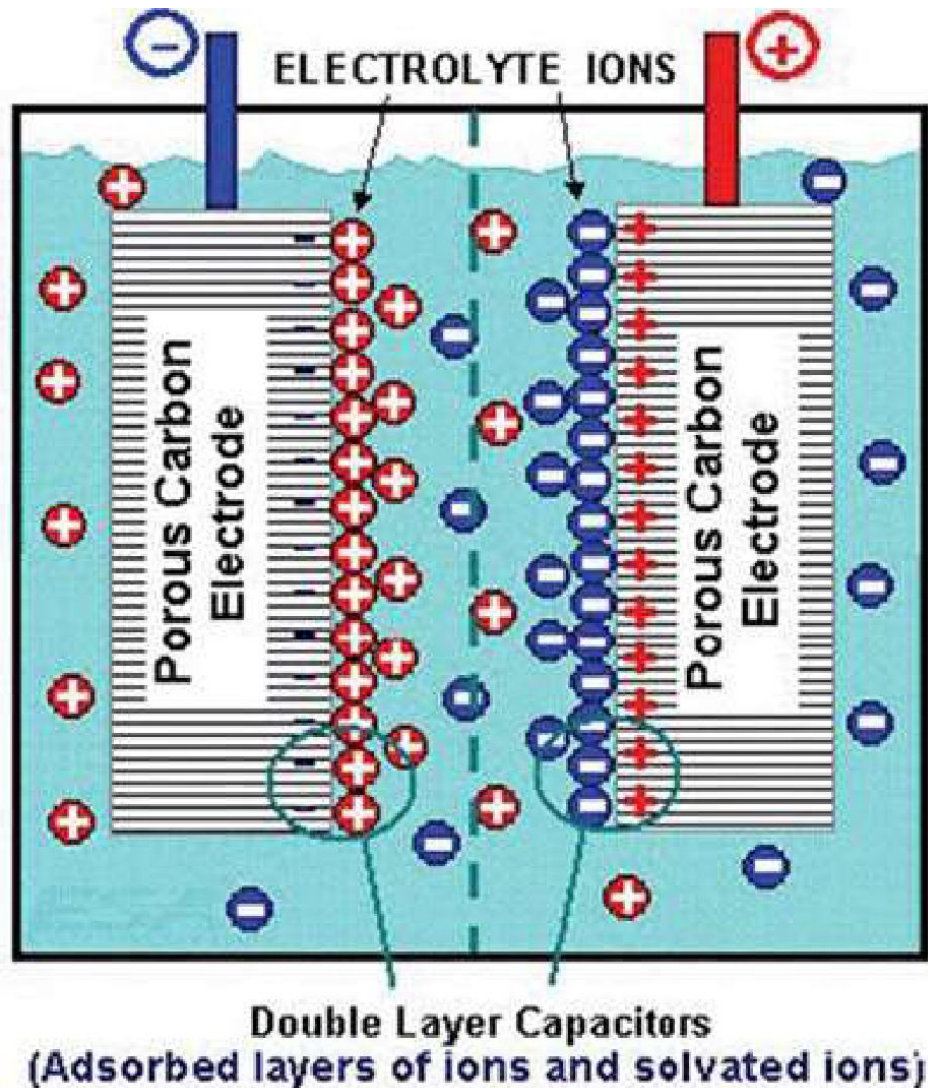


WORKING OF SUPERCAPACITOR



ELECTRIC DOUBLE LAYER

- Electrochemical capacitor has two electrodes, separated by a separator, which are electrically connected to each other via the electrolyte.
- When voltage is applied, and plates get charged, an opposite charge forms on the either side of the separator creating a electric double layer.





Materials/technologies used

TECHNOLOGY USED

- Carbon nano tubes, carbon aerogels are used for supercapacitors plates or electrodes.
- Sodium perchlorate (NaClO_4) or lithium perchlorate (LiClO_4) are used as electrolytes.
- Polyacrylonitrile($\text{C}_3\text{H}_3\text{N}$)_n is used as a separator(thickness 0.3-0.8 nm).
- Aluminium as a packing component.

Why these materials are taken

● Electrodes:-

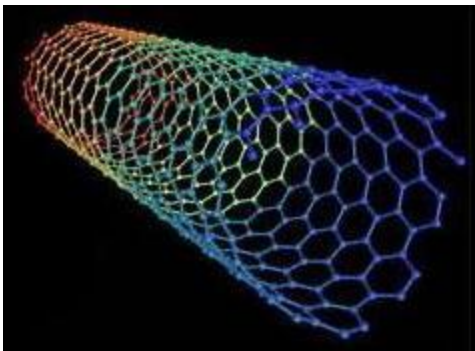
- 1)Carbon nanotubes greatly improve capacitor performance, due to the highly wettable surface area and high conductivity.
- 2)Highly porous.

● Electrolytes:-

- 1) Wide working temperature (-90°C to 400°C).
- 2) Non flammable and low toxic.
- 3)Non-corrosive to electrode & packing components.

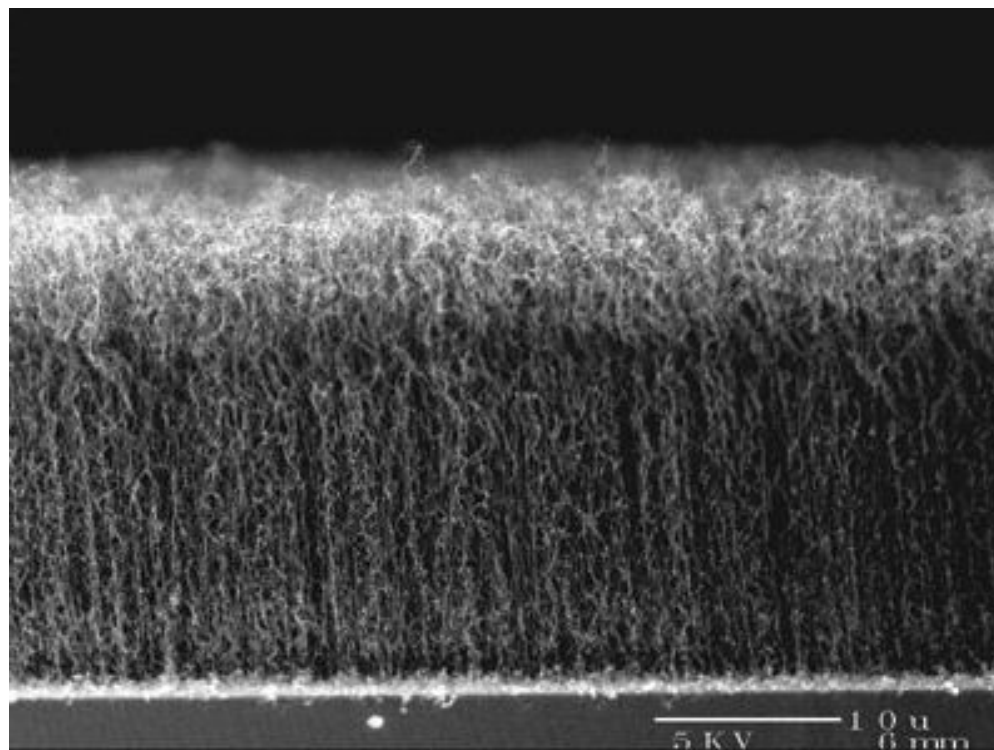
● Separator:-

- 1) Unique tensile strength (10^3 MegaPascals).
- 2)Electrical conductivity (1.5×10^4 S/m).
- 3)Not degraded easily.

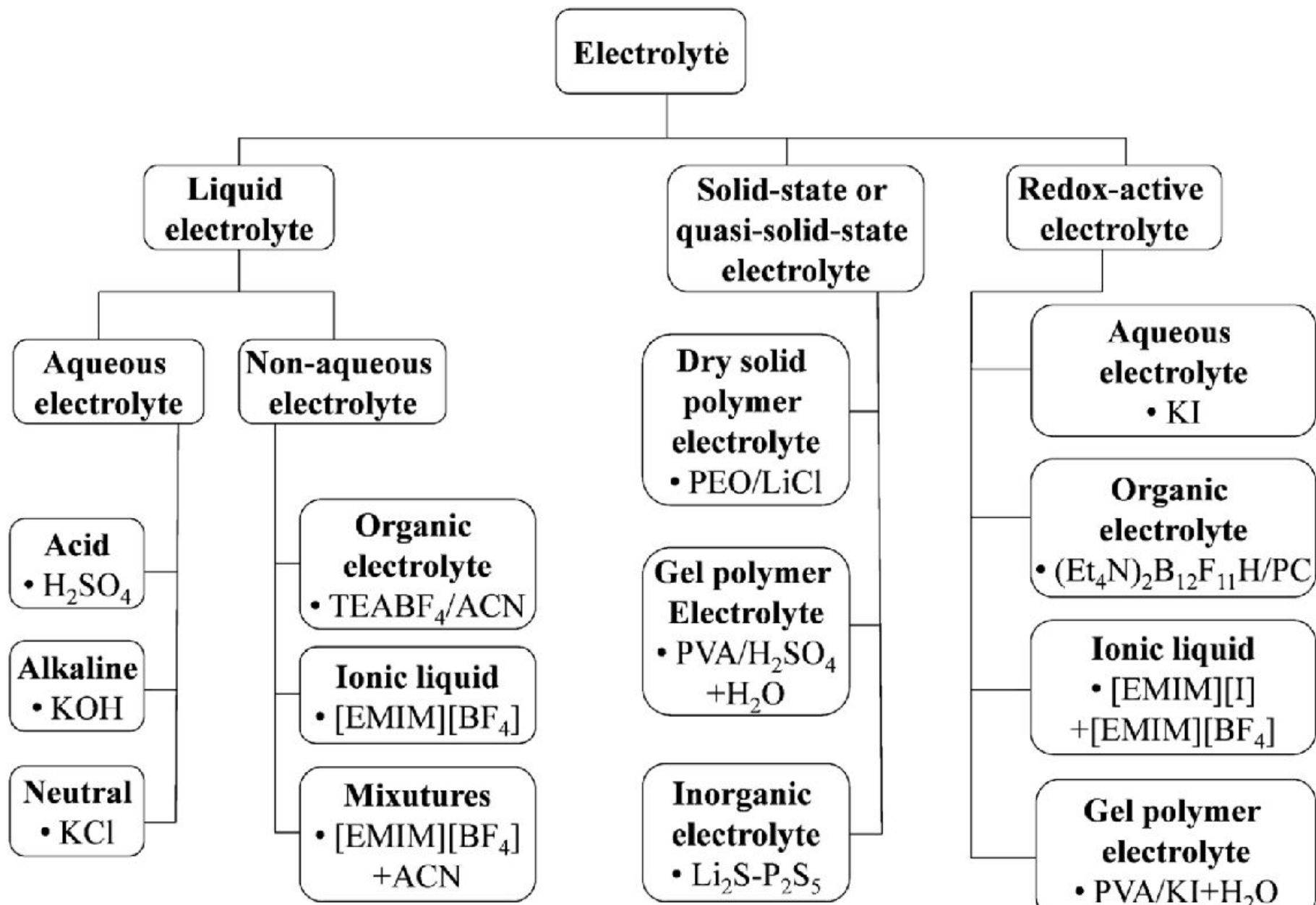


Carbon Nanotubes

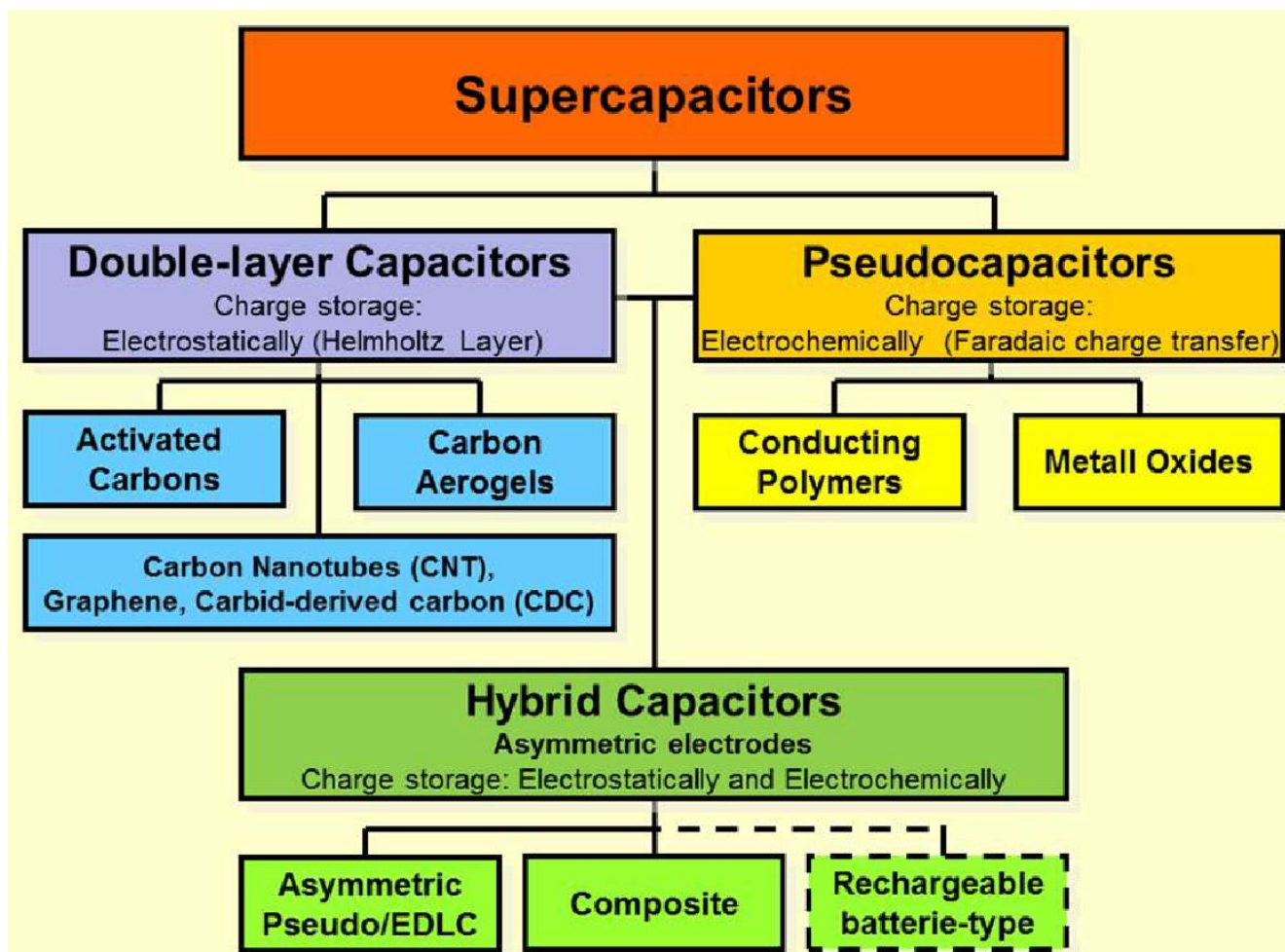
- Approximately 1/50,000th the width of a human hair
- Strongest and stiffest material on earth (>300 X Steel)
- Low density
- Semiconductor



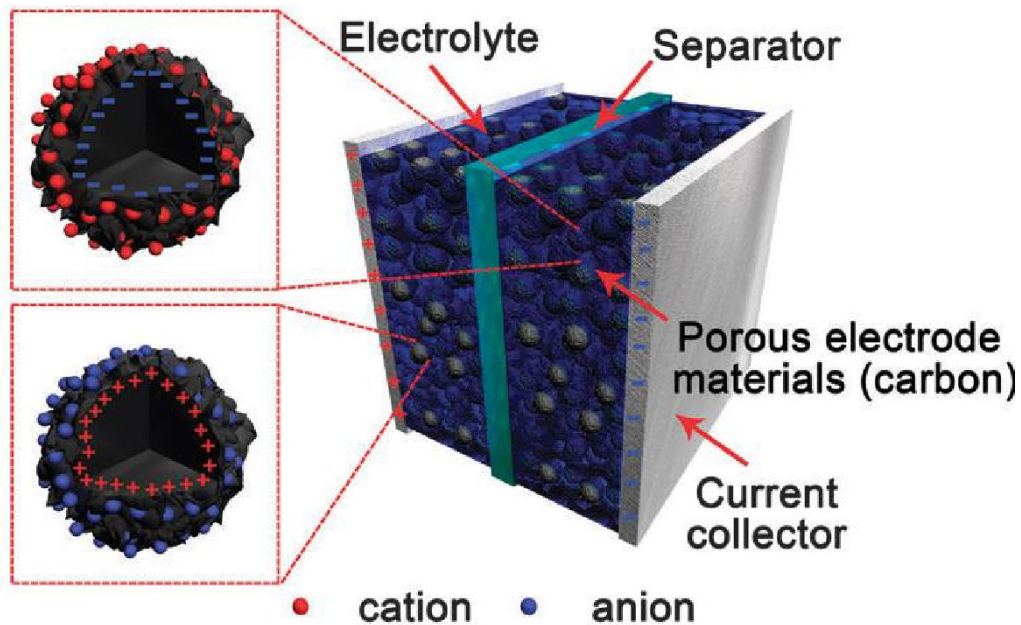
Electrolytes for electrochemical supercapacitors



Types of Supercapacitors



Classification of electrochemical supercapacitors

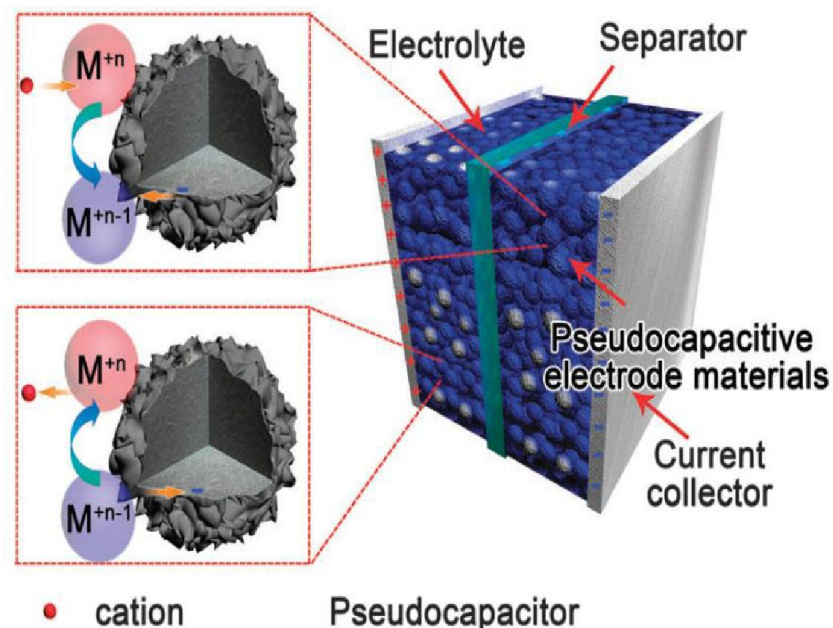


Electrical double-layer capacitor

Carbon-based materials with large surface area, such as activated carbon, carbon nanotubes (CNTs), and graphene.

- high power density
- excellent charge-discharge cycling stability

• low energy density owing to the relatively low specific capacitance



Pseudocapacitor

Transition metal oxides/hydroxides, sulfides, nitrides, and conducting polymers.

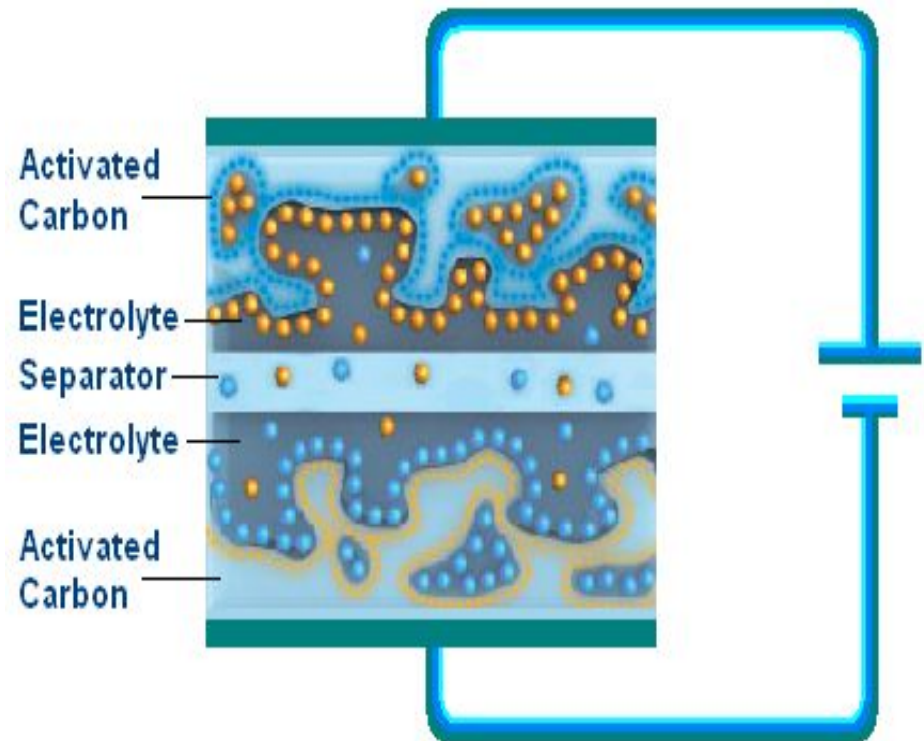
- Higher specific capacitances because the charges are stored on the electrode surface via fast and reversible faradaic reactions.

Advantages Limitation

- Low electrical conductivity
- poor rate capability

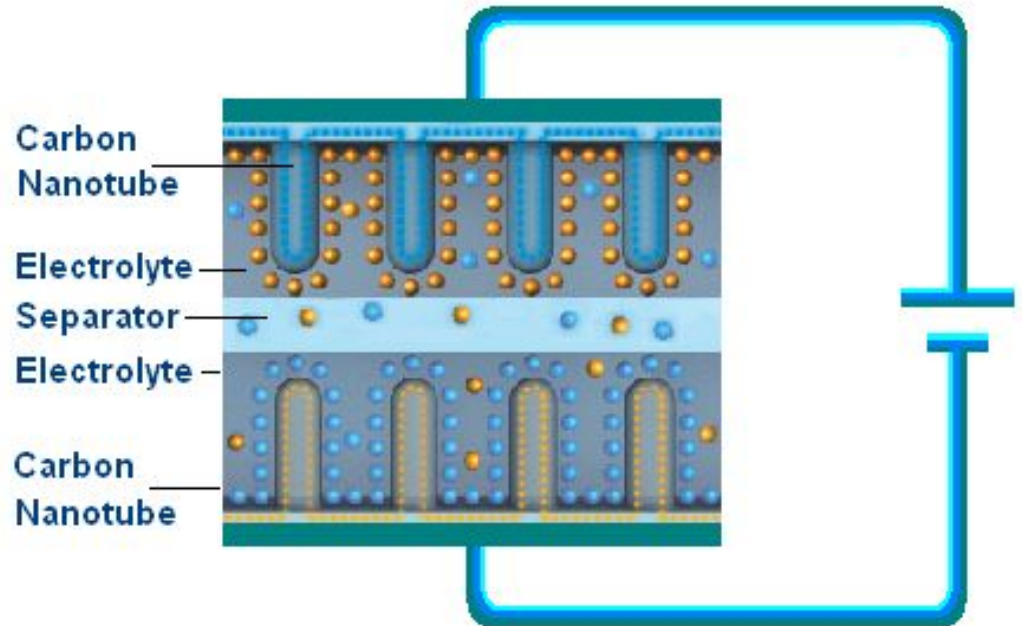
Double Layer Electrolytic Capacitors Using Activated Carbon

- Two layers consisting of nanoporous electrodes
- Separator is impregnated with an organic electrolyte
- Thin separator can only withstand low voltages



Double Layer Electrolytic Capacitors Using Carbon Nanotubes

- Under development at MIT
- Replaces activated charcoal with carbon nanotubes
- Aligned in a regular pattern that exposes greater surface area
- Dramatically increases effective area of electrodes
- Greatly increases power density





Ultra capacitors/super capacitors

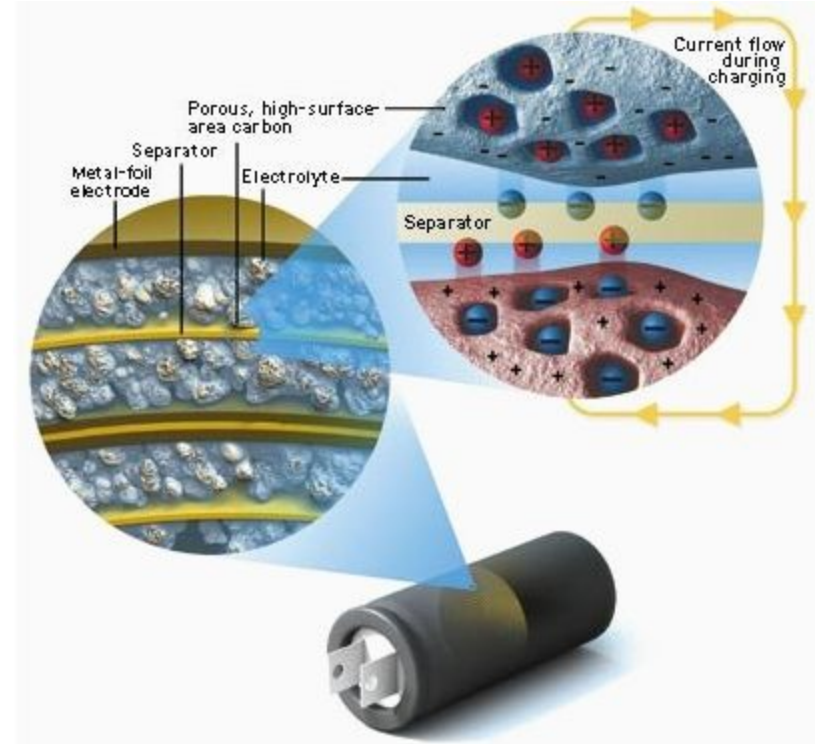
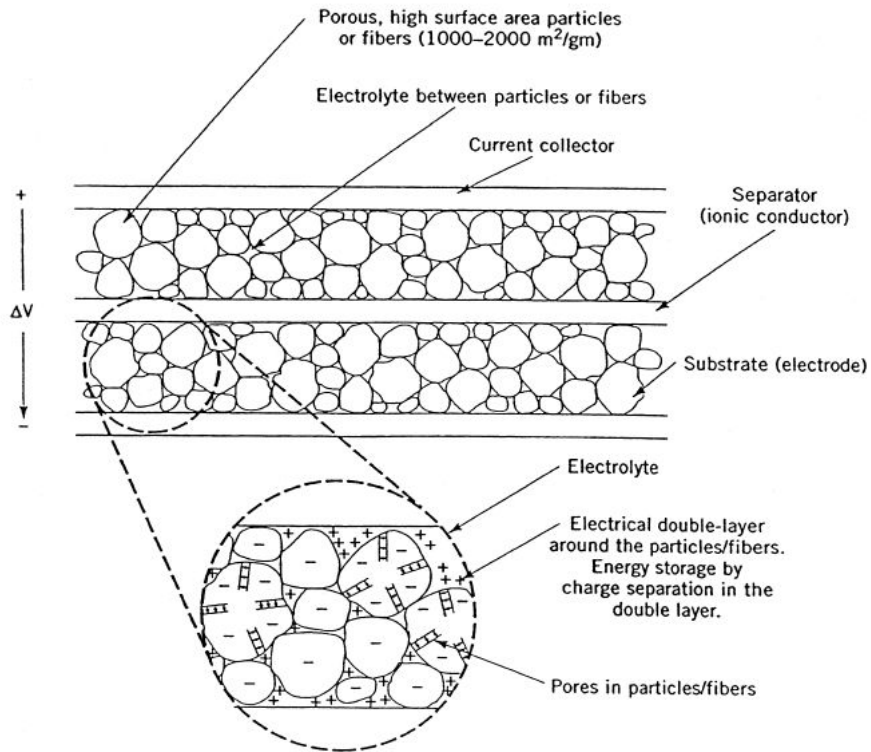
INTRODUCTION

- ⦿ Also known as supercapacitors or double-layer capacitors.
- ⦿ The capacitance can be as high as 2.6 kF(kilo-farad).
- ⦿ First commercial development in the Standard Oil of Ohio Research Center (SOHIO), in 1961. First high-power capacitors were developed for military purposes in Pinnacle Research Institute in early 1980"s.
- ⦿ Attractive for their high energy and power densities, long lifetime as well as great cycle number. Recent developments in basic technology, materials and manufacturability have made these an imperative tool for short term energy storage in power electronics.
- ⦿ Principle:- Energy is stored in ultracapacitor by polarizing the electrolytic solution. The charges are separated via electrode – electrolyte interface.

Ultracapacitors

- Ultracapacitors technology: construction
 - Double-layer technology

A. Burke / Journal of Power Sources 91 (2000) 37–50

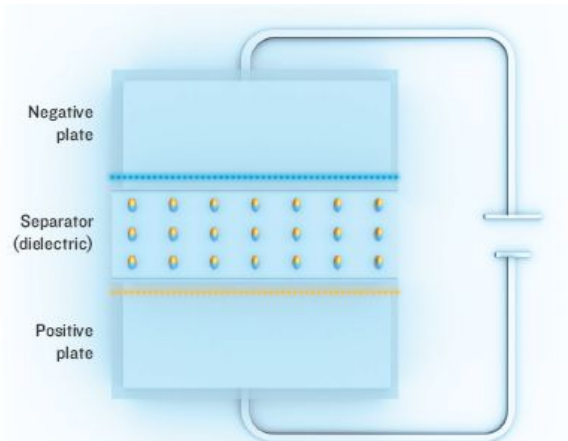


- **Electrodes:** Activated carbon (carbon cloth, carbon black, aerogel carbon, particulate from SiC, particulate from TiC)
- **Electrolyte:** KOH, organic solutions, sulfuric acid.

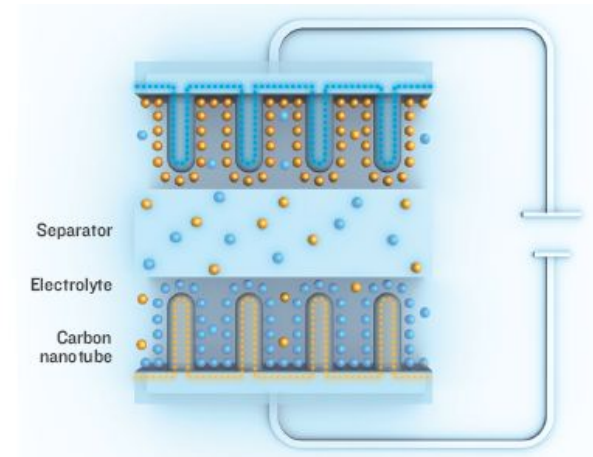
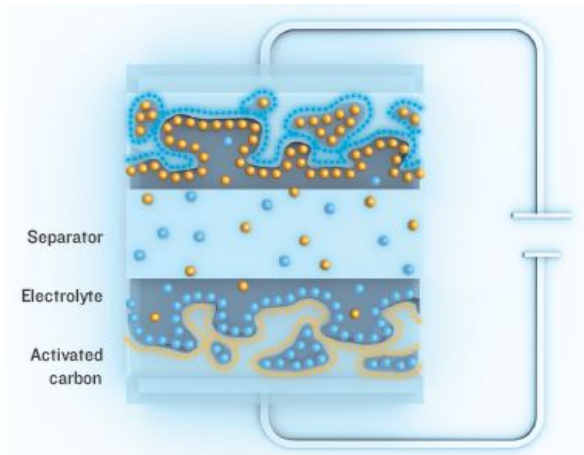
Ultracapacitors

- Ultracapacitors technology: construction

Traditional standard capacitor



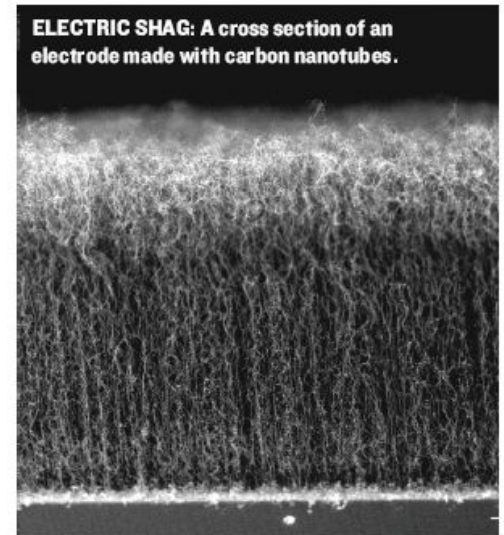
Double layer capacitor (ultracapacitor)



The charge of ultracapacitors, IEEE Spectrum Nov. 2007

$$C = \epsilon \frac{A}{d}$$

- Key principle: area is increased and distance is decreased
- There are some similarities with batteries but there are no reactions here.



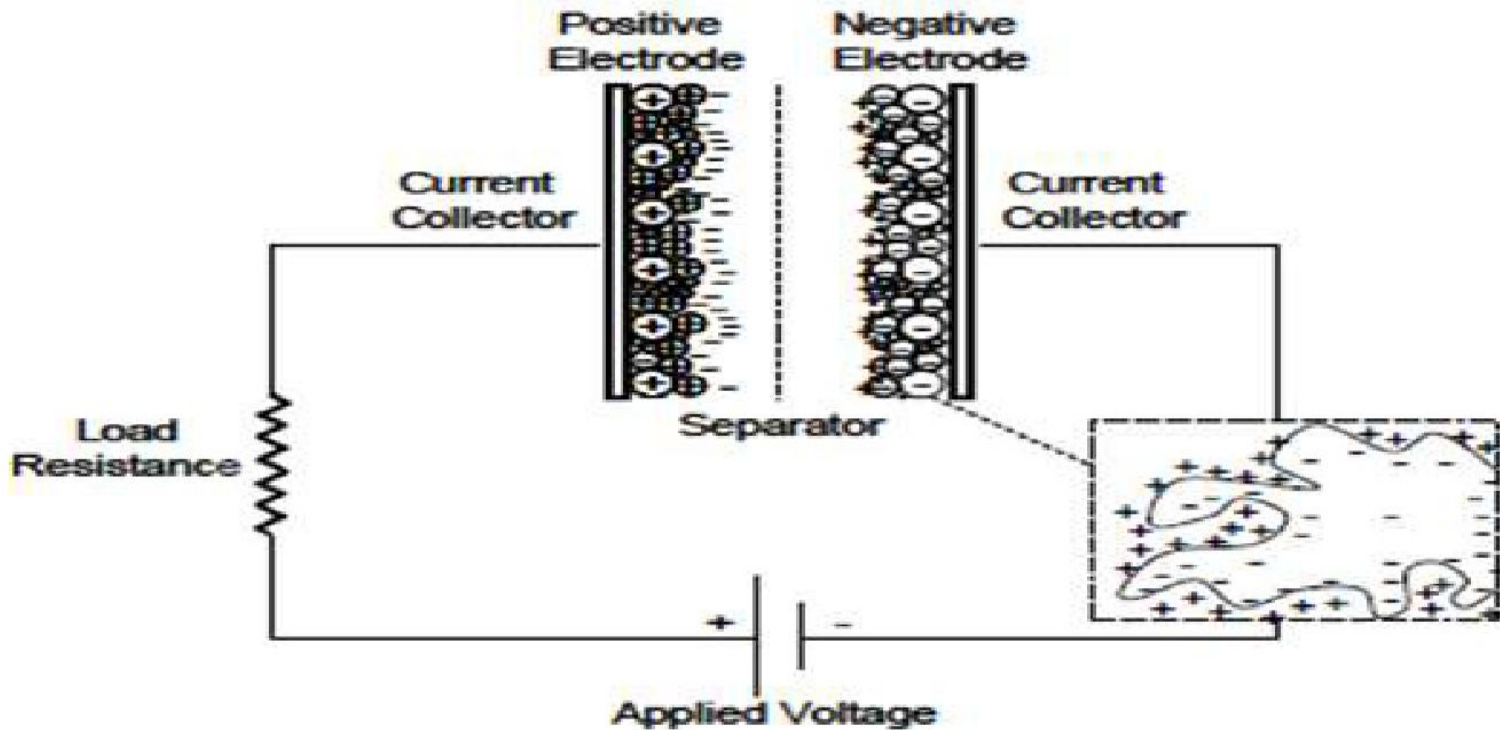


TAXONOMY OF ULTRACAPACITORS

- ❑ Ultracapacitors can be divided into three general classes:
 - ❑ Electrochemical double-layer capacitors
 - ❑ Pseudocapacitors, and
 - ❑ Hybrid capacitors
- ❑ Each class is characterized by its unique mechanism for storing charge. These are, respectively, non-Faradic, Faradic, and a combination of the two.

ELECTROCHEMICAL DOUBLE-LAYER CAPACITORS

- Electrochemical double-layer capacitors (EDLCs) are constructed from two carbon-based electrodes, an electrolyte, and a separator.

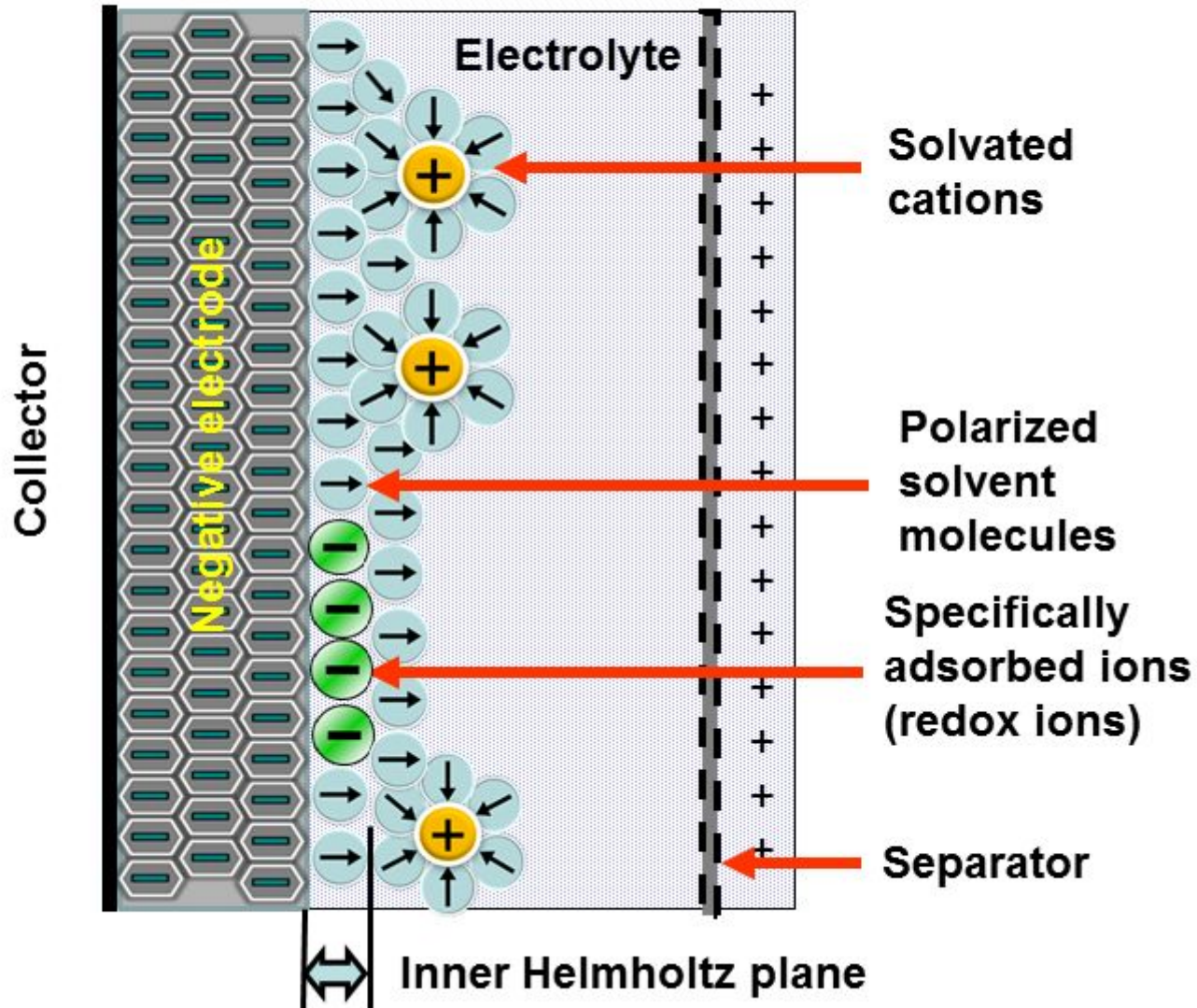




EDLCs(Contd..)

- ❑ EDLCs store charge electrostatically and there is no transfer of charge between electrode and electrolyte.
- ❑ EDLCs utilize an electrochemical double-layer of charge to store energy. As voltage is applied, charge accumulates on the electrode surfaces.
- ❑ These achieve very high cycling stabilities.
- ❑ The subclasses of EDLCs are distinguished primarily by the form of carbon they use as an electrode material.
- ❑ Different forms of carbon materials that can be used to store charge in EDLC electrodes are activated carbons, carbon aerogels, and carbon nanotubes.

Pseudo Capacitor





PSEUDOCAPACITORS

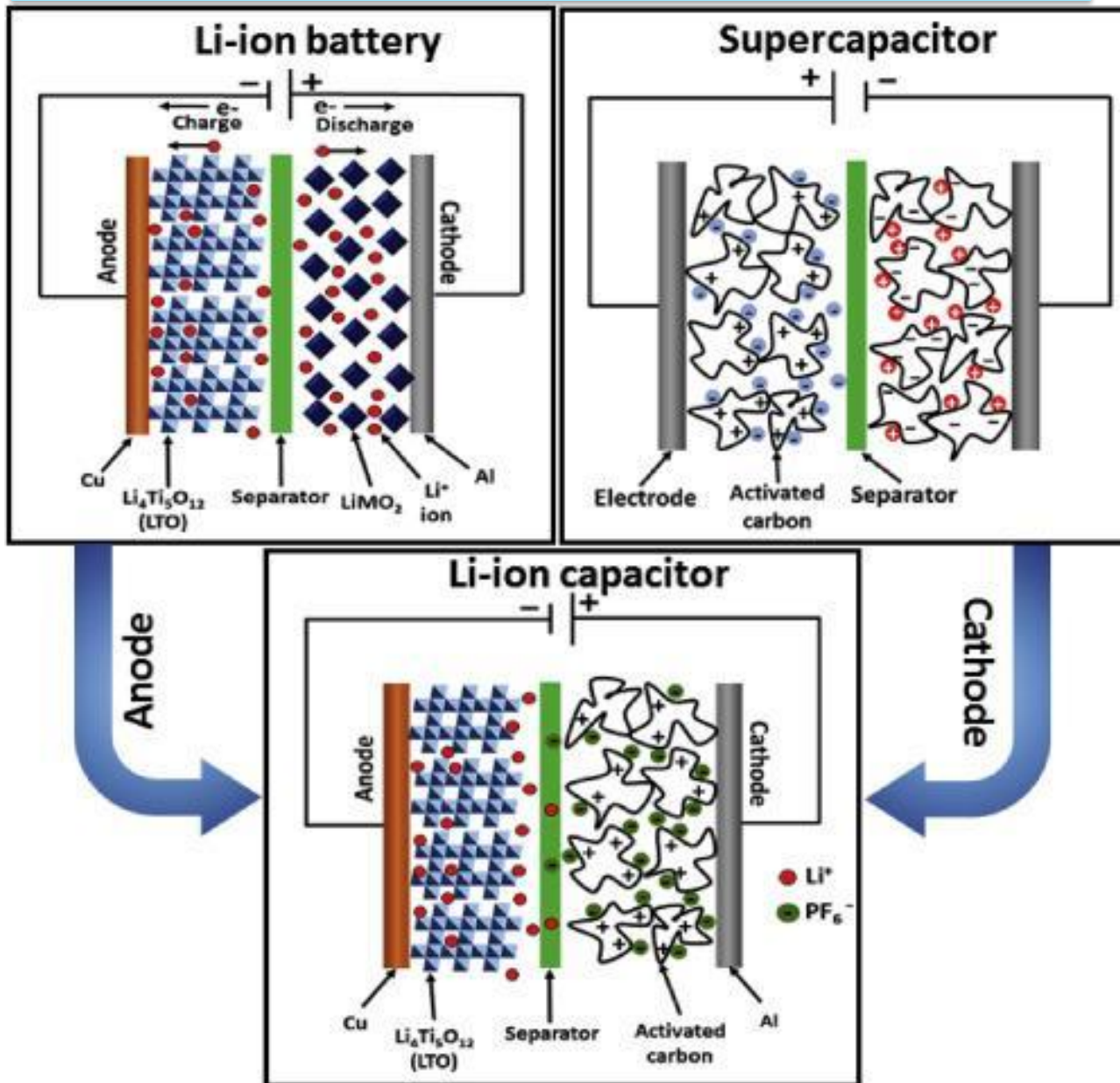
- ❑ In contrast to EDLCs, which store charge electrostatically, pseudocapacitors store charge Faradically through the transfer of charge between electrode and electrolyte. This is accomplished through reduction-oxidation reactions.
- ❑ Faradic processes may allow pseudocapacitors to achieve greater capacitances and energy densities than EDLCs. There are two electrode materials that are used to store charge in pseudocapacitors, conducting polymers and metal oxides.



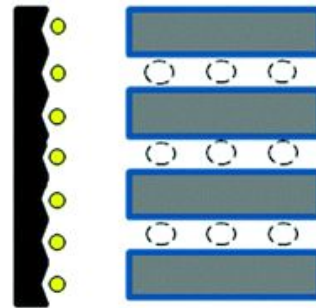
HYBRID CAPACITORS

- ❑ Hybrid capacitors attempt to exploit the relative advantages and mitigate the relative disadvantages of EDLCs and pseudocapacitors to realize better performance characteristics.
- ❑ Utilizing both Faradic and non-Faradic processes to store charge, hybrid capacitors have achieved energy and power densities greater than EDLCs without the sacrifices in cycling stability and affordability that have limited the success of pseudocapacitors.
- ❑ Research has focused on three different types of hybrid capacitors, distinguished by their electrode configuration: composite, asymmetric, and battery-type respectively.

HYBRID CAPACITORS

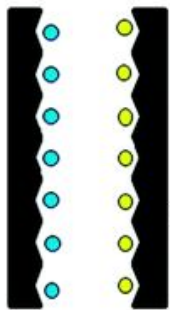


HYBRID CAPACITORS



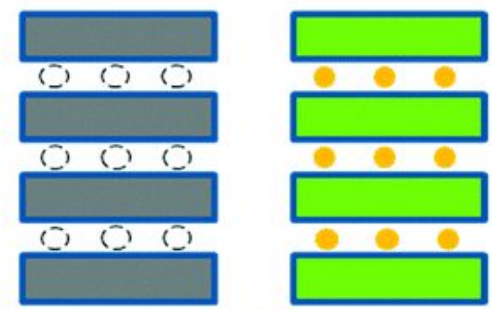
Hybrid device:
one battery-type and
one capacitor type
electrode in a hybrid cell

supercapacitor



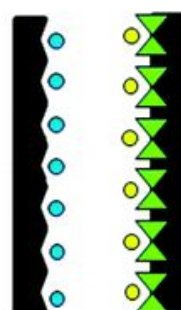
high power but
low energy

battery



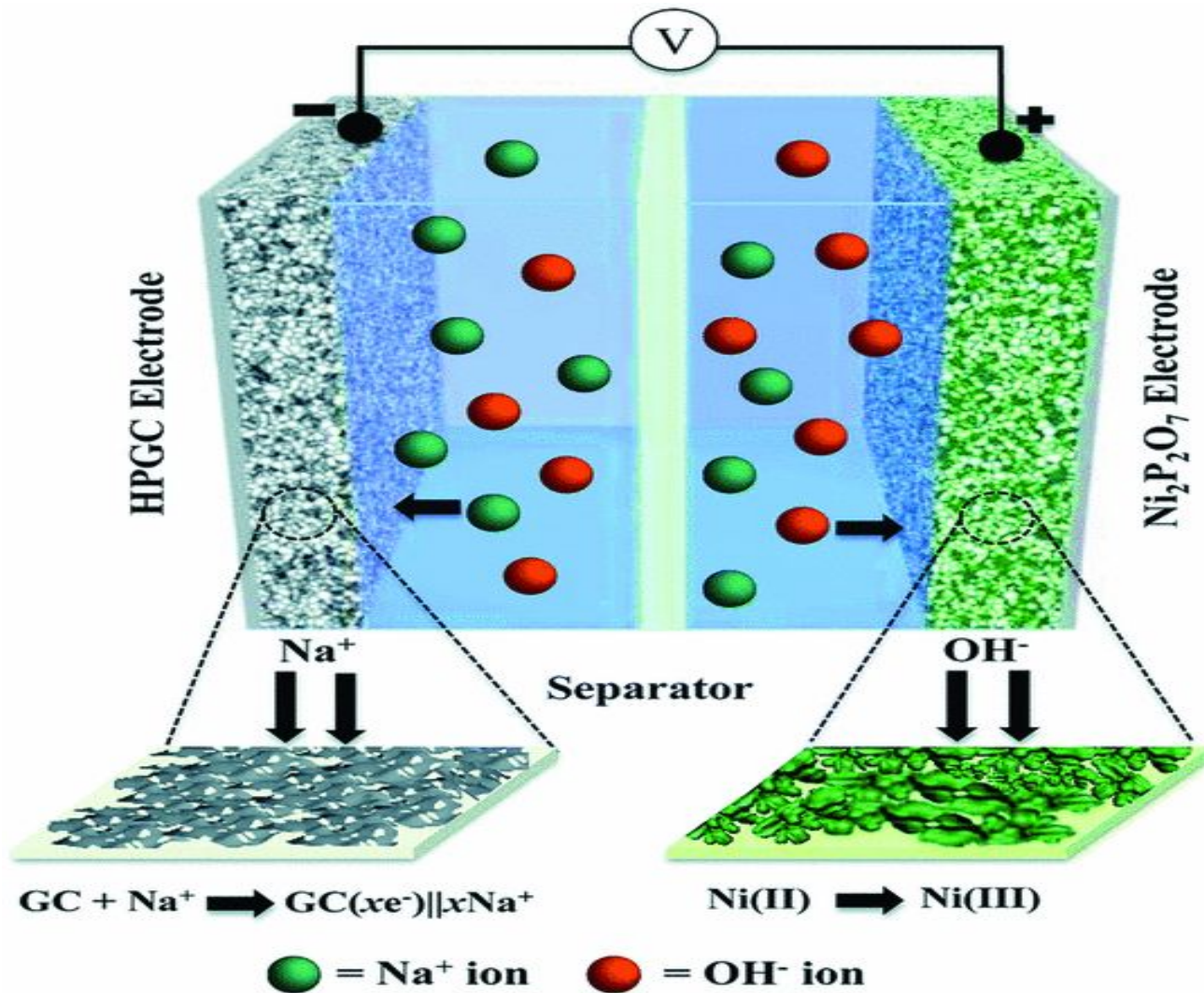
high energy but
low power

**Hybrid material in
a hybrid device**

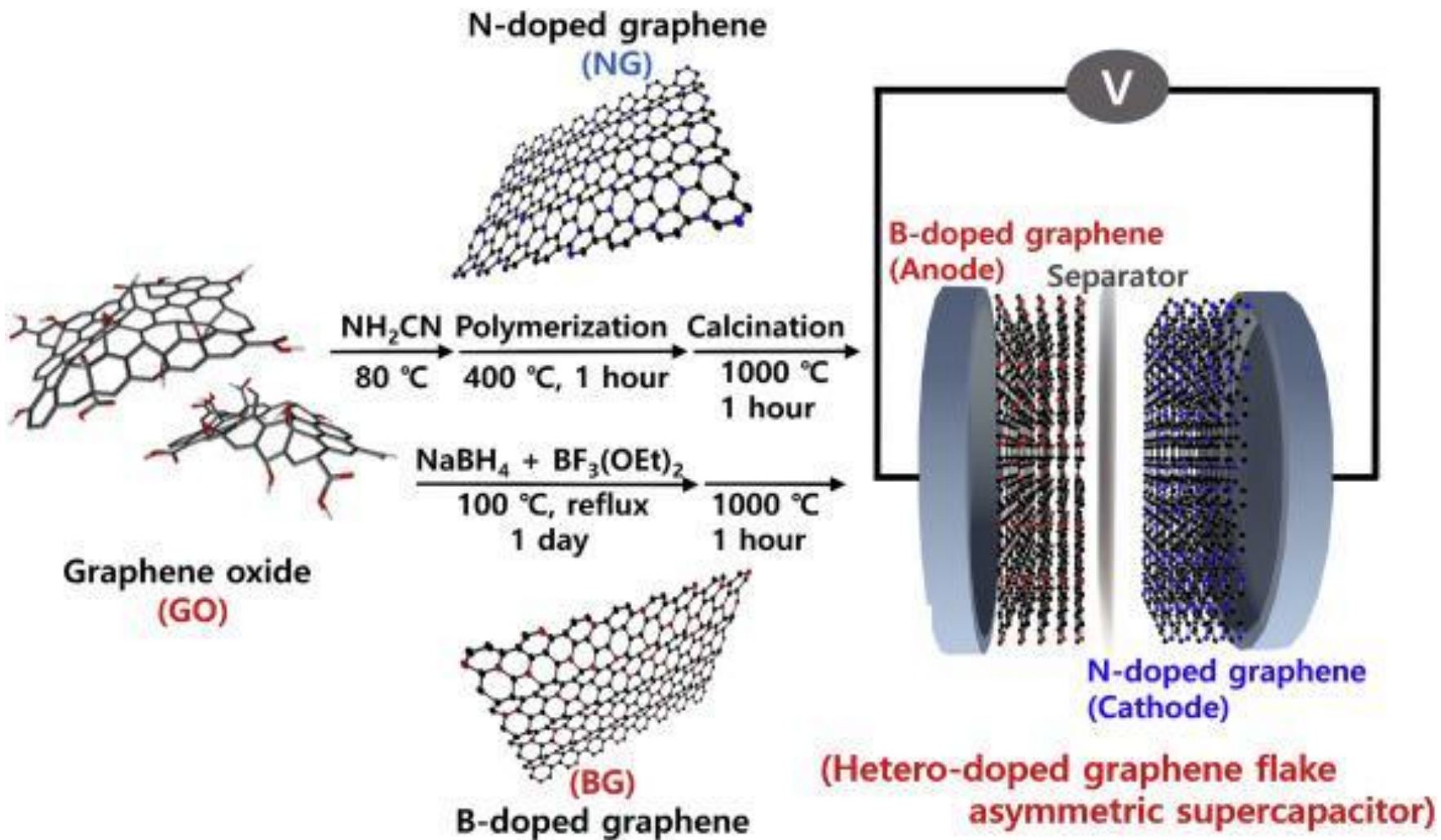


**Hybrid material: battery-type and capacitor type
materials combined in a hybrid electrode**

Asymmetric capacitor



Advanced asymmetric capacitor



Aqueous and Organic super capacitors

Electrolyte	Advantages	Drawbacks
Aqueous (H_2SO_4 , KOH)	- Small size of the ions - Not toxic electrolyte - Good wettability and better impregnation of the materials microporosity. - Additional capacitance (pseudo-capacitance) - Good performance for large temperature range - Low price	- Limit the service voltage - Faradic process - Accelerate the degradation of the component - Chemical modification of the electrode (with oxide electrode) - increasing the self-discharge
Organic (Et_4NBF_4 - Acetonitrile)	- High voltage (3V) - Electrostatic phenomena - No chemical modification of the electrode - Excellent reversibility - Good self-discharge - Low degradation of the component	- Toxic electrolyte - Bad impregnation of the materials microporosity - Narrow temperature range - High size of the ions - Expensive



Aqueous and Organic super capacitors

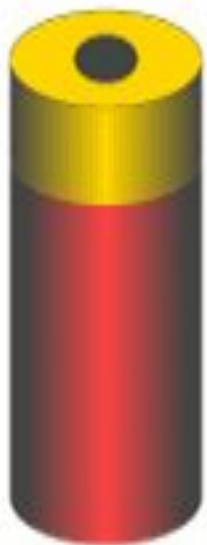
Characteristics	Aqueous technology	Organic technology
weight	45 Kg	12 Kg
Volume	23 ℓ	9,4 ℓ
Electrolyte	Non toxic	toxic
Electrodes	Activated carbon	Activated carbon
Tension max	28V	32V
Capacitance	200F	250F
Résistance HF	2,8 mOhm	3,4 mOhm



COMPARISON WITH BATTERIES

- ❑ Very high rates of charge and discharge Ultracapacitor charges within seconds whereas batteries takes hours.
- ❑ Little degradation over hundreds of thousands of cycles batteries degrade within a few thousand charge-discharge cycles
- ❑ Ultracapacitors can have more than 300,000 charging cycles, which is far more than a battery can handle.
- ❑ Can effectively fulfil the requirement of high current pulses that can kill a battery if used instead Batteries fail where high charging discharging takes place whereas ultracapacitor fares extremely well.
- ❑ Ultracapacitors are much more effective at rapid, regenerative energy storage than batteries.

COMPARISON WITH BATTERIES



Battery

- ✓ Energy
- ✗ Weight
- ✗ Cost
- ✗ Charge speed
- ✗ Lifespan
- ✗ Materials



Capacitor

- ✗ Energy
- ✓ Weight
- ✓ Cost
- ✓ Charge speed
- ✓ Lifespan
- ✓ Materials

www.explainthatstuff.com



COMPARISON WITH CONVENTIONAL CAPACITORS

- ❑ Differ in constructional features with respect to conventional capacitors.
- ❑ Has ability to store tremendous charge.
- ❑ Capacitance ranges up to 5000F!
- ❑ Ultracapacitors are able to attain greater energy densities while still maintaining the characteristic high power density of conventional capacitors.
- ❑ Conventional capacitors have relatively high power densities, but
- ❑ relatively low energy densities when compared to batteries. That is, a battery can store more total energy than a capacitor, but it cannot deliver it very quickly, which means its power density is low.
- ❑ Capacitors store relatively less energy per unit mass or volume, but what electrical energy they do store can be discharged rapidly to produce a lot of power, so their power density is usually high.



Advantages and disadvantages of Ultra/super capacitors

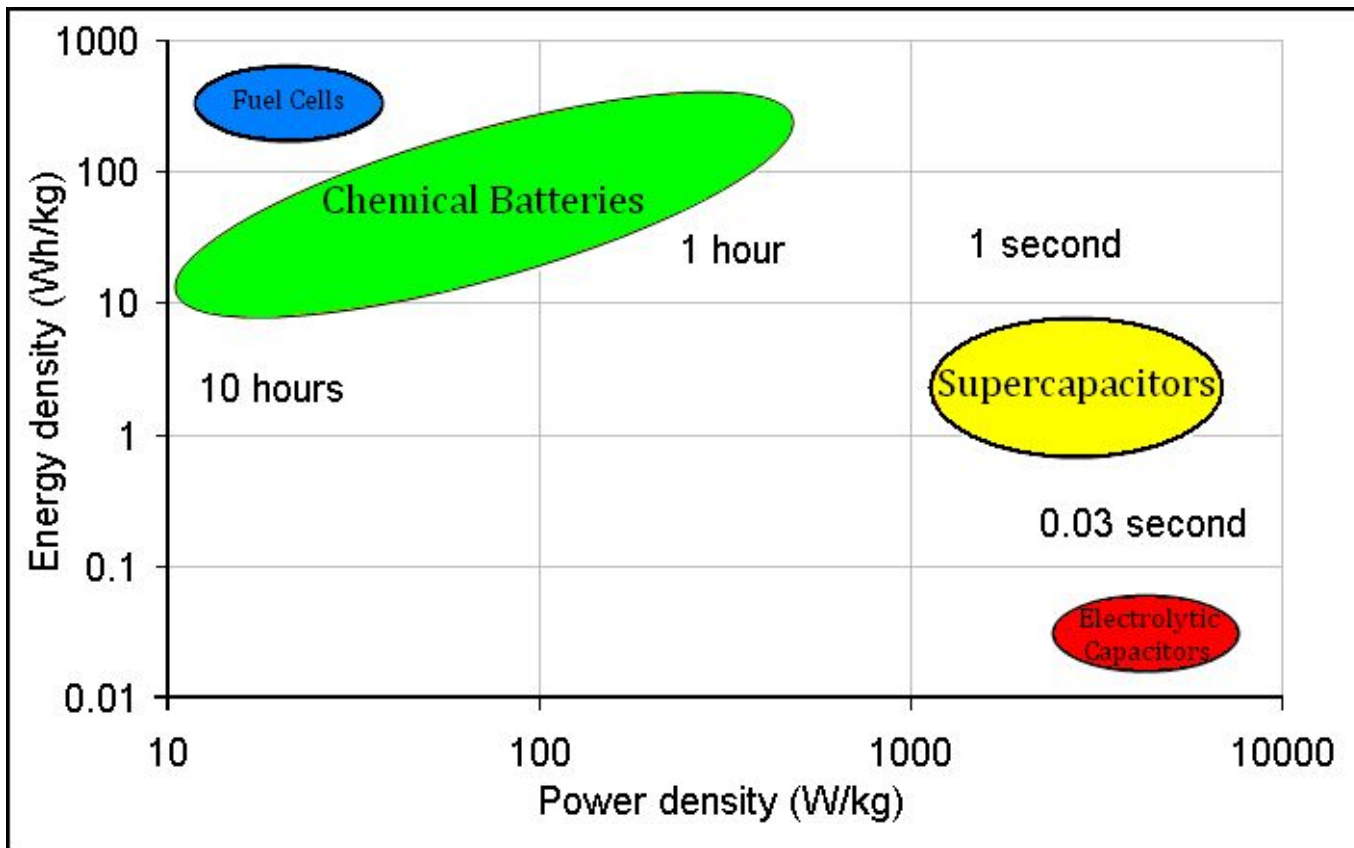
ADVANTAGES

- ☐ Long life: It works for large number of cycles without wear and aging
- ☐ Rapid charging: It takes a second to charge completely
- ☐ High power storage: It stores huge amount of energy in a small volume
- ☐ Faster release: Release the energy much faster than battery
- ☐ Low toxicity of materials used
- ☐ High cycle efficiency (95% or more)¹⁸

DISADVANTAGES

- ☐ High self-discharge
- ☐ The rate is considerably higher than that of a battery
- ☐ The amount of energy stored per unit weight is considerably lower than that of an electrochemical battery (3-5 W.h/kg for an ultracapacitor compared to 30-40 W.h/kg for a battery).
- ☐ The voltage varies with the energy stored. To effectively store and recover energy it requires sophisticated electronic control and switching equipment.
- ☐ Cells have low voltages
- ☐ Series connections are needed to obtain higher voltages
- ☐ Low energy density
- ☐ Typically holds one-fifth to one-tenth the energy of battery

Comparing Batteries & Supercapacitors



- Energy density is the amount of energy stored per unit volume or mass.
- Power density combines energy density with the speed that energy can be drawn out of a device.



Rechargeable Batteries Vs. Supercapacitors

Supercapacitors:

- Higher power density
- Much faster charge and discharge rate
- Environmentally friendly
- Extremely low internal resistance or ESR
- High efficiency (97-98%)
- Over a million charge-discharge cycles

Batteries:

- ⦿ Have higher energy density
- ⦿ Typically 200–1000 charge-discharge cycles
- ⦿ Contain highly reactive and hazardous chemicals
- ⦿ Negatively effected by low temperatures



ADVANTAGES of super capacitor

- High energy storage.
- Wide working temperature(-400c to 600c).
- Eco-friendly.
- Quick charging time.
- Maximum life cycle.
- High cycle efficiency (95%).
- High specific power up to 17 kW/kg.
- Extremely low internal resistance.
- Safe.



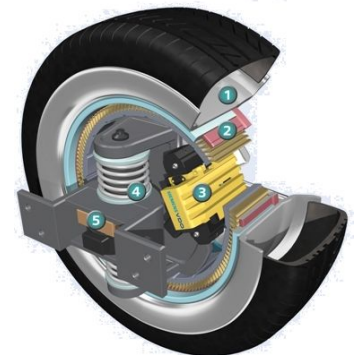
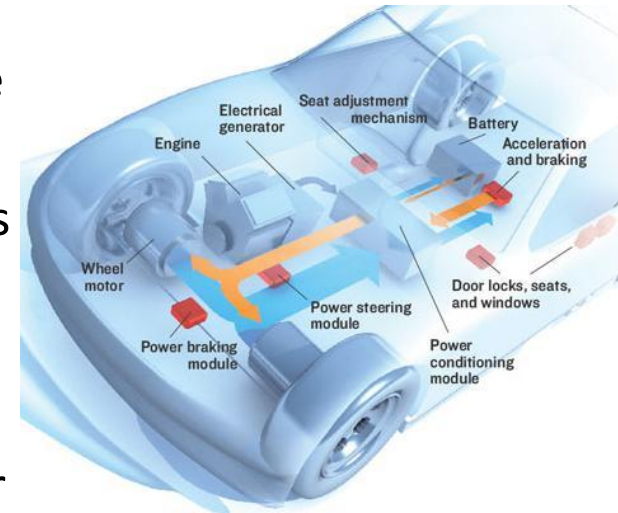
DISADVANTAGES of super capacitor

- ❑ Low energy density; usually holds $1/5 - 1/10$ of a battery.
- ❑ Cannot use the full energy spectrum for some applications.
- ❑ The voltage varies with the energy stored.
- ❑ Have high self-discharge rate.
- ❑ Individual cells have low voltages, and so serial connections are needed to obtain higher voltages.
- ❑ Requires expert electronic control.
- ❑ Cannot be used in AC and high frequency circuits.
- ❑ High cost.

Applications for Supercapacitors



- Back up for uninterruptable power supplies (UPS)
- Light weight power supplies for small aircraft
- Provide short duration power for various vehicle systems such as breaking or steering
- Used to absorb power during short periods of generation such as Regenerative Braking
- Extend range and battery life in Hybrid Electric Vehicles (HEV)



Hybrid Electric Vehicles

- The CSIRO in Australia [national science agency] has developed the UltraBattery, which combines a supercapacitor and a lead acid battery in a single unit
- 4x longer life cycle, 50% more power, 70% cheaper than batteries used in HEV's





APPLICATIONS

- In start up mechanism for Automobiles.
- Used in Diesel engine start up in submarines & tanks.
- China is experimenting with a new form of electric bus that runs without powerlines but runs on power stored in large supercapacitors, which are quickly recharged whenever the electric bus stops at any bus stop, and get fully charged in the terminus. A few prototypes were being tested in Shanghai in early 2005. In 2006 two commercial bus routes began to use supercapacitor buses, one of them is route 11 in Shanghai.
- Backup power system in missiles.
- Power source for laptops, flash in cameras.
- Voltage stabilizer.



APPLICATIONS

To deliver high power and bridge power gaps

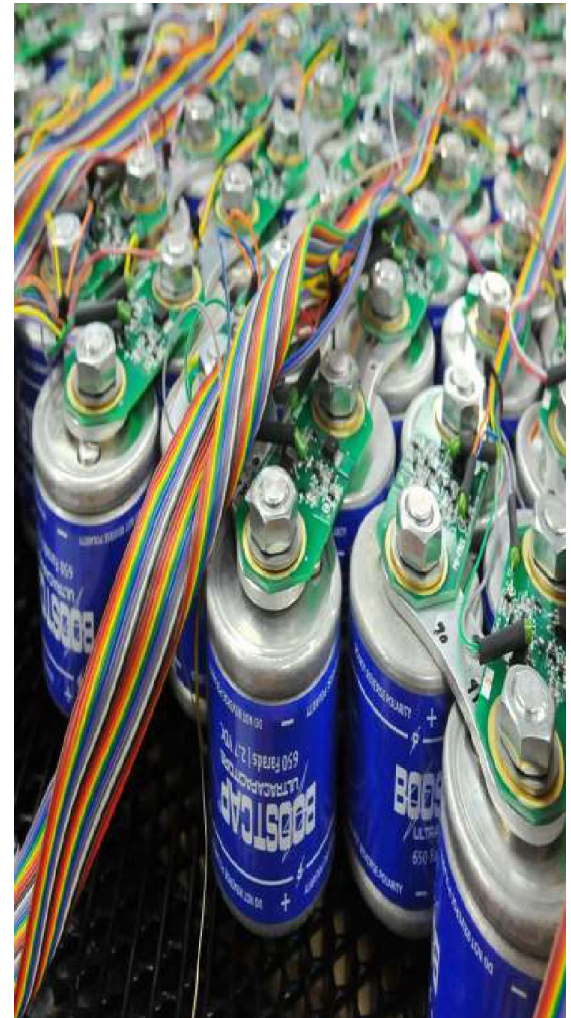
- ❖ Industrial and electronic applications
- ❖ Used in wind turbines, electric and hybrid vehicles
- ❖ Regenerative braking to release the power in acceleration
- ❖ To start power in start-stop systems
- ❖ Regulate voltage in the energy grid
- ❖ To capture and assist the power in lower loads and lifted loads
- ❖ Back-ups the power in a quick discharging state.

Bus operated on Supercapacitor in Shanghai





Supercapacitor installed in bus



Supercapacitors



APPLICATIONS OF ULTRACAPACITORS

- ❑ Considered as environmentally friendly solutions because they can perform reliably in all weather conditions without having to be replaced and disposed to landfills.
- ❑ Function well in temperatures as low as $-40\text{ }^{\circ}\text{C}$, they can give electric cars a boost in cold weather, when batteries are at their worst.
- ❑ Used in military projects such as starting the engines of battle tanks and submarines or replacing batteries in missiles.
- ❑ A bank of ultracapacitors releases a burst of energy to help a crane heave its load aloft; they then capture energy released during descent to recharge.



APPLICATIONS (Contd..)

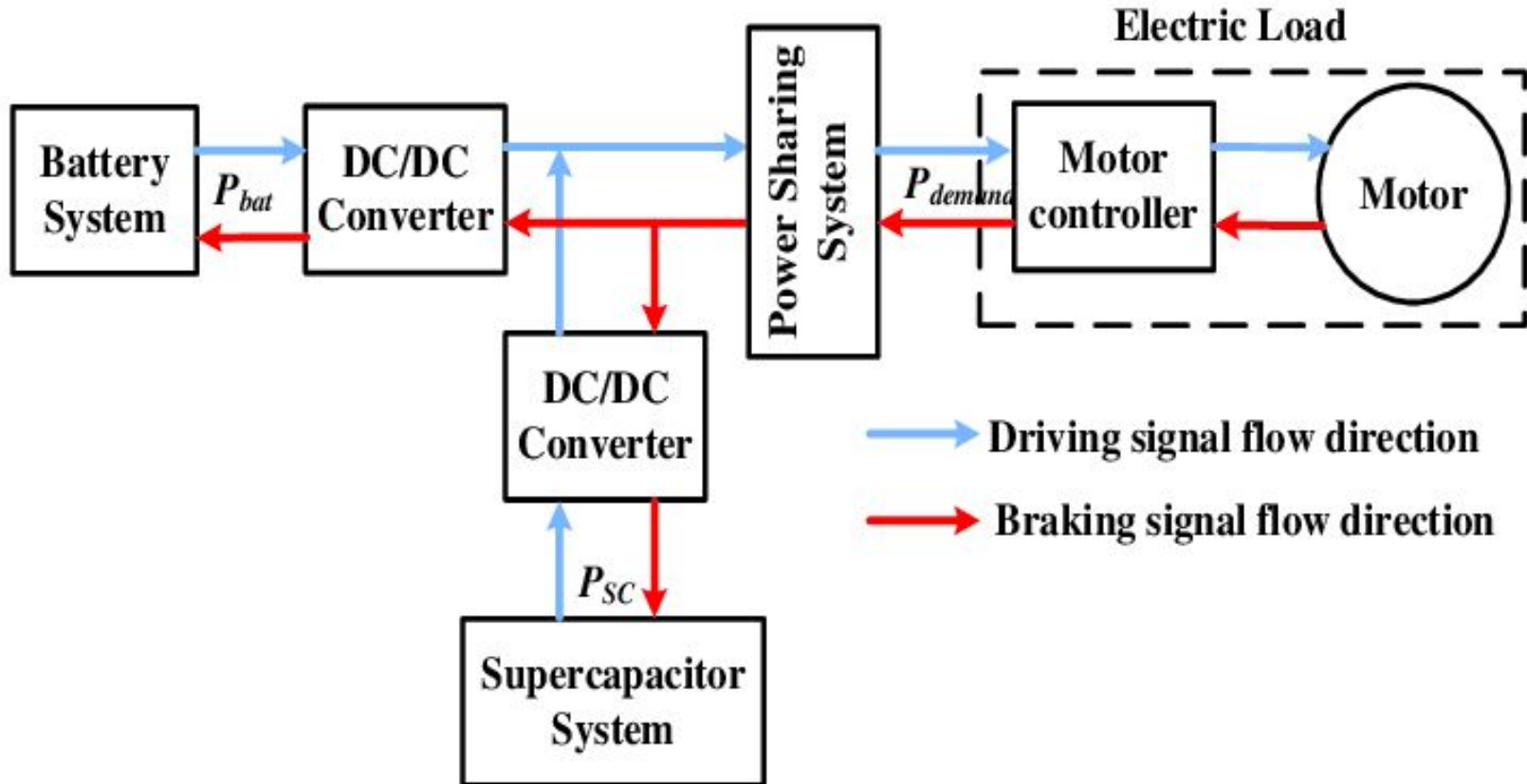
- ❑ In 2001 and 2002, VAG, the public transport operator in Nuremburg, Germany tested a bus which used a diesel-electric drive system with ultracapacitors.
- ❑ Heavy transportation vehicles - such as trains, metros – place particular demands on energy storage devices. Such devices must be very robust and reliable, displaying both long operational lifetimes and low maintenance requirements.
- ❑ Maxwell Technologies solved these issues with its ultracapacitor HTM125 module for braking energy recuperation and torque assist systems in trains, metro transportation vehicles. Ultracapacitors can deliver the peak power for acceleration and store part of vehicle's kinetic energy during deceleration.



APPLICATIONS(Contd..)

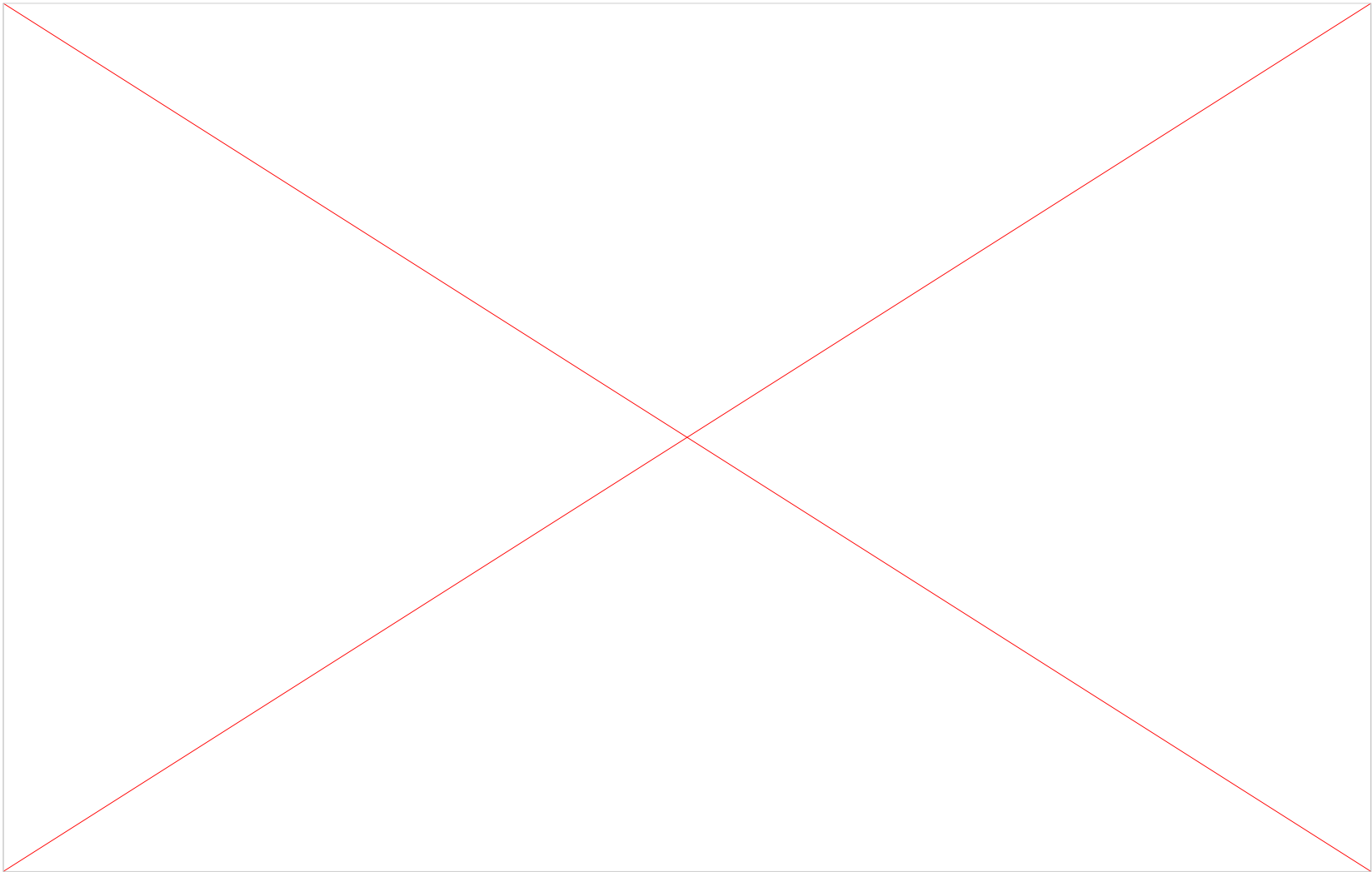
- ❑ China is experimenting with a new form of electric bus that runs without powerlines using power stored in large onboard ultracapacitors. A few prototypes were being tested in Shanghai in early 2005. In 2006, two commercial bus routes began to use ultracapacitor buses.
- ❑ Esma-cap, Russia, developed two experimental vehicles. Electric bus with 50 passengers capacity, maximum speed 20 km.h⁻¹. Electric truck with payload limit 1,000 kg, maximum speed 70 km.h⁻¹. Proton Power Systems has created the world's first triple hybrid Forklift Truck, which uses batteries as primary energy storage and ultracapacitors to supplement this energy storage solution.
- ❑ Delivering or accepting power during short-duration events is the ultracapacitor's strongest suit.

Battery super capacitor hybrid vehicles

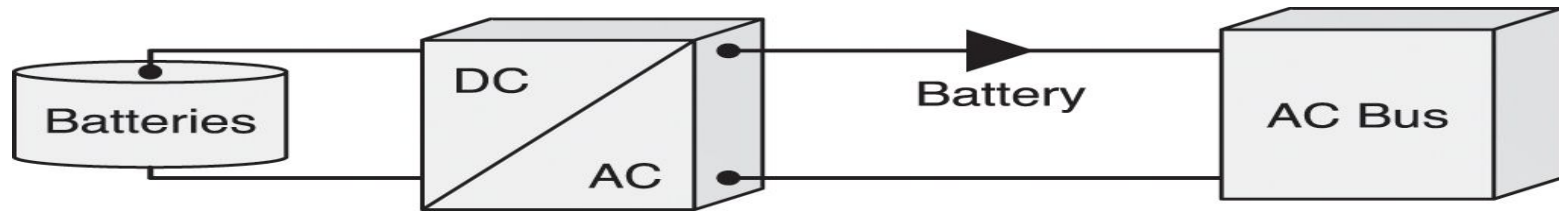




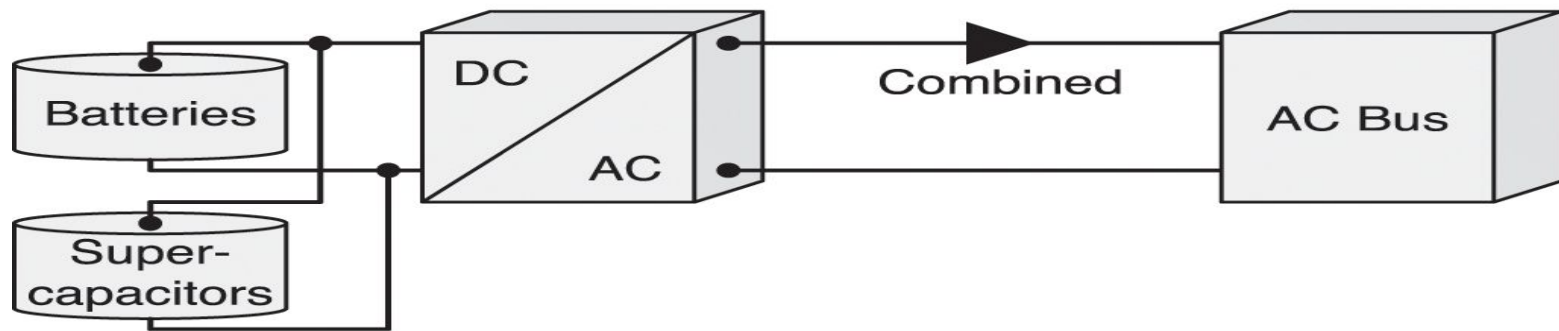
Battery super capacitor hybrid vehicles



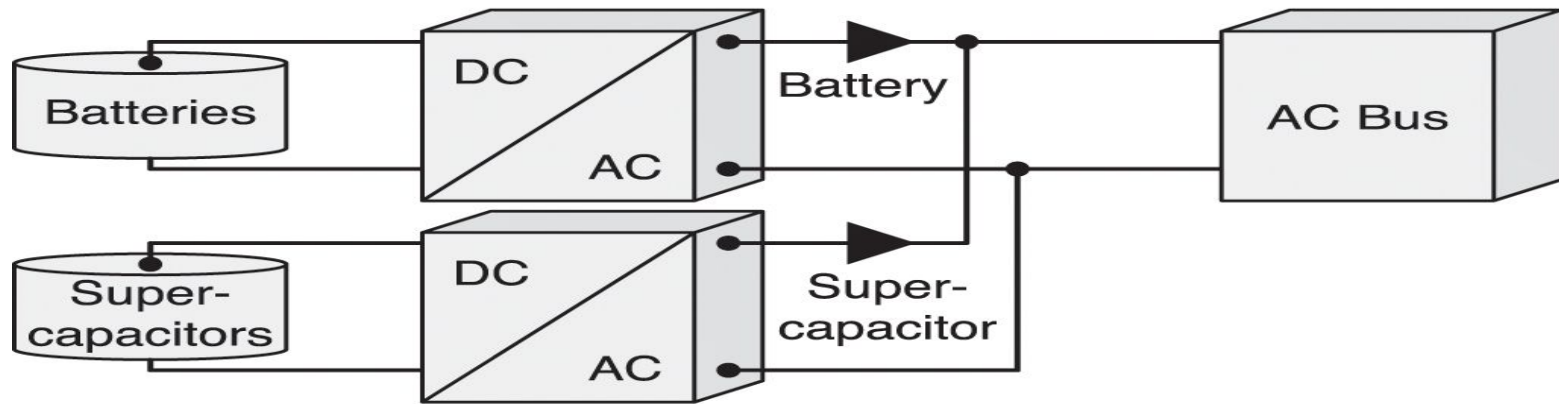
Battery super capacitor hybrid vehicles



(a) Battery-only homogeneous EES system



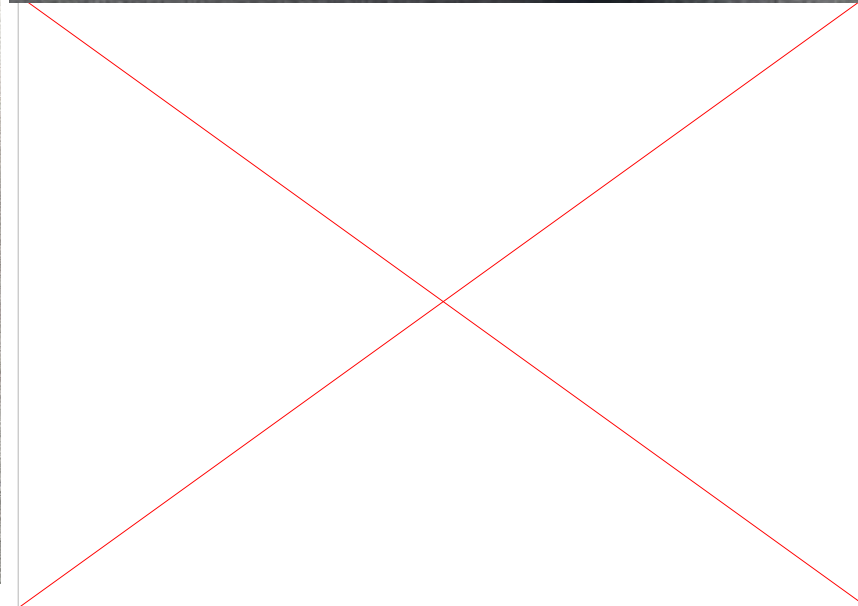
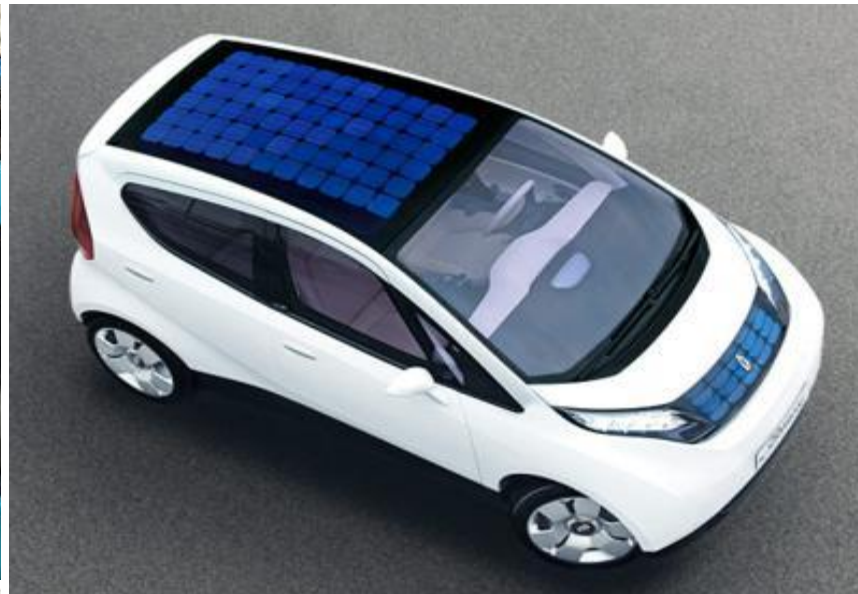
(b) Battery-supercapacitor passive HEES system



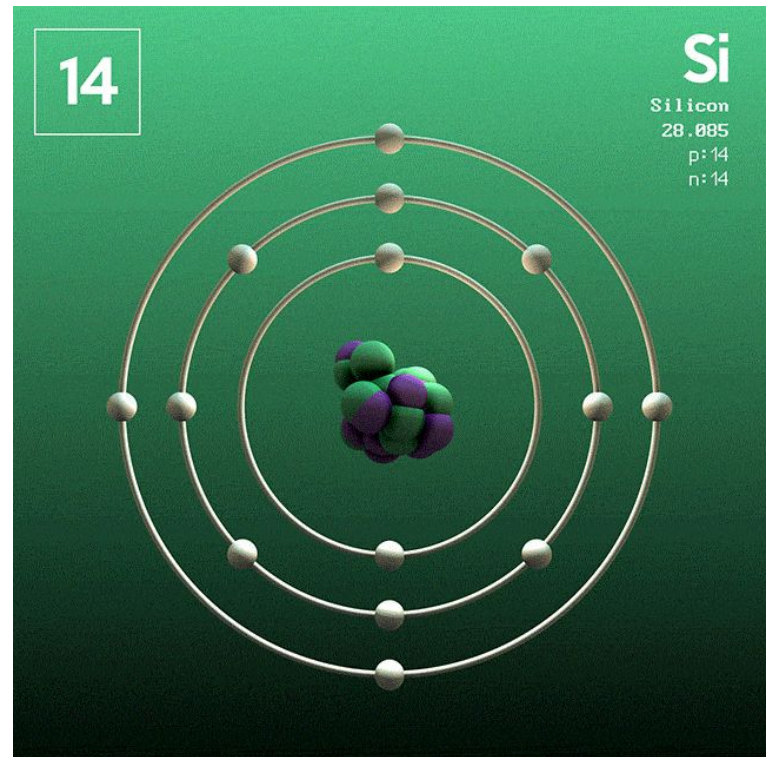
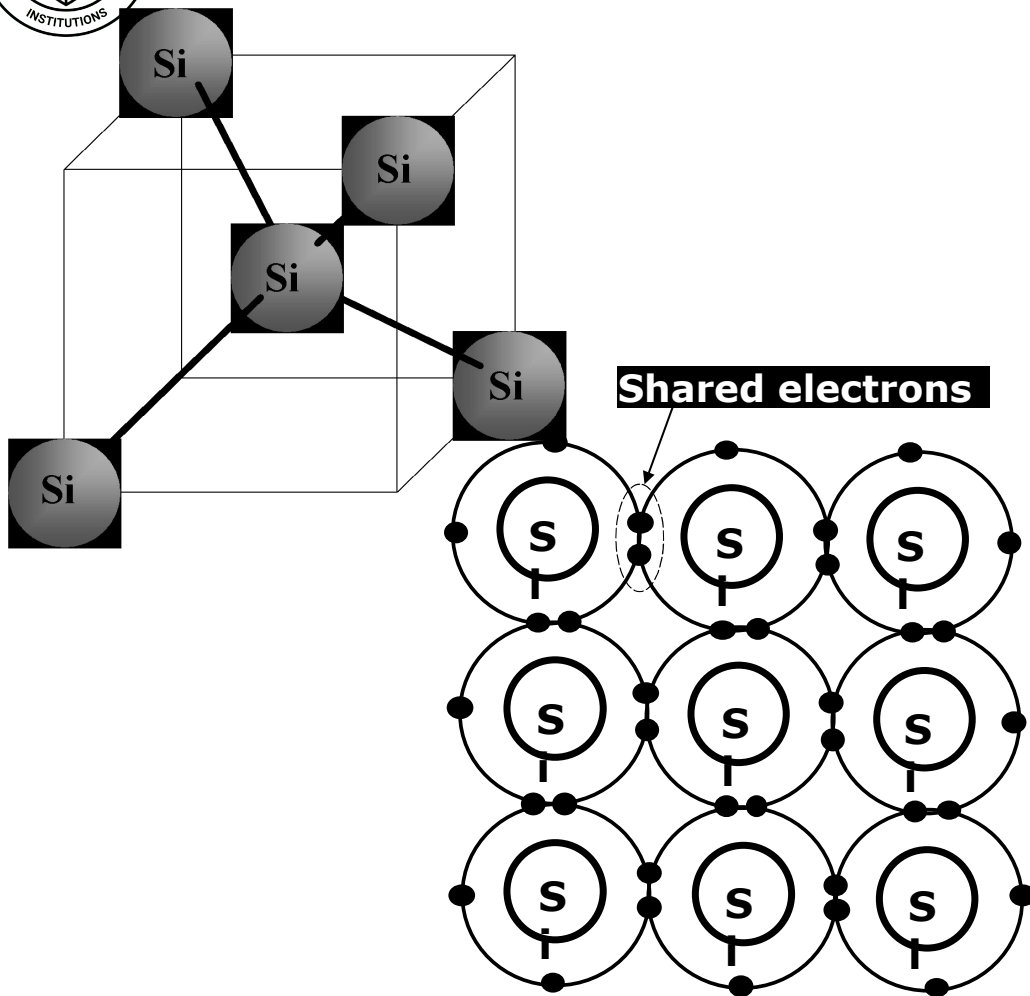
(c) Battery-supercapacitor active HEES system



Solar cell hybrid vehicles

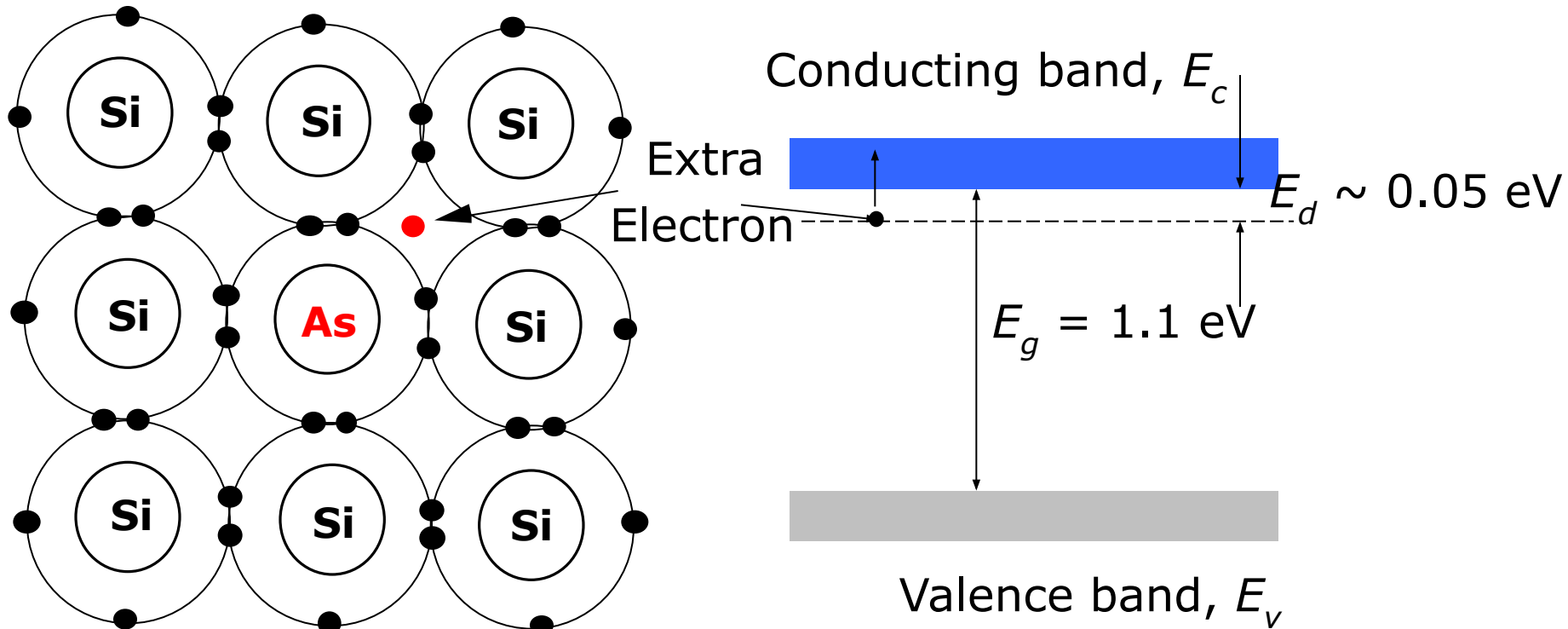


Silicon



- ❑ Silicon is group IV element – with 4 electrons in their valence shell.
- ❑ When silicon atoms are brought together, each atom forms covalent bond with 4 silicon atoms in a tetrahedron

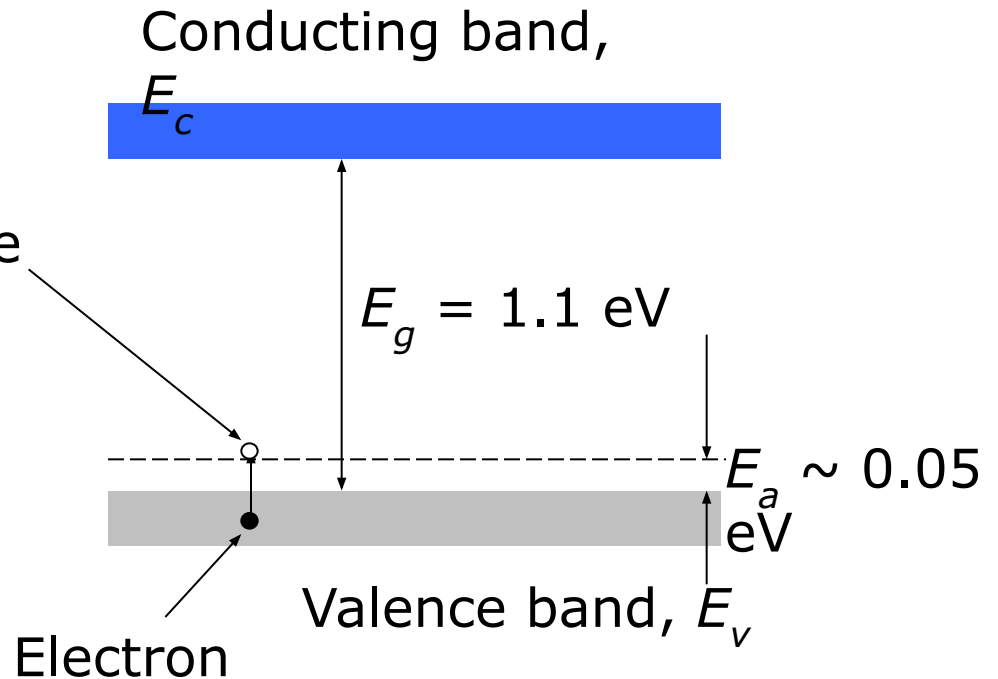
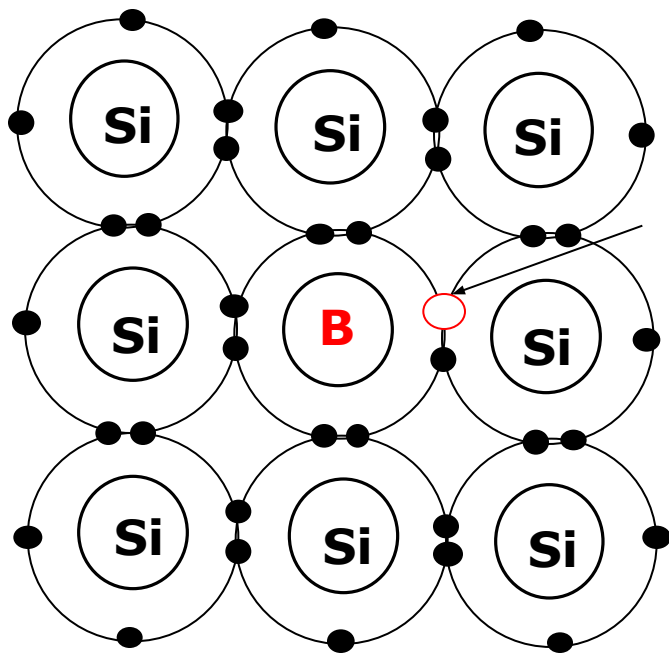
n-type Doping



Doping silicon lattice with group V elements can create extra electrons in the conduction band — **negative charge carriers (n-type), As- donor.**

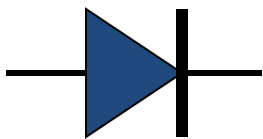
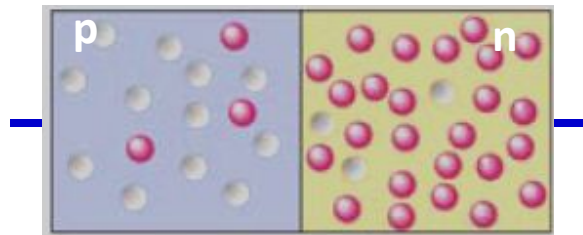
Doping concentration $\#/\text{cm}^3$ ($10^{16}/\text{cm}^3 \sim 1/\text{million}$).

p-type doping



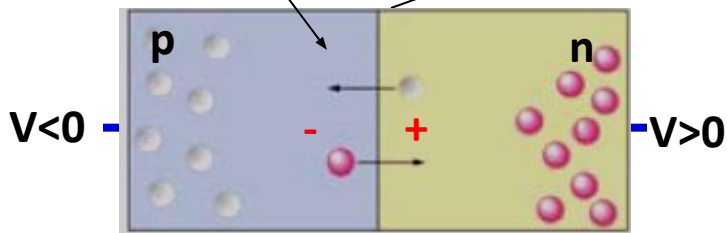
□ Doping silicon with group III elements can create empty holes in the conduction band — **positive charge carriers (p-type), B-(acceptor).**

p-n Junction (p-n diode)

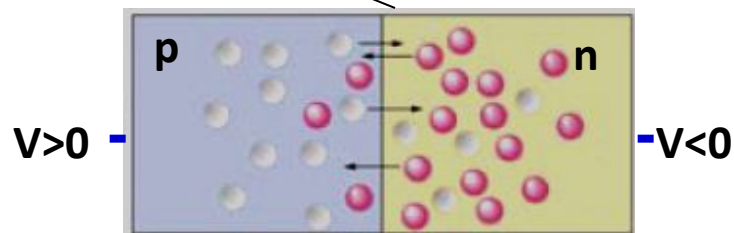


i

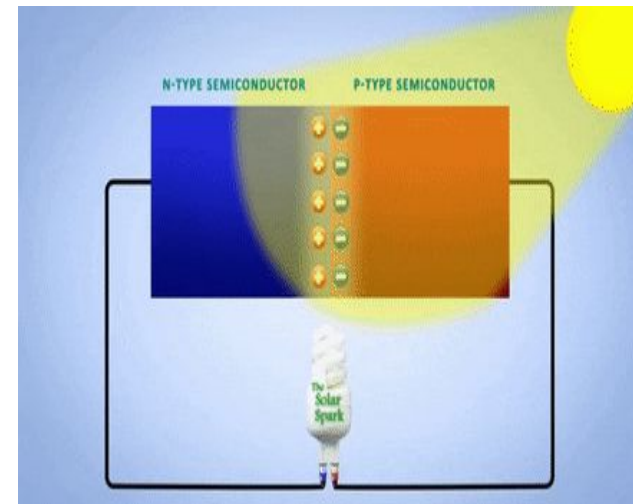
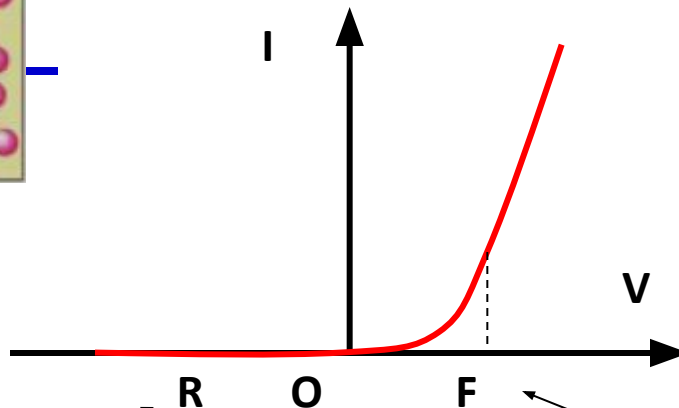
depletion layer



Reverse bias



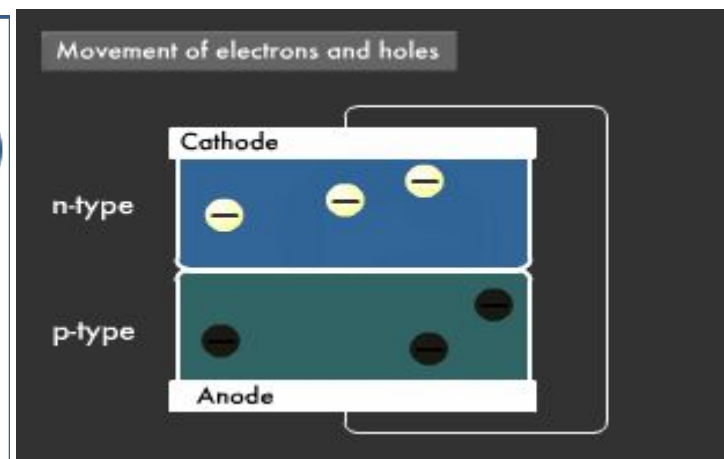
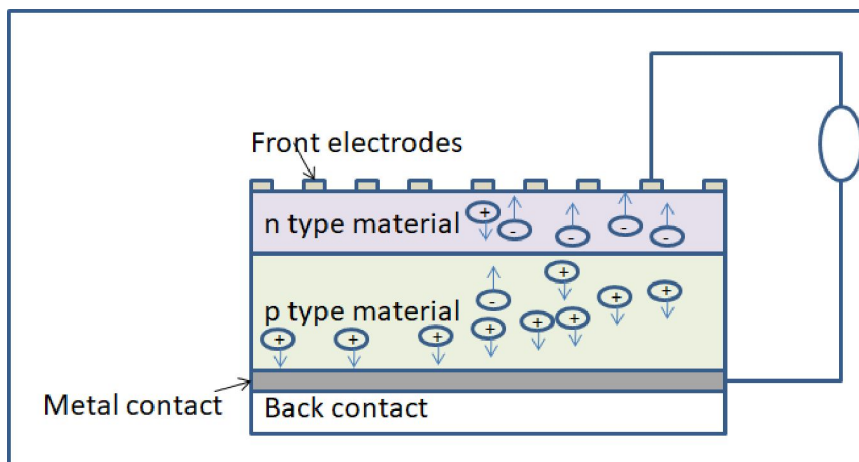
Forward bias



- ❑ A p-n junction is a junction formed by combining p-type and n-type semiconductors together in very close contact.
- ❑ In p-n junction, the current is only allowed to flow along one direction from p-type to n-type materials.

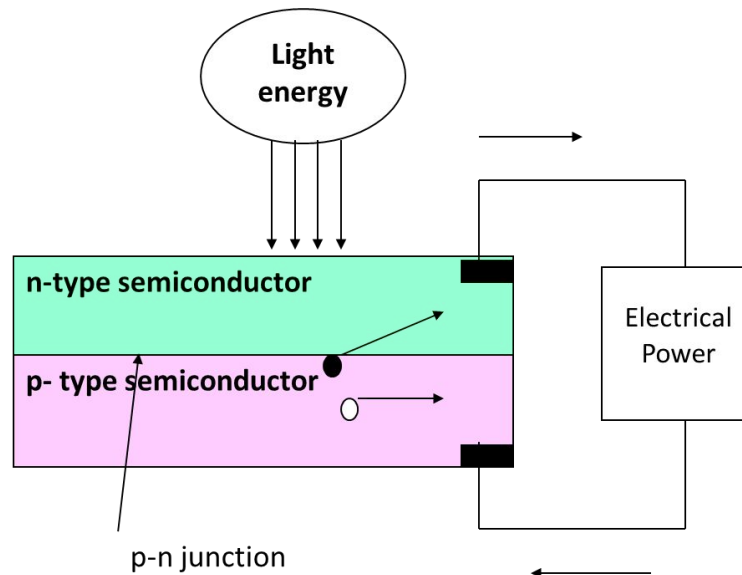
Construction of amorphous silicon solar cells

- A typical silicon solar cell consists of n-type material (Phosphorus doped silicon) and a p-type (Boron doped silicon) material made of silicon.
- In the fabrication of device these n-type and p-type materials are joined to form a p-n junction.
- On top of n type connecting front electrodes are placed.
- The metal back contact is attached to p-type layer.
- An antireflective coating is filled in between the front electrodes to avoid the reflection of sunlight.
- Finally p-type and n-type layers are joined externally to the circuit.
- The complete solar cells are sealed to avoid the environmental contact.



Working of silicon solar cells

- When sunlight is incident on solar cells photon strikes on p-n junction.
- Then electron hole pair will be created at the junction by the absorption of photon.
- This electron hole pair diffuses towards respective layer based on affinity.
- The electrons diffuse towards n-type later and holes are diffused towards p-type layer.
- These layers are connected externally through a circuit, hence current will be generated.





Advantages of silicon solar cells

- ❑ **Solar cell is renewable energy which can be continuously drawn from the sun.**
- ❑ **It is cost effective energy because once it installed there will be minimum maintenance charges for small usage.**
- ❑ **Solar energy is environment friendly and green energy because it doesn't produce any greenhouse gasses and no pollutants.**
- ❑ **It doesn't involve any combustion reaction or radioactive residue hence no pollution.**



Dis-advantages of silicon solar cells

- ❑ **Space utilization: Solar cells required large area for installation.**
- ❑ **It requires high investment cost for a big power plant.**
- ❑ **During the rainy season power production is less hence grid maintenance will be difficult.**
- ❑ **Solar energy is produced only in the day time hence storage will be a big challenge.**



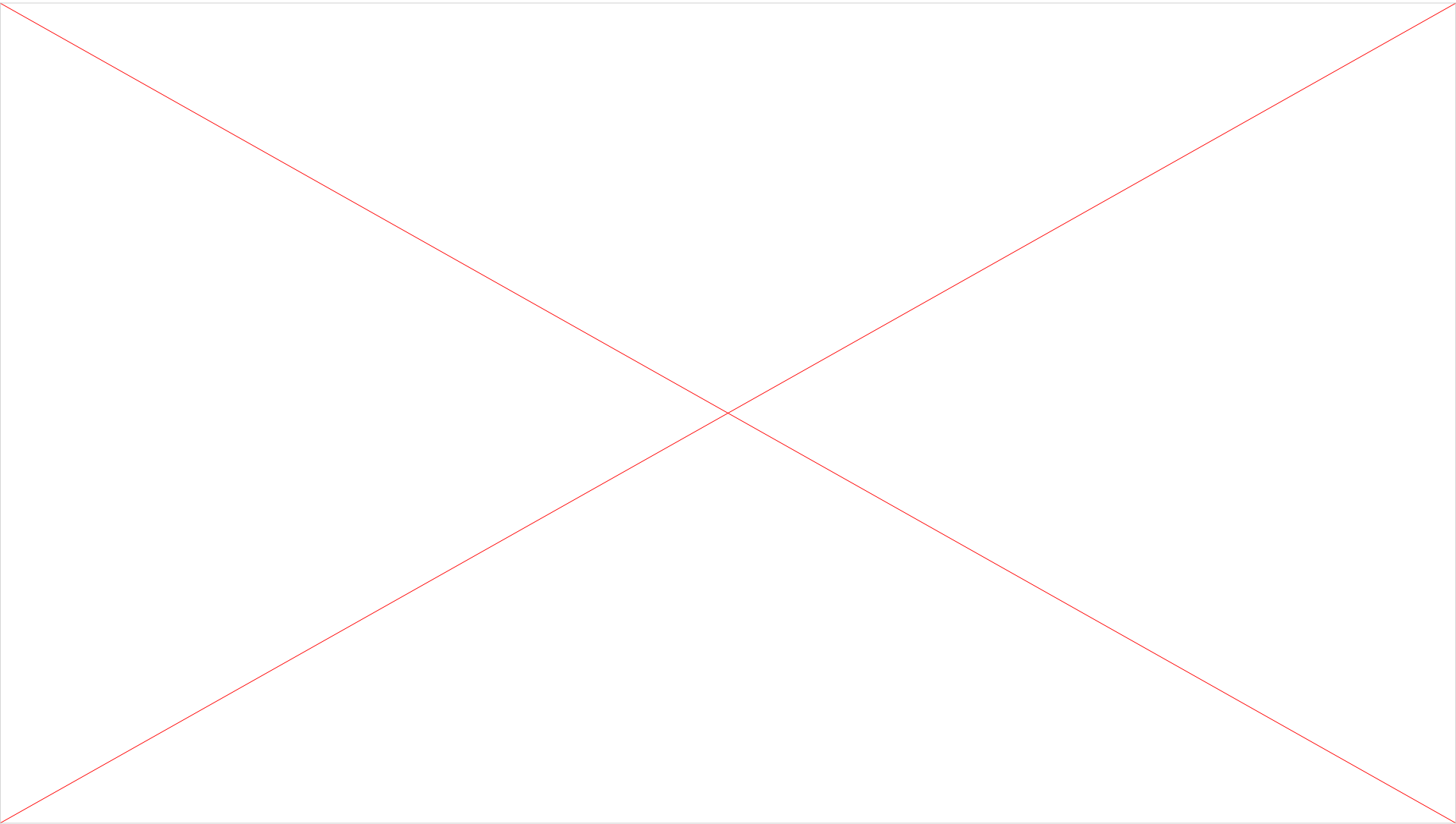
Solar cells hybrid vehicles

4 ways of using solar-energy to power cars

- Using solar electricity to charge batteries
- Using solar electricity to make H_2 for use in a fuel cell
- Using solar energy to grow biomass that is converted to methanol and used in a fuel cell
- Using solar energy to grow biomass that is converted into ethanol and used in an ICE



Solar cell hybrid electric vehicles



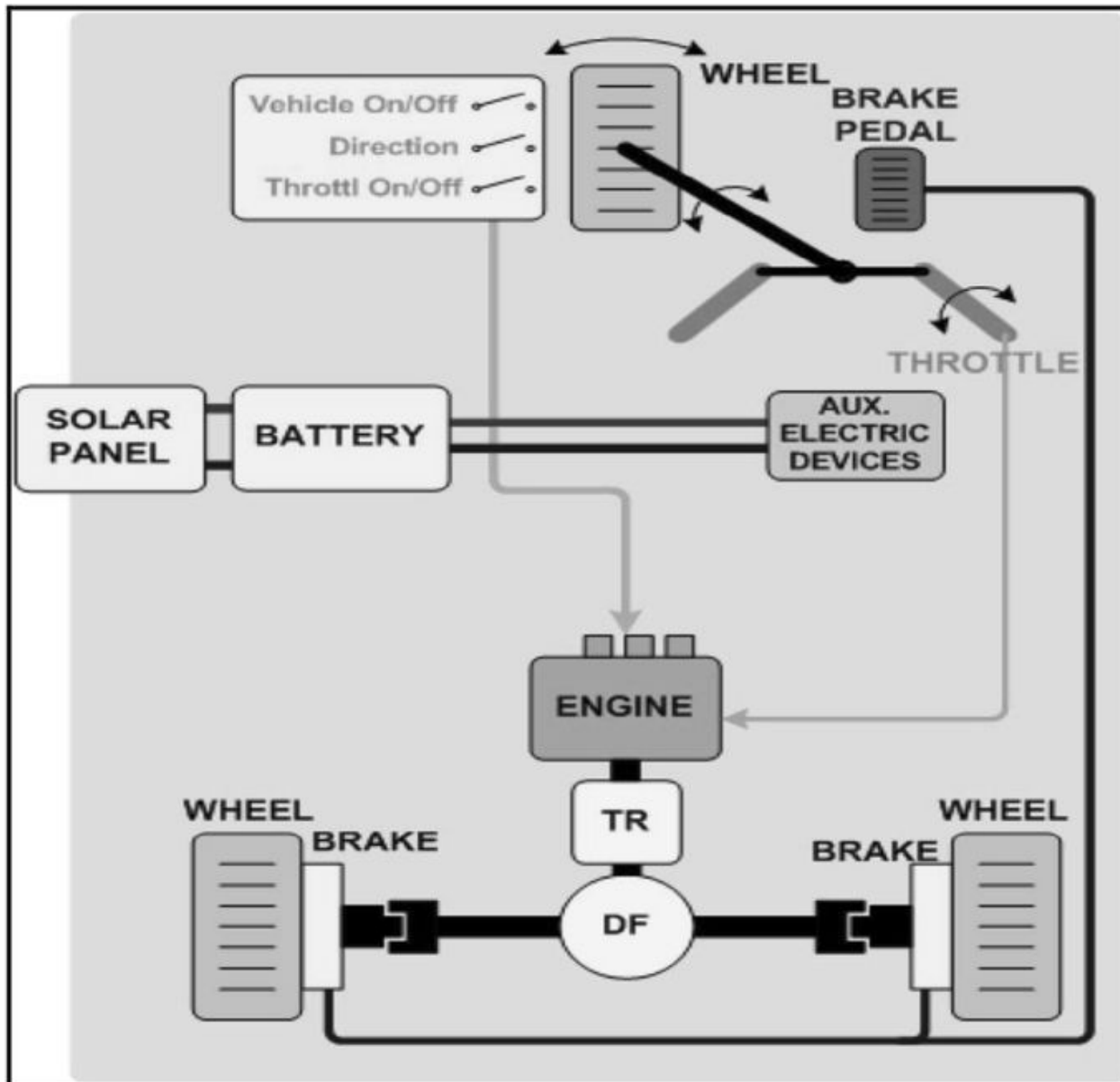


Steps, solar energy to battery

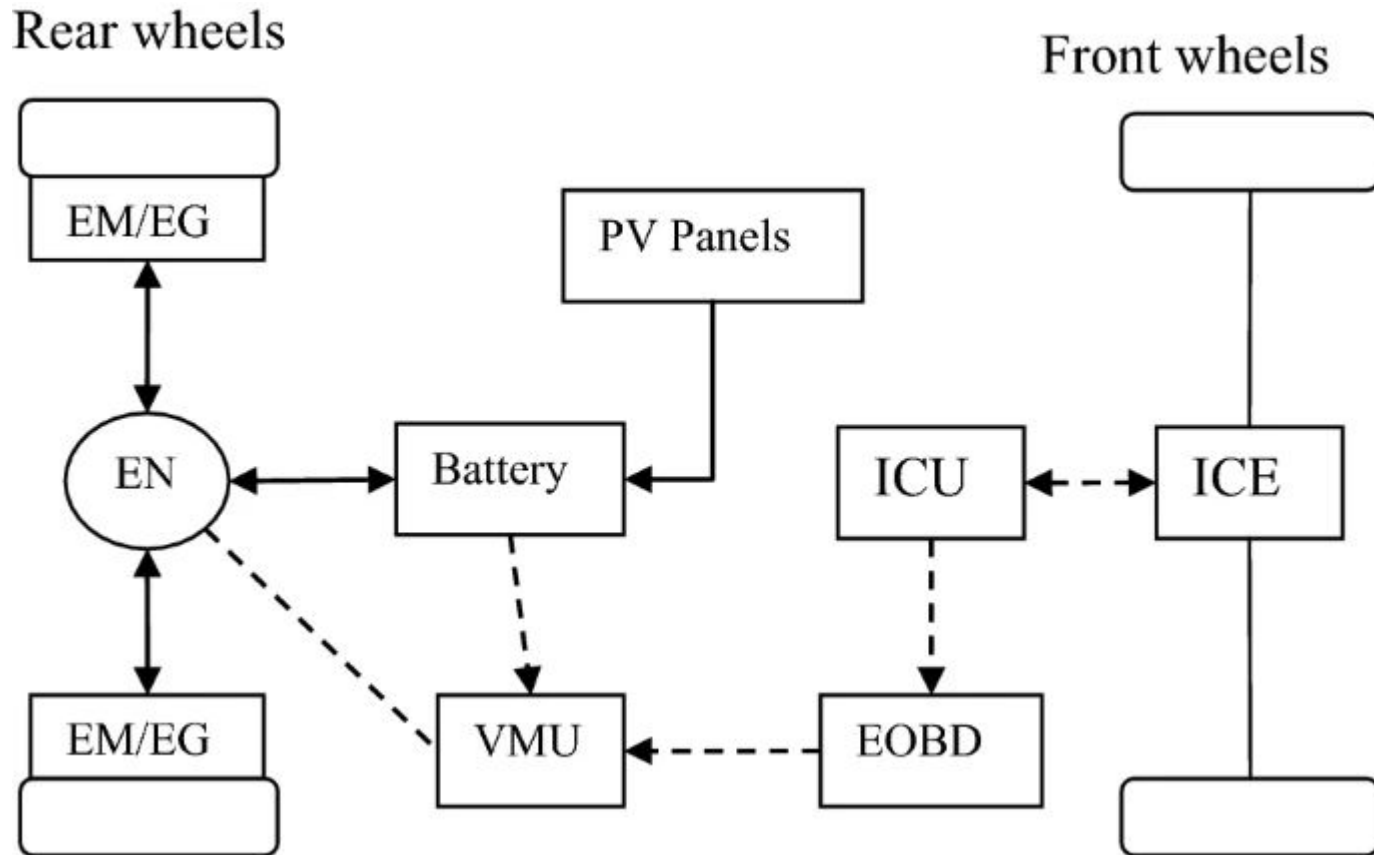
- PV modules, 15% efficiency
- DC to AC conversion, 85%
- Transmission of electricity, 96% (say)
- Battery charging, 95%
- Drive train (motor, transmission), 87%

Overall sunlight to wheels energy transfer: **10.1%**

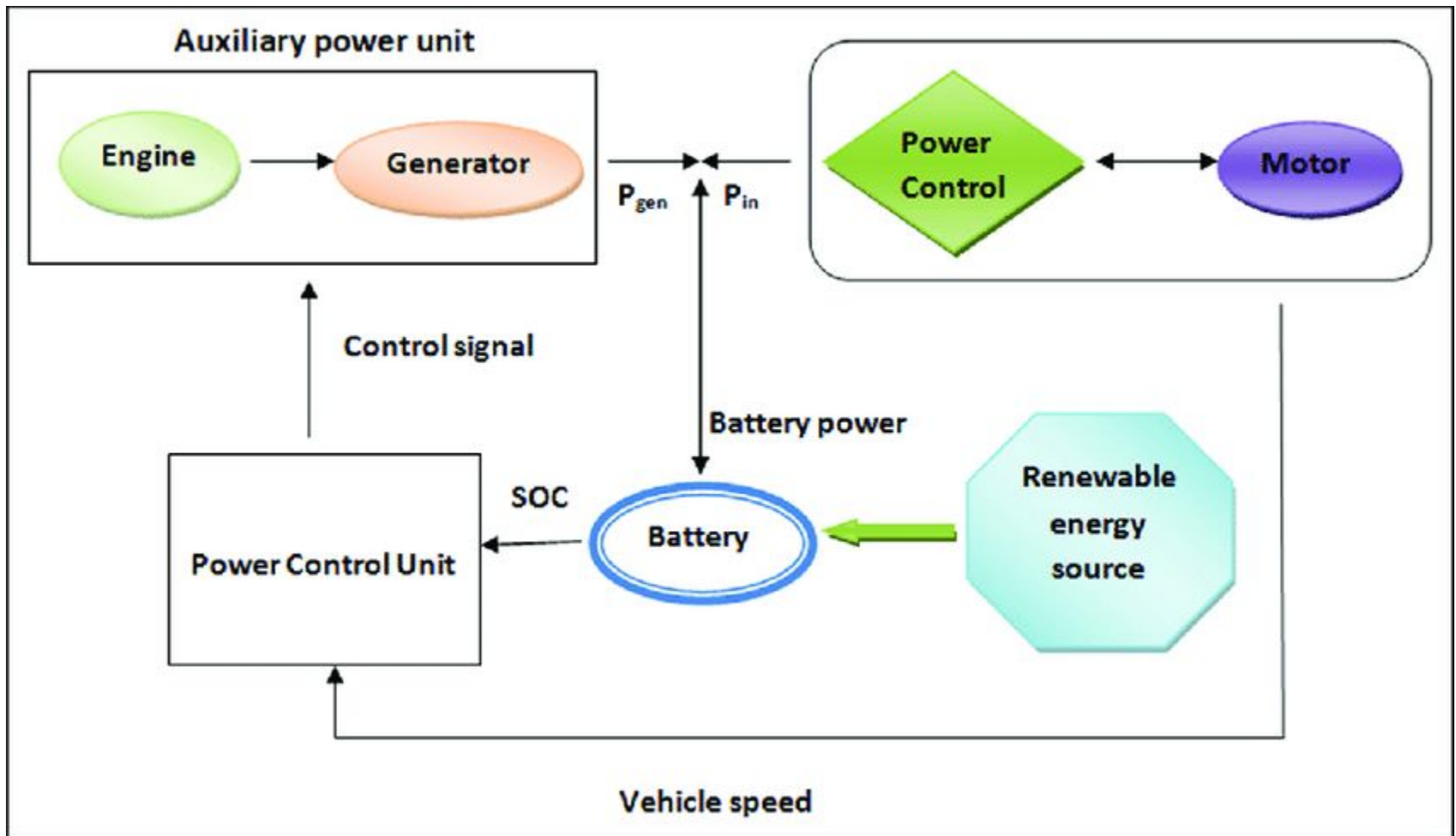
Solar cell hybrid electric vehicles



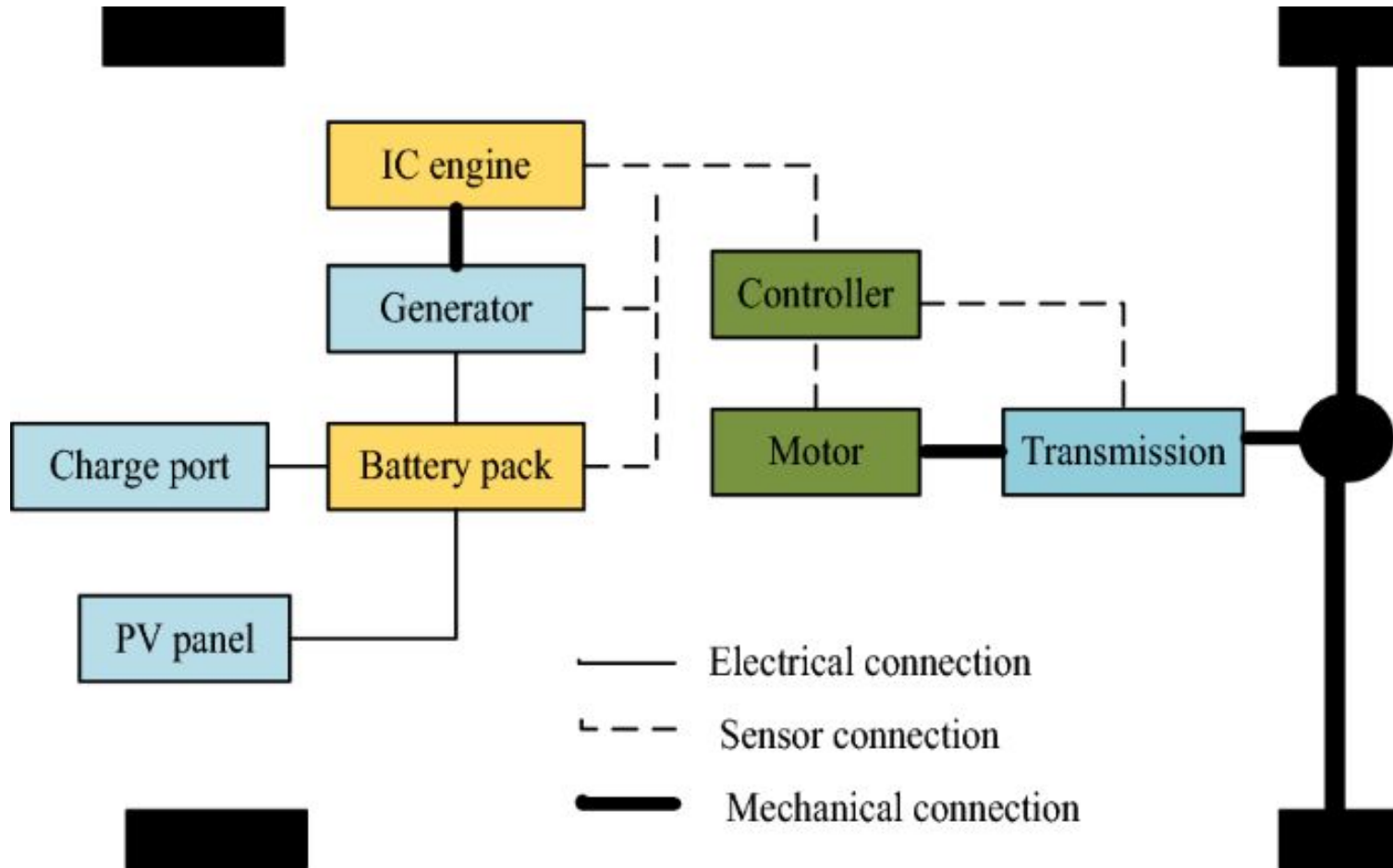
Solar cell hybrid electric vehicles



Solar cell hybrid electric vehicles



Solar cell hybrid electric vehicles with conventional IC Engine





Solar cell hybrid electric vehicles: Advantages

Eco-friendly and Quiet: Solar-powered vehicles have zero emission level, as they don't utilize non-renewable resources and burn fuel. The electric motors generate electricity that doesn't emit any greenhouse gases or any other pollutants. These cars are quieter than the vehicles powered by conventional fuels, which don't cause noise pollution as well.

Energy Availability: Solar cars derive their power from the sun, indirectly, that always shines and provides endless energy. The efficient solar panels can produce and store more horsepower for the vehicle.

No Fuel Costs: Unlike the conventionally fueled vehicles, solar vehicles have no fuel costs and a low cost of maintenance.

Driving Comfort: Having aluminum and lightweight components, the solar-powered cars run faster and more smoothly than petrol and diesel engine vehicles.



Solar cell hybrid electric vehicles: Limitations

Design Challenge: The solar vehicles require large surface area on roof for mounted solar panels, have low wind resistance and space only for two passengers.

Poor Practicality: These green cars don't have any driver safety features and other equipment such as wiper blades, headlights and rear view mirrors. Aspects like suspension, chassis strength, steering, brakes, secured solar panels and batteries arrangement also need to be taken seriously.

Expensive Batteries: The efficient solar panels and batteries and their replacement are way too expensive that need to be changed so often. This is what makes the solar vehicles a costly affair.

Energy Storage Capacity: The photovoltaic cells or solar panels can convert 15-30% of sunlight into electricity, depends on the material used, which is quite limited.

Fuel Cells hybrid electric vehicles

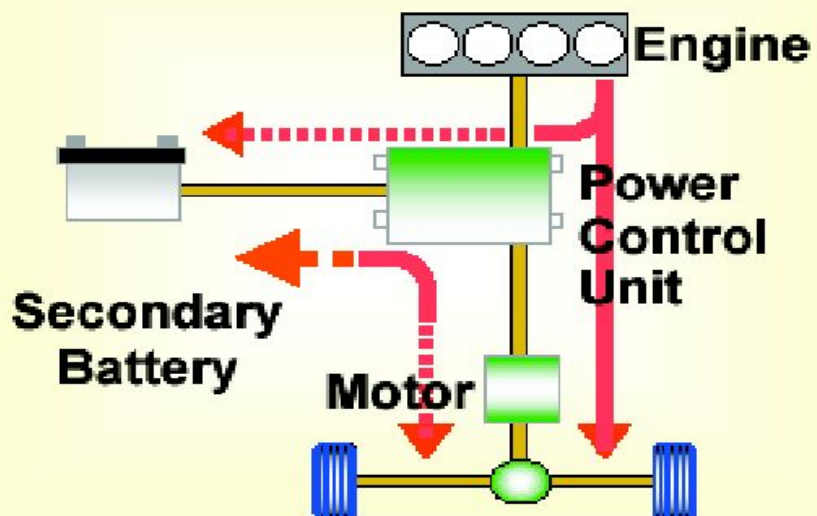
Prius



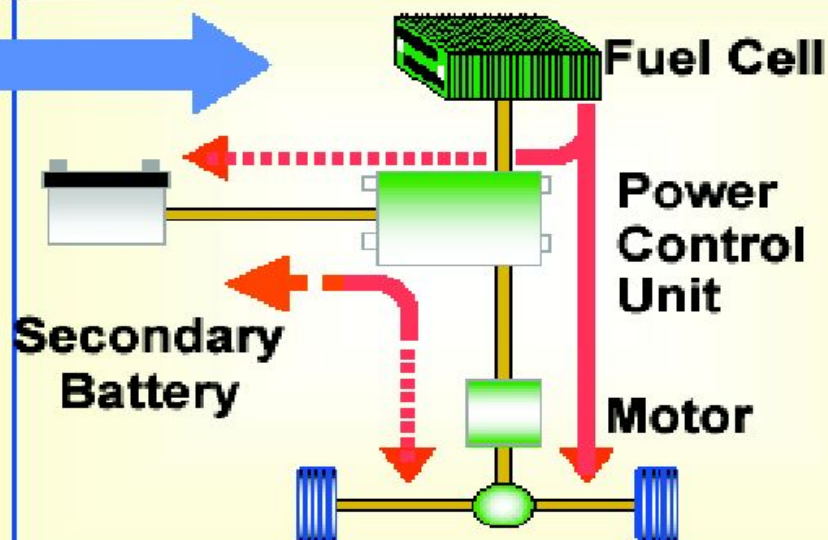
FCHV



ICE Hybrid Vehicle (Prius)



Toyota Fuel Cell Hybrid Vehicle

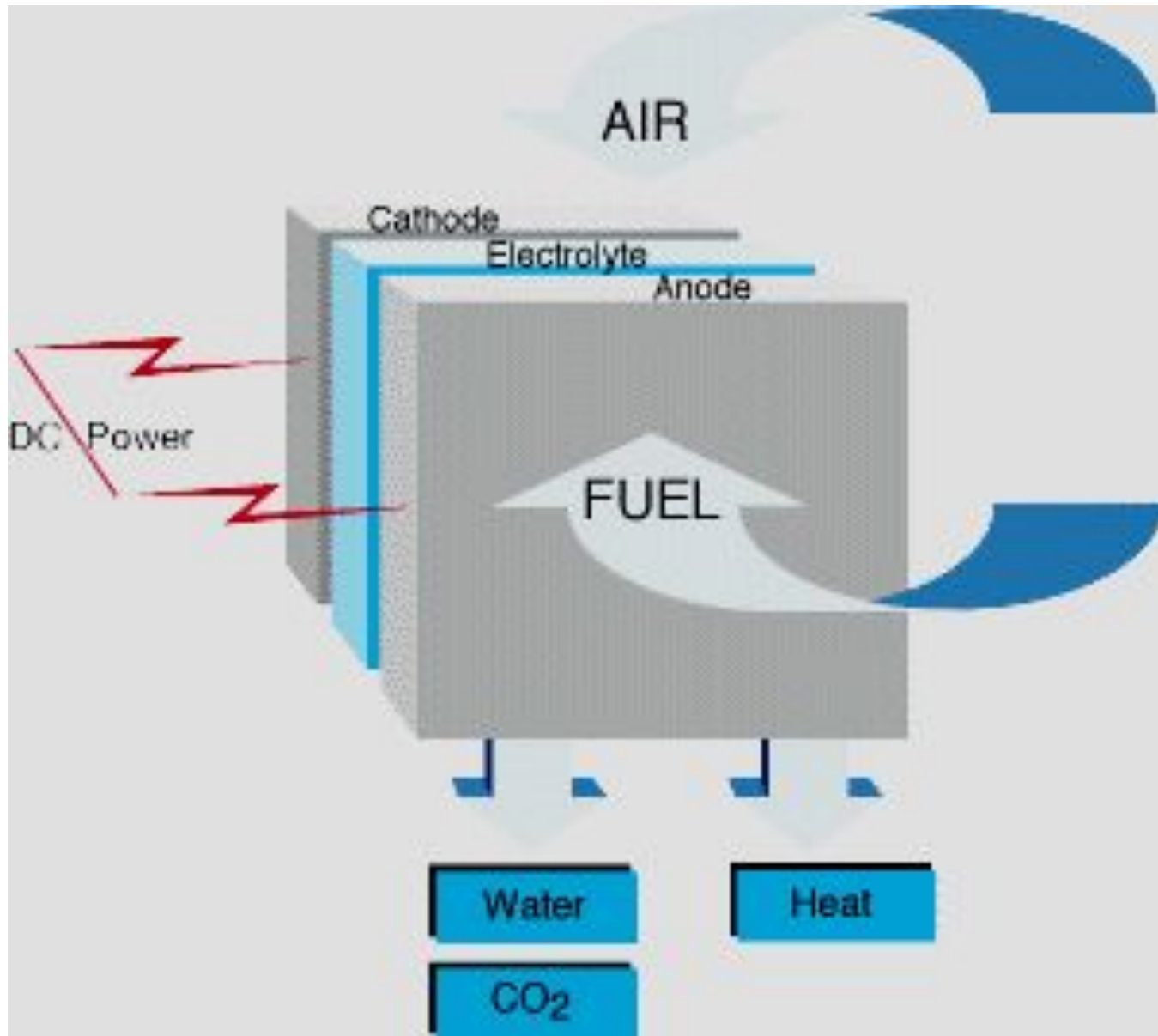




Attractive characteristics of Fuel Cell

- High energy conversion efficiency
- Modular design
- Very low chemical and acoustical pollution
- Fuel flexibility
- Cogeneration capability
- Rapid load response

Basic Configuration





CELL REPRESENTATION

Fuel / Electrode / Electrolyte / Electrode / Oxidant

CELL REACTION

The general electrochemical reaction of fuel cell is redox reaction between fuel and oxidant

Anode (-Ve): **Fuel $\square$ Oxidized product + ne**

Cathode(+Ve): **Oxidant + ne $\square$ Reduced product**

Overall reaction: **Fuel + Oxidant $\square$ Oxidized product + Reduced Product**

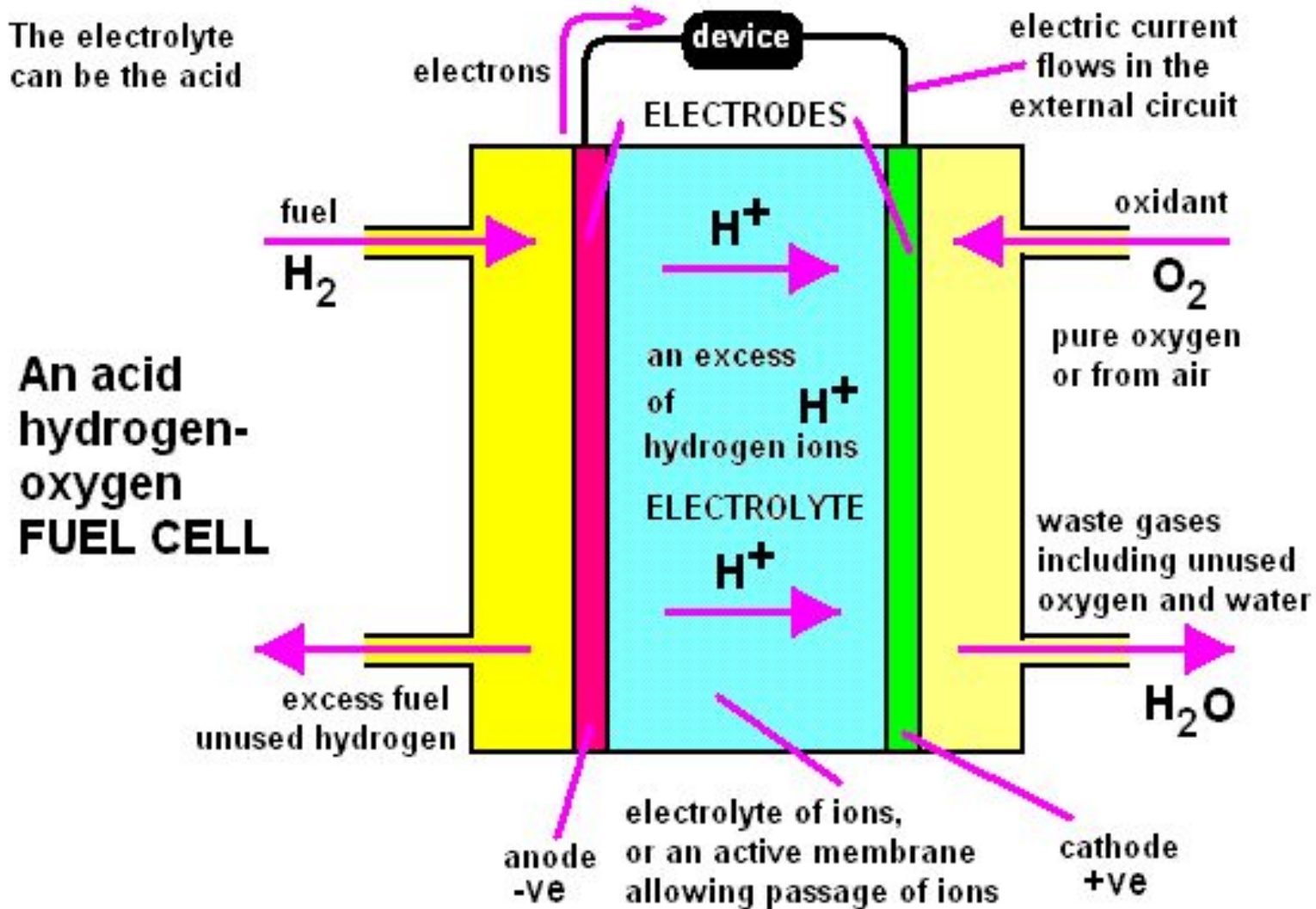


Classification of fuel cells

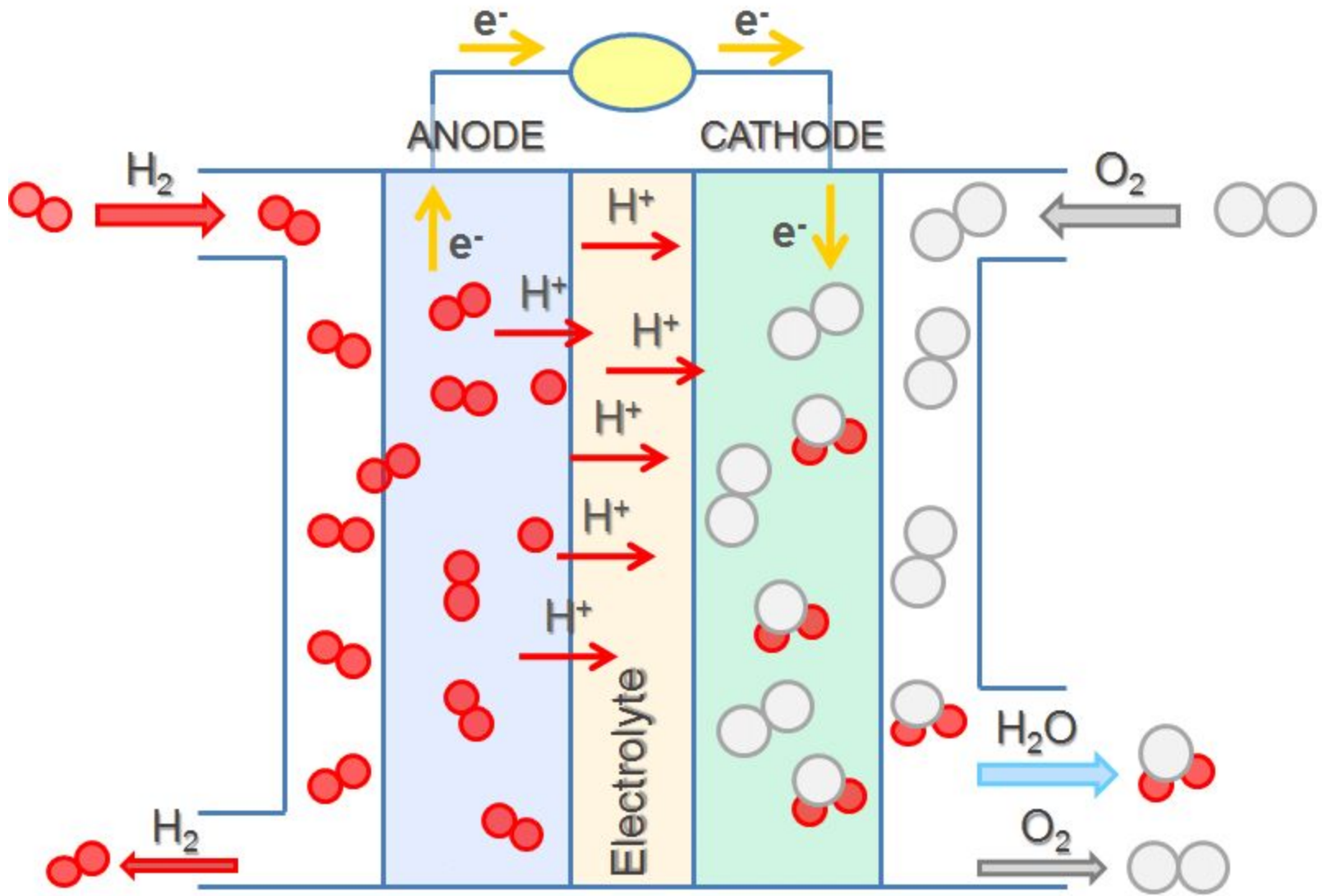
□ Classification based on electrolyte.

Sl No	Type	Electrolyte	Anode	Cathode	Temp. range
1	Alkaline fuel cells	KOH	Porous C/Pt	Porous C/Ag	80°C
2	Phosphoric acid fuel cells (PAFC)	Phosphoric acid	Pt	Pt	160-220°C
3	Molten carbonate fuel cells (MCFC)	Molten metal carbonates	Ni/Cr(2-10%)	Ni/Li(2-10%)	650°C
4	Solid oxide fuel cells (SOFC)	Solid metal oxides	Porous NiZnO_2	Porous LaMnO_3	650-1000°C
5	Solid Polymer electrolyte fuel cells (SPEFC)	Polymer membrane containing Fluorocarbon with SO_3H group	Pt	Pt	80°C

Hydrogen Oxygen fuel cells

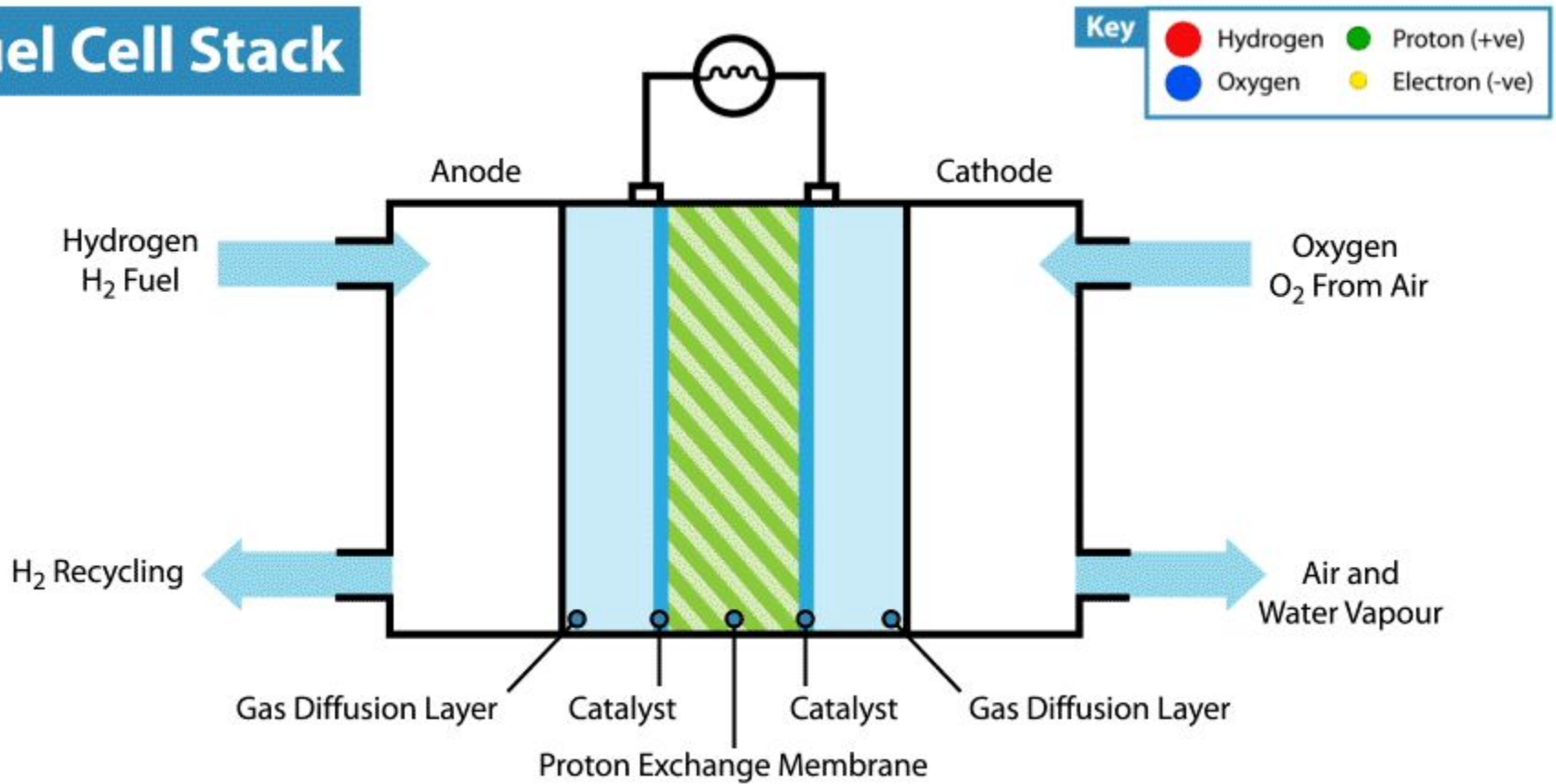


Hydrogen Oxygen fuel cells



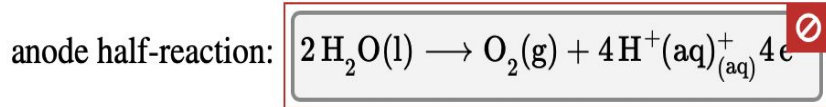
Hydrogen Oxygen fuel cells

Fuel Cell Stack

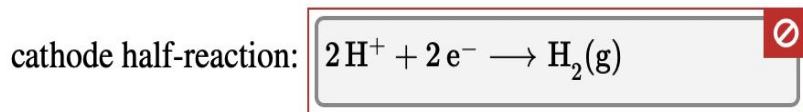


Hydrogen Oxygen fuel cells: Working

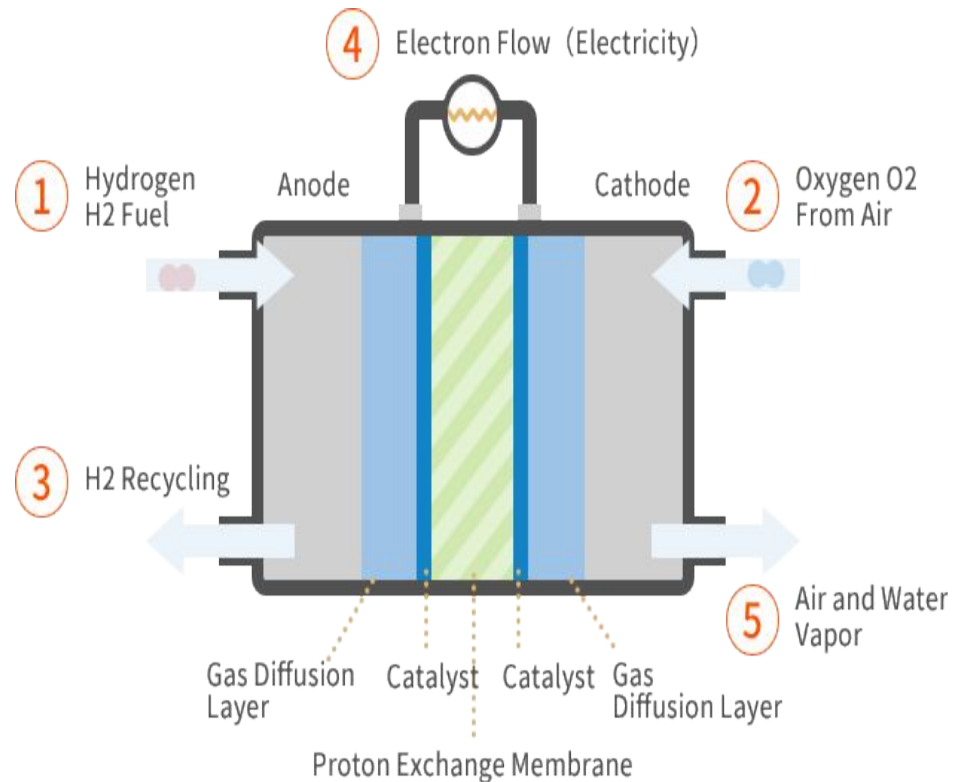
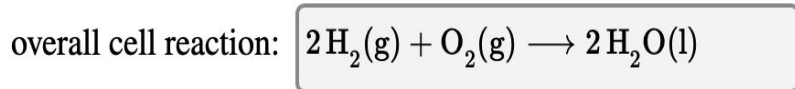
Write the balanced half-reaction that occurs at the anode in a hydrogen–oxygen fuel cell in which an alkaline electrolyte is used.



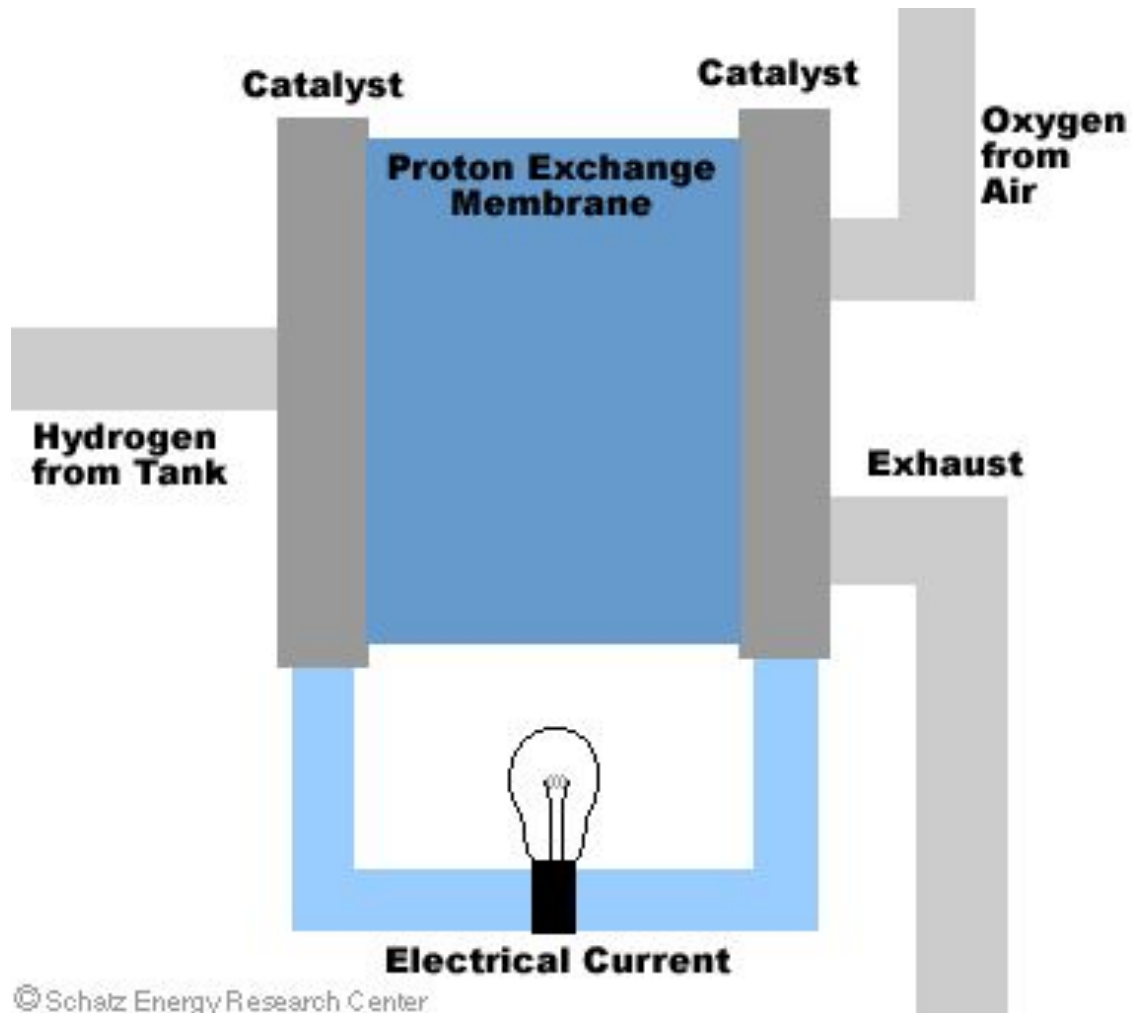
Write the balanced half-reaction that occurs at the cathode in a hydrogen–oxygen fuel cell in which an alkaline electrolyte is used.



Write the balanced overall cell reaction.



Animation of PEMFC



© Schatz Energy Research Center



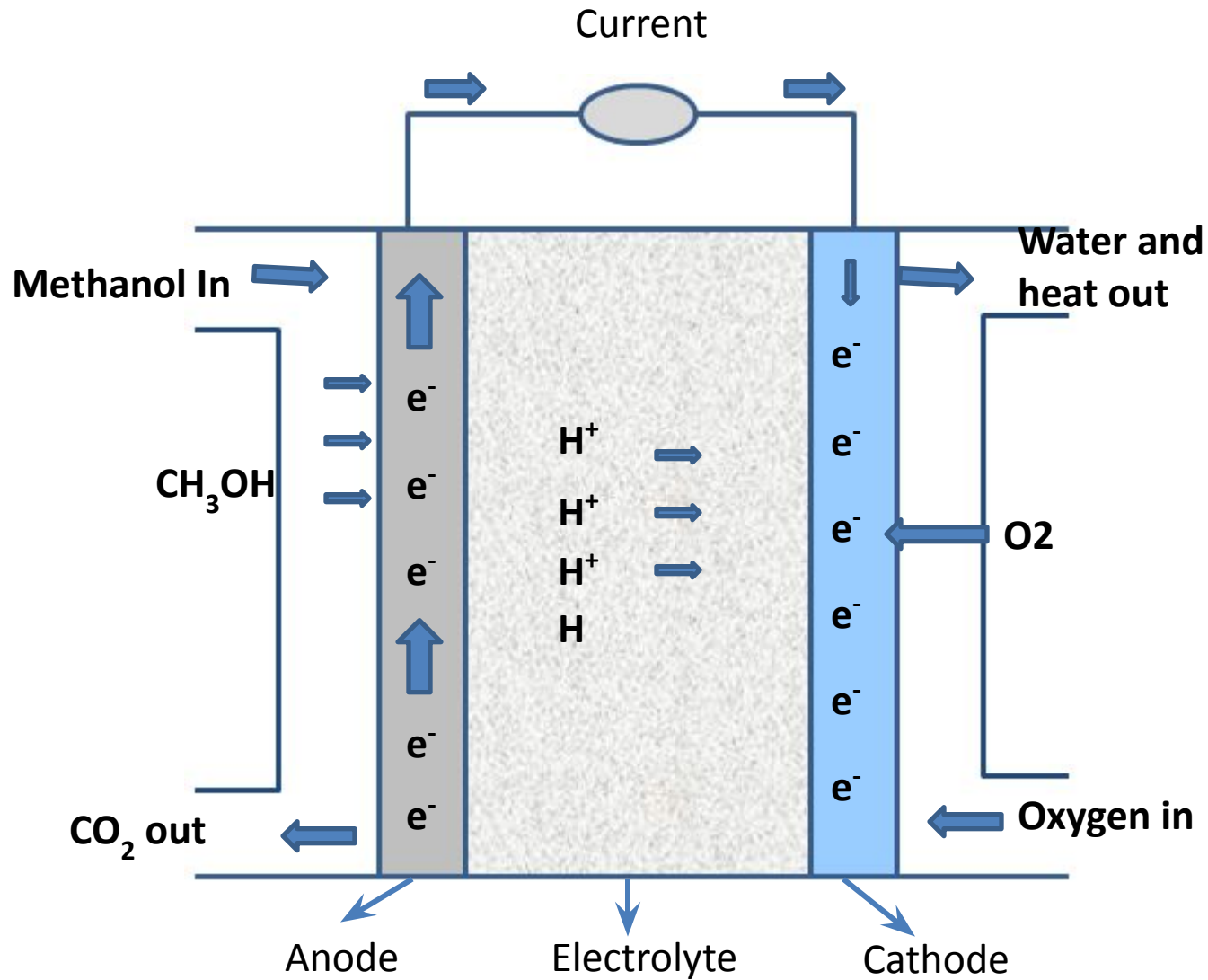
Direct Methanol Fuel Cell (DMFC)

- Sub-category of PEMFC
- Fuel at anode: Methanol ; Oxidant at cathode: Oxygen
- Membrane used: Proton exchange membrane (PEM)
- Operating temperature: 50-120°C
- Power density: 240 mW/cm²
- Fuel cell efficiency: ~60%
- Power output: 0.1 – 15W

Why methanol is preferred over hydrogen fuel ?

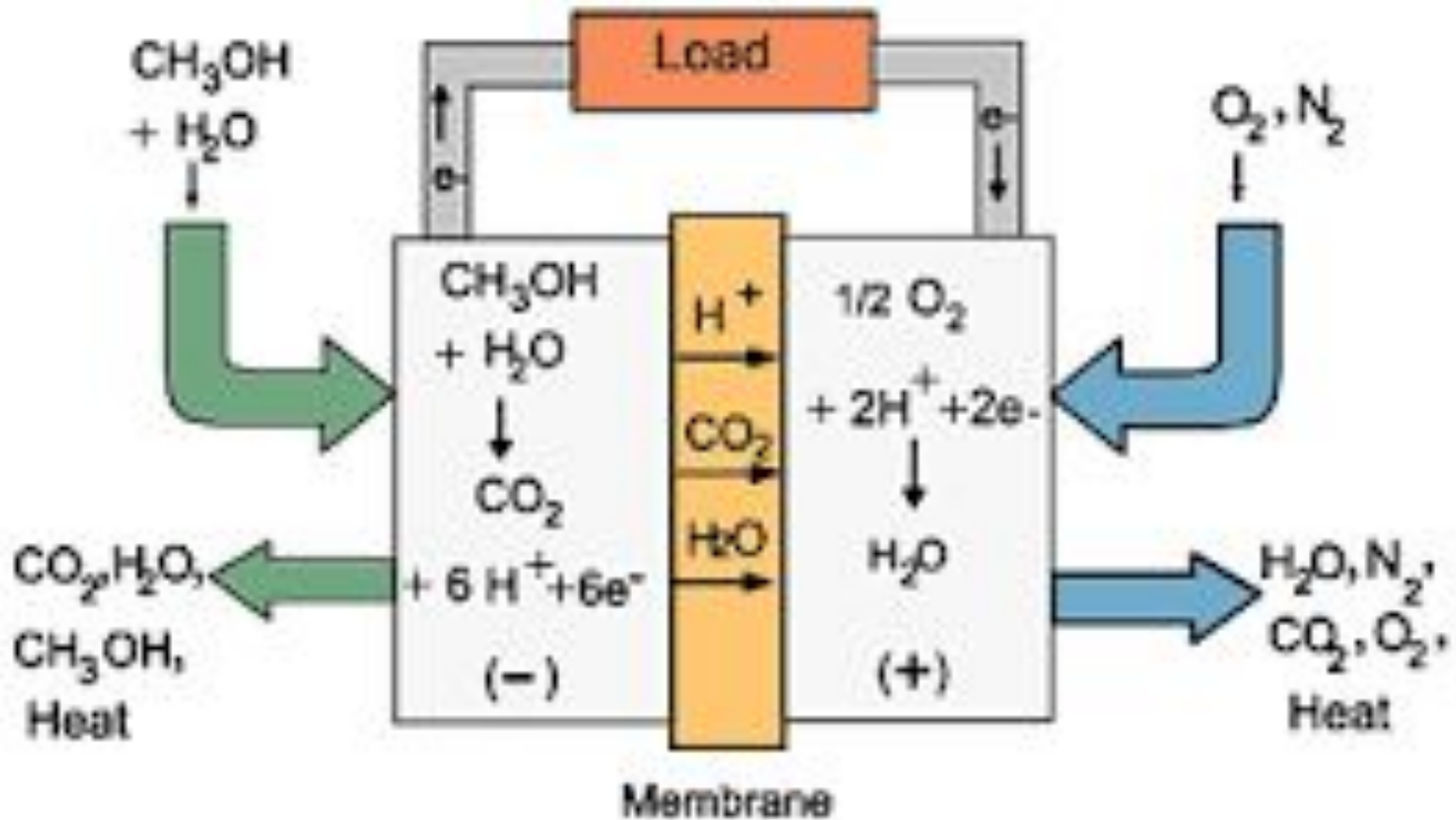
- Energy density: Methanol: 4.8 Wh/cm³
Hydrogen: 2.7 Wh/cm³
- Easy transportation and handling
- Readily available, relatively lesser cost
- Stable at all atmospheric conditions

Direct Methanol Fuel Cell (DMFC)



Methanol- Oxygen fuel cell

Methanol oxygen fuel cells

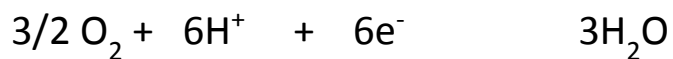


Electrochemical reactions involved in DMFC

Anodic reaction(Oxidation): 0.03 V



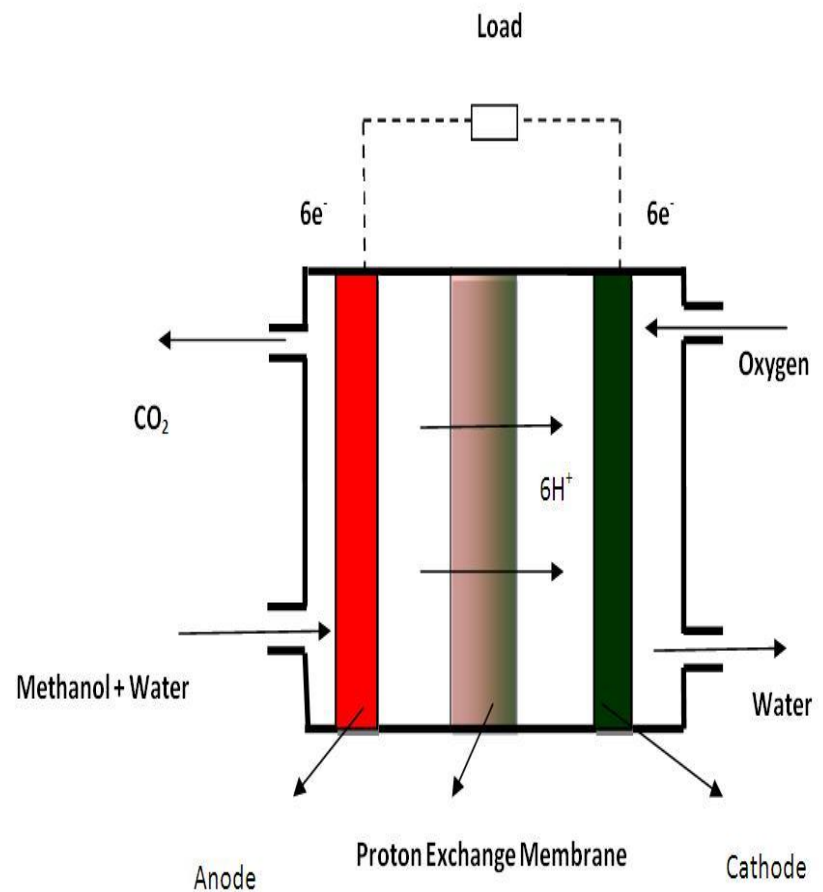
Cathodic reaction (Reduction): 1.22 V



Overall reaction: 1.19 V



(Silva at al. 2005)



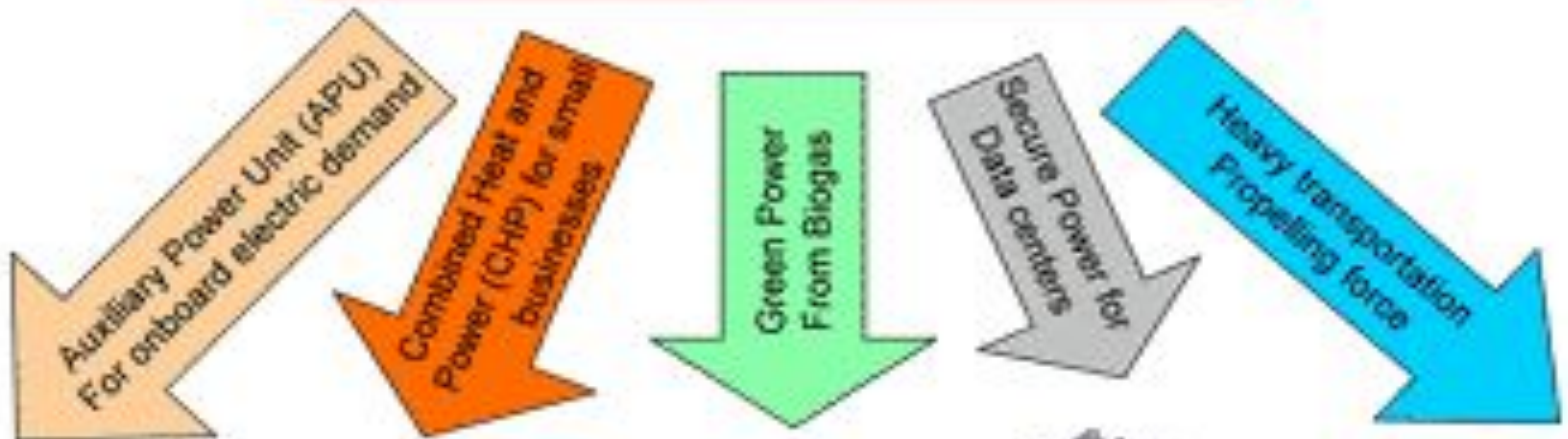
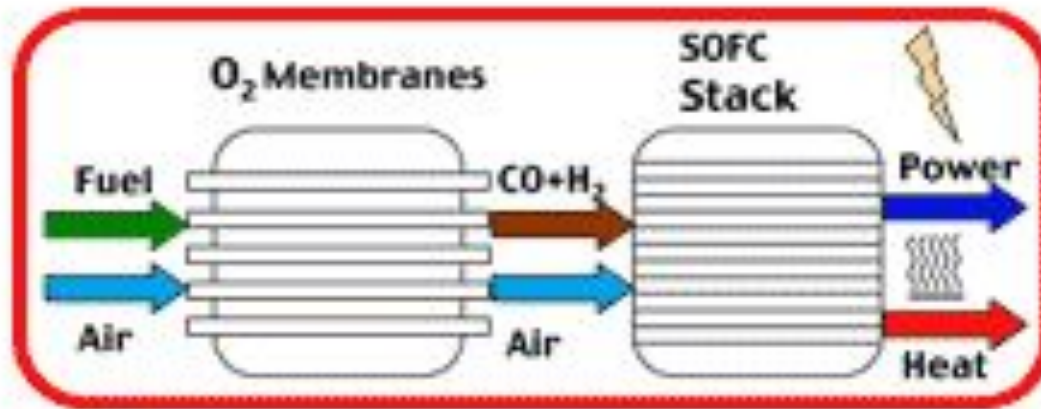


Applications of DMFC

All kinds of portable, automotive and mobile applications like,

- Powering laptop, computers, cellular phones, digital cameras
- Fuel cell vehicles (FCVs)
- Spacecraft applications
- Any consumables which require long lasting power compare to Li-ion batteries

Applications of Fuel cells





Attractions of hydrogen FCVs

- Zero pollution emissions
- Much lower noise
- The hydrogen could be produced by electrolysis (splitting) of water using electricity supplied from renewable wind, solar or hydro sources of energy
- Thus, zero greenhouse gas emissions and sustainable energy supply



Fuel cell vehicles

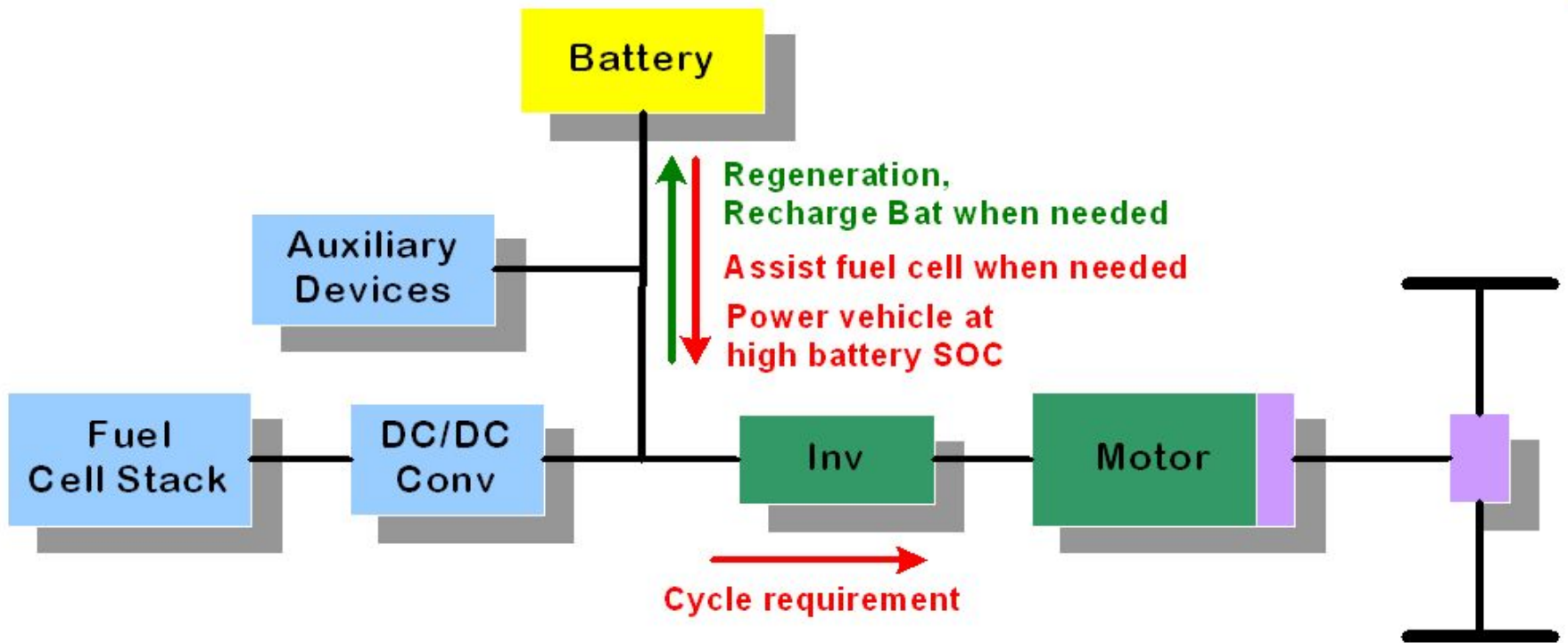
- A hydrogen FCV would operate at a typical efficiency of $\sim 60\%$, which is **more than three times** the efficiency (17%) of a typical ICE gasoline-engine vehicle today
- This in turn reduces the amount of energy (as H_2) that needs to be stored on the vehicle for a given driving range by over a factor of 3
- This in turn is critical because any system of onboard hydrogen storage will be bulky and/or heavy in relation to the amount of energy stored
- The high efficiency also greatly reduces the amount of wind or solar power that would need to be installed in order to produce enough hydrogen to replace petroleum for transportation



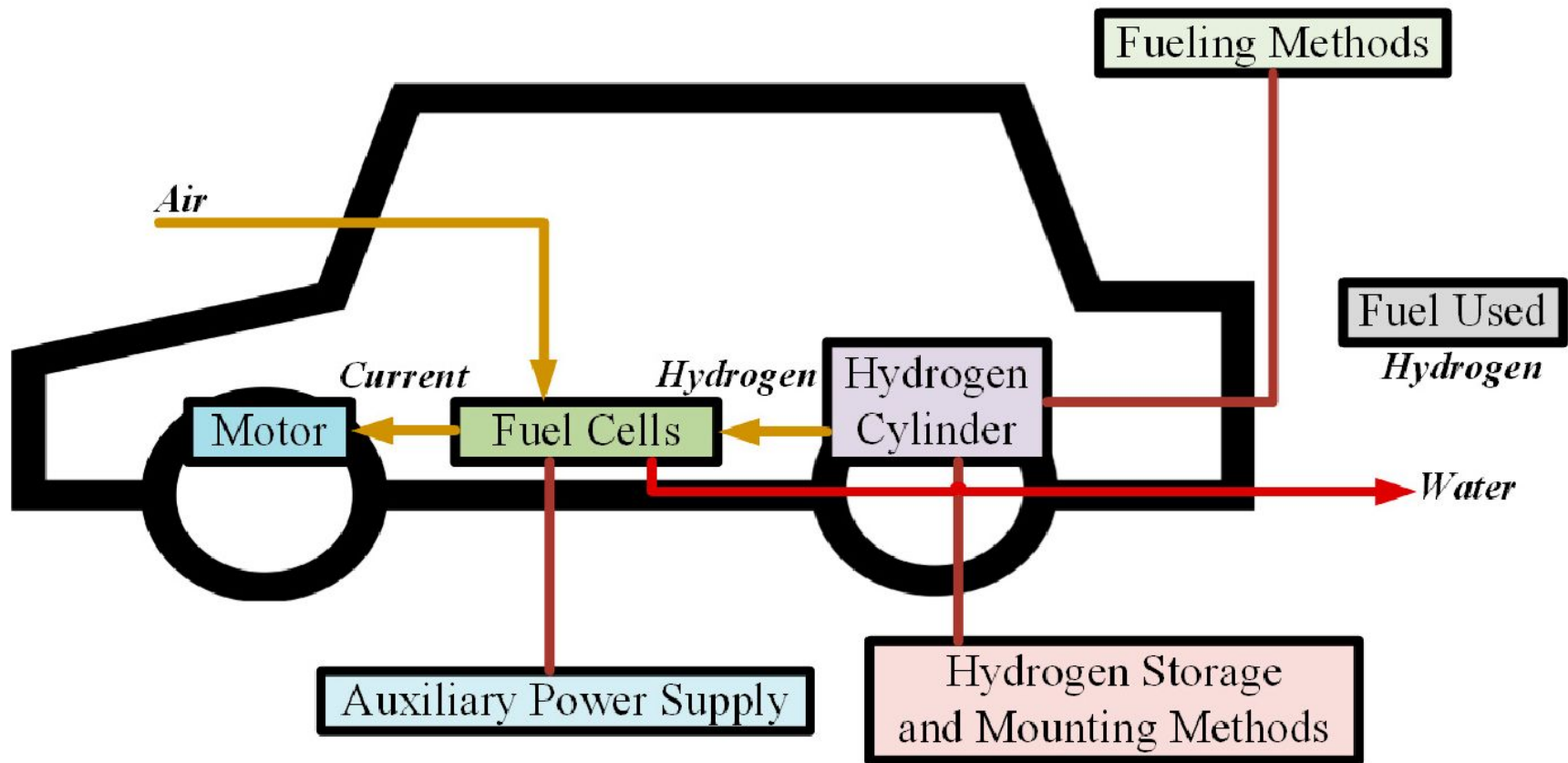
Problems

- Fuel cells suitable for use in cars need to be able to operate at low temperature (120°C)
- Low-temperature fuel cells require precious-metal catalysts (Pt and ruthenium) in order to operate (these catalysts are also needed in 3-way catalytic converters, but would not be needed for such in H₂ FCVs)
- Supplies of Pt are quite limited – the availability of Pt could be a significant constraint on the long-term viability of H₂ FCVs

Battery fuel cell hybrid



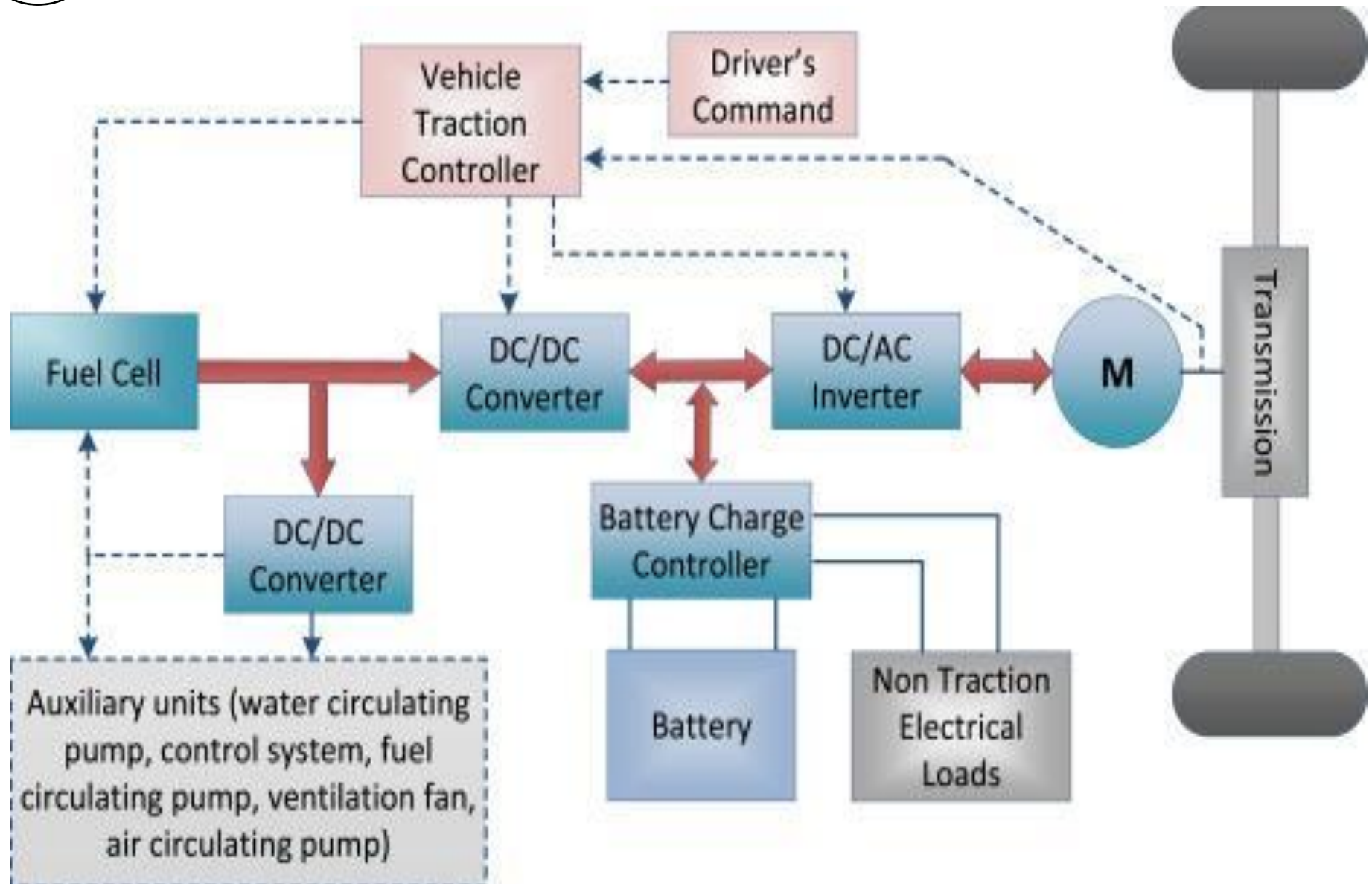
Battery fuel cell hybrid



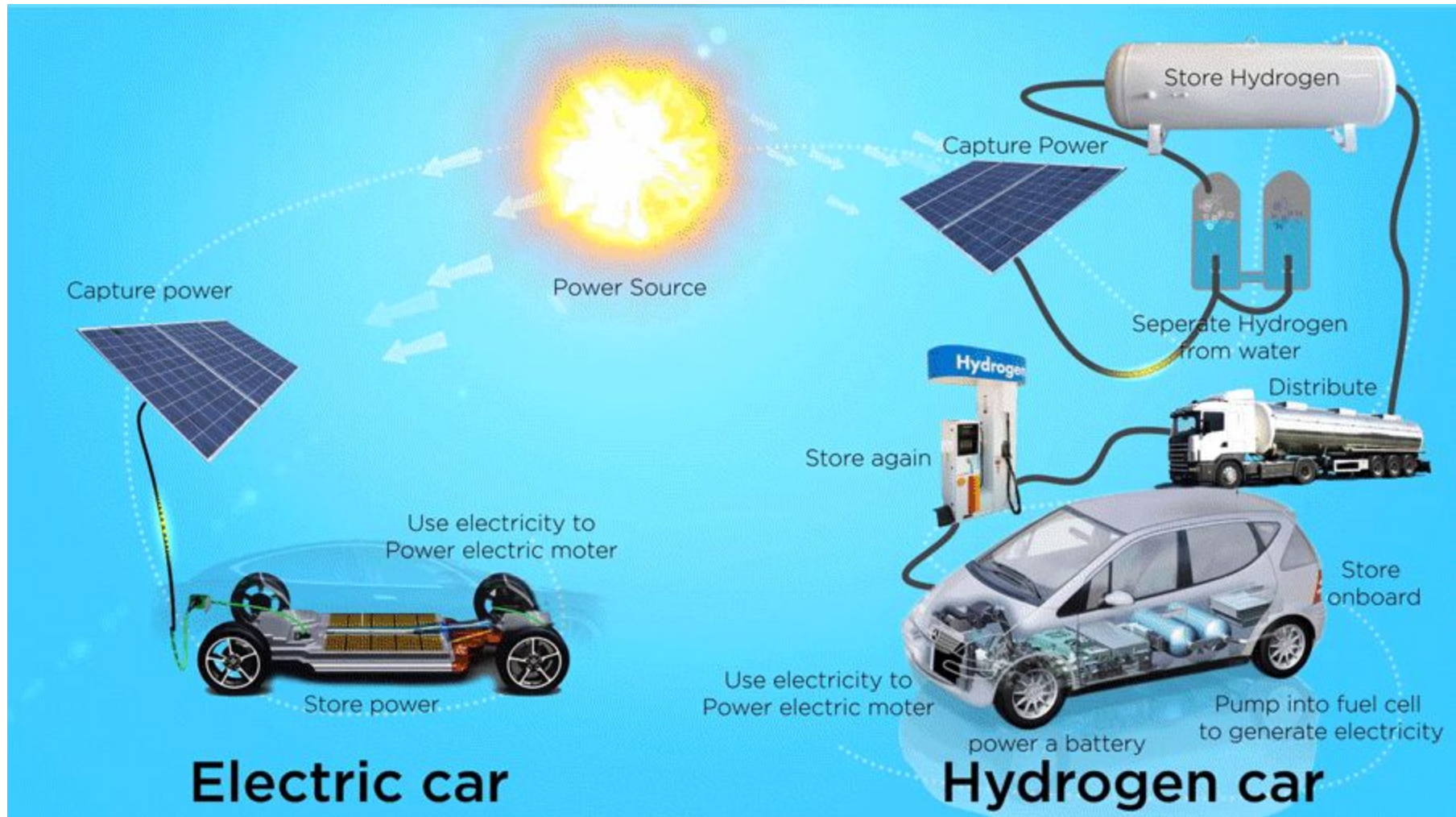
Green GT

Pininfarina H2 Speed (0-100 in 3s)

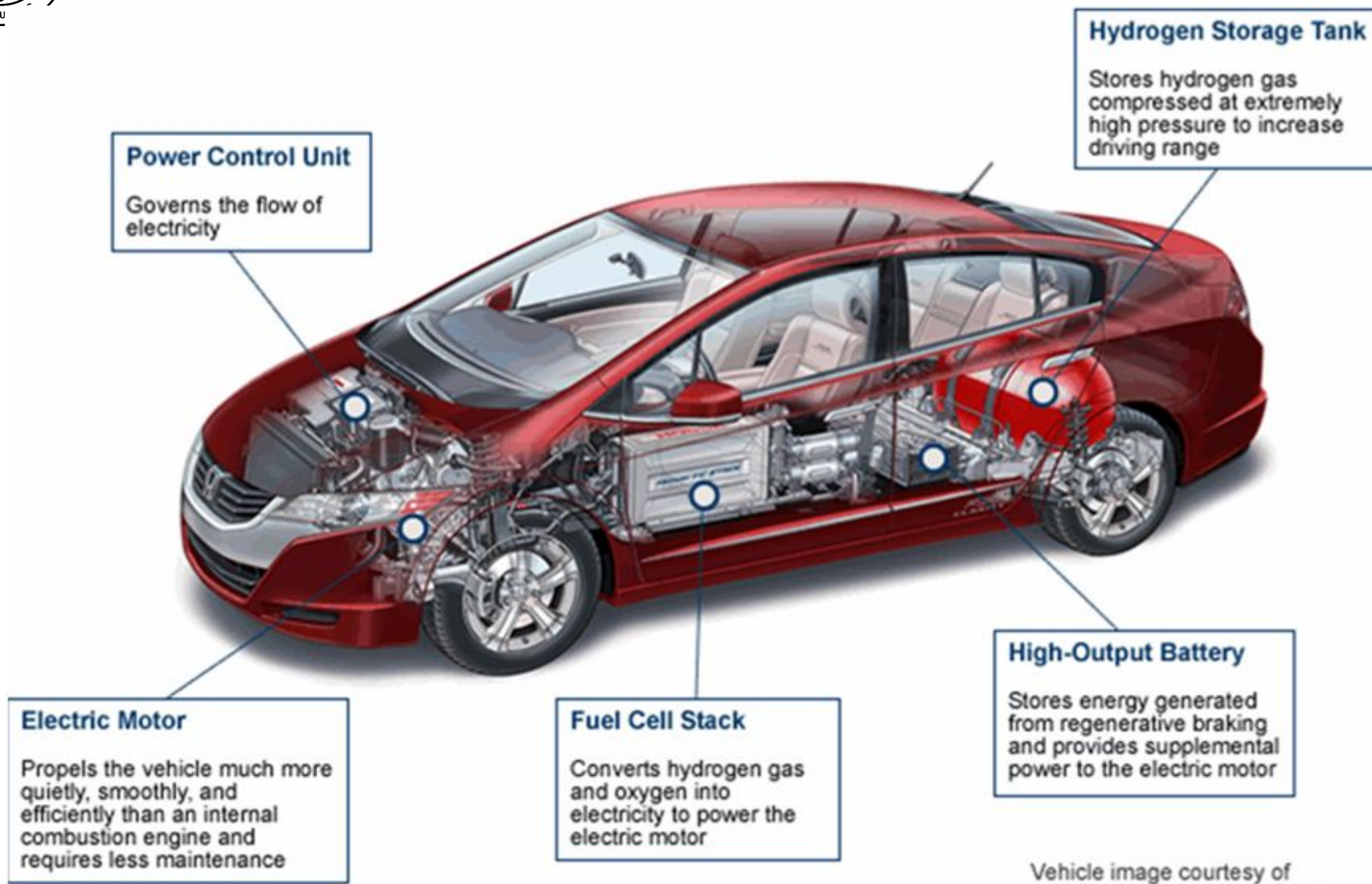
Basic infrastructure of Fuel cell hybrid electric vehicles



Production of hydrogen



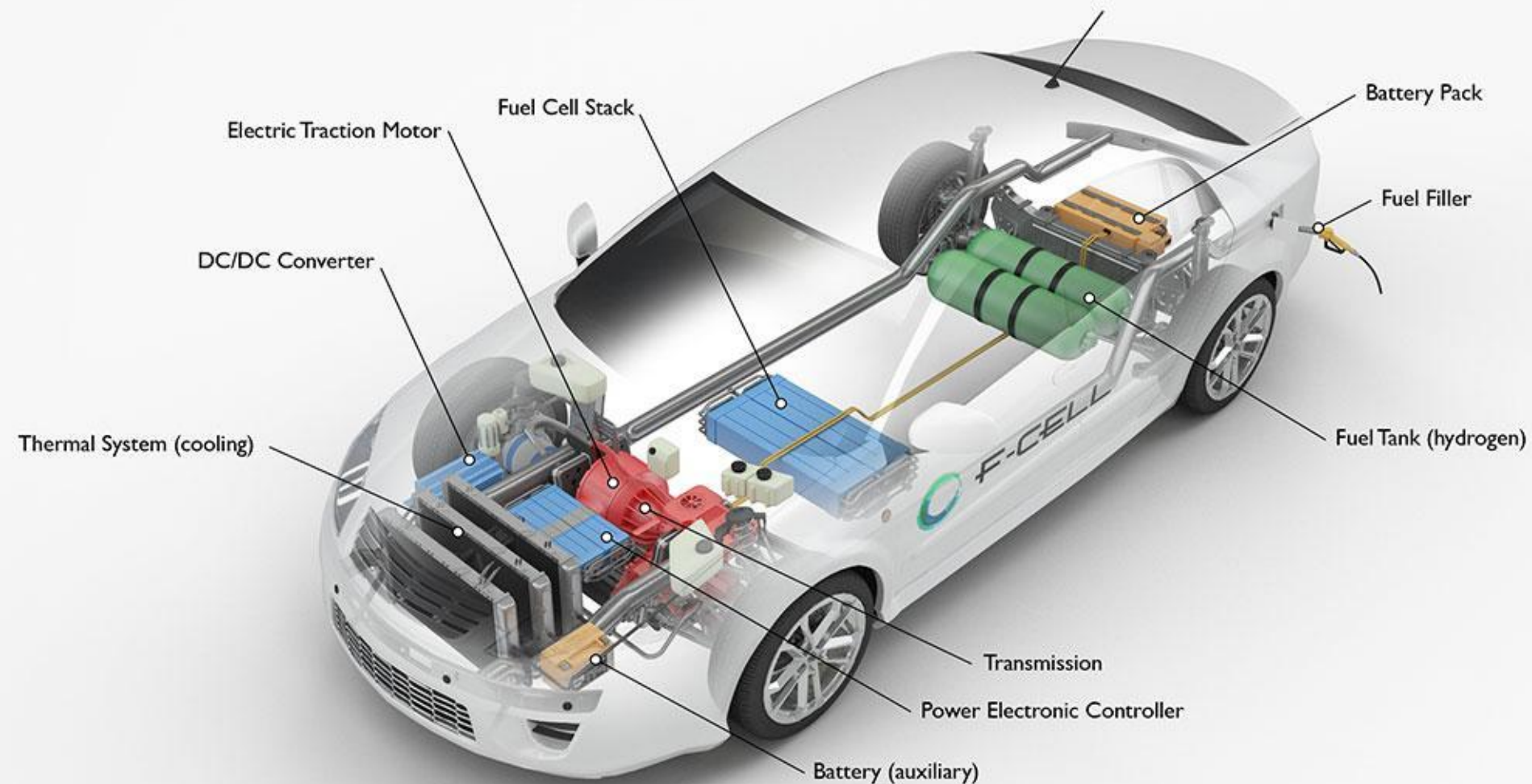
Battery fuel cell hybrid



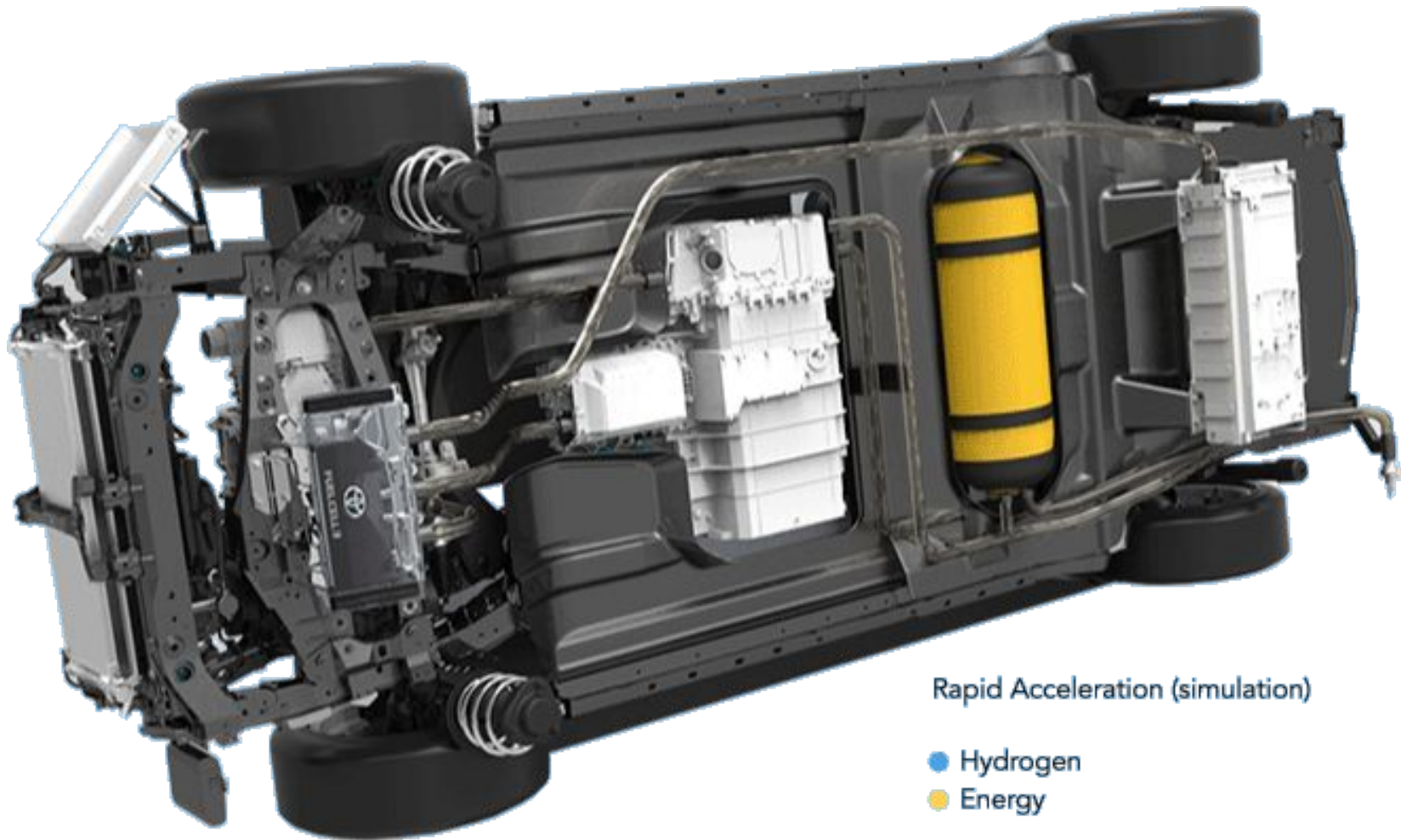
Vehicle image courtesy of
American Honda Motor Co., Inc.

Battery fuel cell hybrid

Hydrogen Fuel Cell Vehicle



Battery fuel cell hybrid working mechanism





Steps, solar energy to H₂ Fuel cell battery hybrids

- PV modules, 15% efficiency
 - PV to electrolyzer coupling, 85%
 - Production of hydrogen by electrolysis and compression to 30 atm, 80%
 - Transmission 1000 km at 30 atm pressure, 98%
 - Compression from 30 to 700 atm pressure, 90%
 - Fuel cell (generates electricity), 50% or better
 - Drive train (motor, transmission), 87%

Overall sunlight to wheels energy transfer: **3.9%**



Steps, solar energy to methanol to fuel cell (methanol is another candidate as a fuel for fuel cells)

- Photosynthesis, 1% efficiency
- Biomass to methanol, 67%
- Transport of methanol, 98% (say)
- Fuel cell to produce electricity,
45% (less than using H_2)
- Drive train (motor, transmission), 87%

Overall sunlight to wheels energy transfer: **0.26%**



Conclusion:

- Direct use of renewably-based electricity to recharge batteries makes far better use of the renewable electricity than using it to make H_2 to for use in a fuel cell (extra steps mean extra losses)
- The land area required to convert sunlight to H_2 and drive a given distance is ~ 20 times less than growing biomass to make methanol for use in a fuel cell, or ~ 40 times less than growing biomass to make ethanol
- This is largely because the efficiency of PV modules ($\sim 15\%$ or more) is vastly greater than the efficiency of photosynthesis ($\sim 1\%$)